

Module Introduction

Hello Dear Learners,

Well come to your second module on cost and management accounting. This module is a continuation of your pervious semester course in cost and management accounting. In the first module you studied the fundamentals of cost and management accounting and how cost data are processed and presented for financial statement preparation and managerial decision making purpose.

The focus in this module is on the application of cost and management techniques and tools for managerial decision making process. The management process refers to the different cyclical functional activities and all that activities in the process could not be performed without the help of management accounting information. Therefore to be successful in you carrier as an accountant who process and cost and management information and or taking part in various operational and strategic management decisions you need to critically go through the topics in this module, do the exercise and problems. The module contains seven units. On the first chapter you will be acquainted with the concept of management accounting and how it helps in managerial decisions making. The second unit deals with how budget as management accounting is used in planning in the organization. The third and fourth unit will expose you to the application of budget and variance analyses as a management control function. On the last three units, the cost volume profit analysis, relevant information and transfer pricing are discussed.

MODULE OBJECTIVES

After completing the course in this module you will be able to:

- Describe how cost and management accounting is useful in the management decisions
- To apply the concept, techniques and tolls of management accounting in managerial decision at various functional and level of management an organization

- Understand and use of budget in planning and controlling organizational performance
- Understand the concept of standard costing and variance analysis
- Perform variance analysis and find out reason for variance for budgeted and actual performance
- Understand and apply the concept of cost-volume-profit analysis in profit planning
- Able to differentiate relevant and irrelevant information in decision making
- Understand transfer pricing

CHAPTER ONE

Introduction to Management Accounting

UNIT OUTLINE

- ✓ Unit objectives
 - 1. Introduction
 - 1.1. Meaning and Role of Management Accounting
 - 1.2. The Management Process and Accounting
 - 1.3. Financial and Managerial Accounting
- ✓ Unit Summary
- ✓ Model Exam Questions

Unit Objectives:

After completing this unit, you will be able to:

- Explain the meaning and role of management accounting
- Explain how management accounting information system facilitate functions in the management process
- Explain the difference and relationship between managerial and financial accounting.

1. Introduction

To manage diverse operations of organizations managers need monetary and nonmonetary information that helps them to analyze and solve problems by reducing uncertainty. Accounting, often referred to as the language of business, provides much of that necessary information. Accounting language has two primary “variations.” financial accounting and management accounting. Cost accounting is a bridge between financial and management accounting.

Accounting information addresses three different functions: (1) providing information to external parties (stockholders, creditors, and various regulatory bodies) for investment and credit decisions; (2) estimating the cost of products produced and services provided by the organization; and (3) providing information useful to internal managers who are responsible for planning, controlling, and decision making, and evaluating performance.

Financial accounting is designed to meet external information needs and to comply with generally accepted accounting principles. Management accounting attempts to satisfy internal information needs and to provide product costing information for external financial statements.

Activity1. List and explain the three major functions of accounting information

1.1. Meaning and Role of Management Accounting

Management accounting can be defined as the process of identification, measurement, accumulation, analysis, preparation, interpretation, and communication of financial as well as non financial information used by management to plan, evaluate, control within the organization and to assure appropriate use and accountability for its resources.

The management accountant is expected to provide timely, accurate information-including budgets, standard costs, variance analysis, support day-to-day operating decisions, and analyses of expenditures.

The management accounting consists of accounting techniques and procedures of gathering and reporting financial, production, and distribution data in order to meet management's information needs.

The importance or role of management accounting can be explained in the following points:

1. **Planning:** business planning is a basic function of management. For example, deciding what products to make, and where and when to make them, determining the materials, labor, and other resources that are needed to achieve desired output. In not-for-profit organizations, deciding which programs to fund. During this planning function management needs the help of the Management Accounting system.
2. **Effective Control:** Accounting information is useful in control and as a means of communication and motivation. Periodically, management needs to evaluate as to how well the employees are doing their jobs. Such as an appraisal of performance results in corrective action of various kinds.

In general management accounting greatly assists the management in achieving better results by making a clear shift in emphasis from mere recording of transactions to their analysis and interpretation to give a new vista to the management.

1.2. Management Process and Accounting

The management process summarizes the major activities performed by management in leading an organization. A manager's work generally involves a cycle:

- (a) Setting organizational objectives,
- (b) Formulating an operating plan,
- (c) Implementing the plan,
- (d) Measuring the results,
- (e) Evaluating the results to see if the plan was properly implemented and the objectives of the organization are being accomplished.

Organizational objectives are not changed frequently, although slight modifications may be made annually to keep them current and at a realistic level of aspiration.

The **operating plan** is formulated for a specific period of time, such as one year, and is prepared in detail. Central to each of the planning activities, there is the accounting system. Financial resources available in the business, as indicated in accounting reports, are frequently a limiting factor in the development of organizational objectives. The planning process results in a budget representing the formal plan of operations for the coming year. The accounting function **measures the results** of operations and compares them with the planned level of operations. Differences between the actual results and planned results help management to **evaluate** each area of the business and to identify areas in which corrective action is required

Activit1.2:

1. *Define management accounting and explain its role in an organization.*

2. *Explain how management accounting assists managers under the management process.*_____

1.3. Financial and Managerial Accounting

The primary differences between these two accounting disciplines are described as follows: Financial accounting must comply with the generally accepted accounting principles (GAAP) or International Financial Reporting Standards (IFRS) depending upon the accounting and reporting practice adopted by a country. The information used in financial accounting is typically historical, quantifiable, monetary, and verifiable. These characteristics are essential to the uniformity and consistency needed for external financial statements. Financial accounting information is usually quite aggregated and related to the organization as a whole. In some cases, a regulatory agency such as the Inland Revenue and some agencies having the mandate of supervising specific sectors such as National Banks may mandate financial accounting practices. In other cases, financial accounting information is required for obtaining loans, preparing tax returns, and understanding how well or poorly the business is performing.

By comparison, management accounting provides information for internal users. Because managers are often concerned with individual parts or segments of the business rather than the whole organization, management accounting information commonly addresses such individualized concerns rather than the “big picture” of financial accounting. Management accounting is not required to adhere to generally accepted accounting principles in providing information for managers’ internal purposes. It is, however, expected to be flexible in serving management’s needs and to be useful to managers’ functions. A related criterion is that information should be developed and provided only if the cost of producing that information is less than the benefit of having it. This is known as cost-benefit analysis. These two criteria, though, must be combined with the financial accounting information criteria of verifiability, uniformity, and consistency, because all accounting documents and information (whether internal or external) must be grounded in reality rather than whim.

The objectives and nature of financial and management accounting differ, but all accounting information tends to rely on the same basic data system and set of accounts. The accounting system provides management with a means by which costs are accumulated from input of materials through the production process until completion and,

ultimately, to cost of goods sold. Although technology has improved to the point that a company can have different accounting systems designed for different purposes, some companies still rely on a single system to supply the basic accounting information. The single system typically focuses on providing information for financial accounting purposes, but its informational output can be adapted to meet most internal management requirements.

*Activity 1:3. Explain the difference and similarities between financial and managerial accounting.*_____

Model Examination Questions

PART I: MULTIPLE CHOICE QUESTIONS

- (1) Pick the wrong statement from the following about management Accounting.
 - A) It uses information from cost accounting.
 - B) Solely manufacturing businesses uses it.
 - C) It provides information for decision-making.
 - D) Only [A] and [C]
 - E) None of the above.
- (2) The management process in an organization involves the following, except
 - A) Setting organizational objectives.
 - B) Formulating an operating plan.
 - C) Measuring and evaluating results.
 - D) All of the above.
 - E) None of the above.
3. Managerial accounting:
 - A) involves the financial history of an organization.
 - B) Is governed by GAAP.
 - C) Focuses primarily on the needs of personnel within the organization.
 - D) Provides information for parties external to the organization.
 - E) Focuses on financial statements and other financial reports.
4. Managerial accounting differs from financial accounting because financial accounting is:
 - A) more oriented toward the future.

- B) Primarily concerned with external financial reporting.
 - C) Concerned with qualitative information.
 - D) Heavily involved with decision analysis and implementation of decisions.
 - E) None of the above.
5. Which of the following statements represents a similarity between financial and managerial accounting?
- A) Both are useful in providing information for external users.
 - B) Both are governed by GAAP.
 - C) Both draw upon data from an organization's accounting system.
 - D) Both use similar methods of reporting financial statements.
 - E) Both are solely concerned with historical transactions.

Answer to Model Examination Questions

- 1. C 2. E 3. C 4. B 5. C**

UNIT TWO

MASTER BUDGET AND RESPONSIBILITY ACCOUNTING

UNIT OUTLINE

- ✓ Unit objectives
 - 2.1. Introduction
 - 2.2. Budget in general
 - 2.2.1. Strategic planning and its implementation
 - 2.2.2. Budget and the budget Cycle
 - 2.2.3. Purpose of budget
 - 2.2.4. Types of budget and Budgeting techniques
 - 2.3. Budget in business organization
 - 2.3.1. Introduction
 - 2.3.2. Process of developing a master budget
 - 2.3.2.1. Process of developing an operating budget
 - 2.3.2.2. Financial budget
 - 2.4. Responsibility Accounting
 - 2.5. Budget Administration
 - 2.5.1. Budget Manual
 - 2.5.2. Behavioral Implication of Budget
 - 2.5.3. Budgetary Slack
- ✓ Unit summary
- ✓ Model Examination Questions

UNIT OBJECTIVES

After completing this unit, you will be able to:

- Explain the purpose of budgeting
- Explain how strategic planning is related to budgeting
- differentiate the different types of budget budgeting techniques
- prepare a master budget
- Explain the importance of cash budget in the master budgeting process
- Explain how adopting responsibility accounting facilitate management control

2.1. Introduction

This unit contains two different topics Master Budget and Responsibility Accounting. The first portion is about master budget which is an important management accounting tool for planning future activities and controlling current operation in the organization.

Budgets are crucial to the ultimate financial success of any organization. Budgets are so important, mainly because they serve as road map towards achieving organizational goals. Budgets as a management accounting tool helps management in planning, controlling and performance evaluation. In this unit you will study how budget is used in planning the operation of an organization.

2.2. Budgeting in General

There are different types of organizations in today world. Generally these organizations can be divided as profit making organization and not for profit organizations. The main objective of profit making organization is making profit. There for in a for profit oriented company, decisions made by management are intended to increase or at least maintain profit. Success is measured to a significant degree by the amount of profit the organizations earn. A not for profit organization is an organization whose goal is something other than earning a profit for its owners. Usually its goal is to provide service. In not for profit organization, decisions made by management ordinarily are intended to produce the best possible service with the available resources.

Success in a not for profit organization is measured primarily by how much service the organization provided and by how well these services are rendered. Most basically, the success of a not for profit organizations is measured by how much it contributes to the public well being.

Since service is vague and less measurable concept than profit, it is more difficult to measure performance in not for profit organization. Despite these complications, management must do what it can do to assure that resources are used efficiently and effectively.

Despite the difference in the objective of organization, all of them have to plan what they want to achieve. Planning is the process of establishing enterprise objective. There should

be agreement among and all levels of management as to the objective of the company and the proposed means of accomplishing them.

Developing a budget is a critical step in planning any economic activity. This is true for business, for government agencies and for individuals. We must all budget our money to meet day to day expense and plan for the major expenditure, such as buying a car or paying for college tuition. Similarly, business of all types and government units at every level must make financial plans to carry out routine operation, to plan for major expenditure and to help in making financing decision.

Most people associate the word “budget” with the approving, rejecting or arguing over various budgets. Tax payers demand that governments plan the effective use of their hard earned tax dollars and budget not only allow government to plan spending , but also allow tax payers to see exactly where and how their money is being spent. Government and government agencies, however, tend to use budget only as a means of limiting spending. In contrast, most business organizations use budget to focus attention on company operation and financial not just to limit spending. Budget highlights potential problem and advantage early, allowing management to take steps to avoid these problems or use the advantages wisely. Thus, a budget is a tool that helps managers in both their planning and control function. A budget is a formal written summary (statement) of management plan for a specific future time period expressed in financial terms. It normally represents primary means of communicating agreed up on objectives throughout the business organization. Once adopted, a budget becomes an important basis of for evaluating performance. Thus, it promotes efficiency and serves as a deterrent to waste and inefficiency

<i>Activity 2.2: Define budget and explain why organizations prepare budget.</i> _____
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2.2.1. Strategic Planning and Its Implementation

Planning is the first function of management. It is performed continuously because the passages of time demand both re-planning and making new plans. More over current feedback often necessitates newly planned action to,

- Improve current performance deficiency
- Cope with unanticipated events that are unfavorable, and
- Take advantage of new development

Management planning is a process that includes the following five stapes

1. Establish enterprise objective and goals
2. Developing premise about the environment of the entity
3. Making decision about course of action
4. Initiating actions to activate the plans
5. Evaluating performance for re planning

The development of organization objectives is the most fundamental level of the planning process. Objective states the desired, broad, long range future state of the organization .For example, the objective for a manufacturing Company should relate to such basic issue as breadth of the product line, quality of a product, growth expectations etc. The next planning level is known as goals, which represent the broad objective brought in to sharper focuses by explicitly specifying

- The time dimension for attainment
- Quantitative expression and
- Subdivision of responsibility

For example, goals would explicitly state such items as the following. Three years from now the new product being developed will be introduced. The return on investment goal for the next year will be 15% and the profit goal of product A is 5% of sales for next year. To establish the foundation for the attainment of the enterprise objective and specific goals, management must develop strategies to be pursued by the entity.

Strategy specify the “how”; they detailed the plan of attack to be used in pursuing the goals operationally. For example, the strategy for a company may include expanding the current sales territory, reducing the selling price to attract higher volume, increasing the advertising and financing the expansion with debt rather than equity.

Finally, the most detailed level of planning occurs when management operationallize the objective, goals and strategies already established by incorporating them in the budget. A budget is a financial and narrative expression of the expected result from the planning decision.

2.2.2. Budget and the Budget Cycle

Most people associate the word “budget” with the approving, rejecting of resource spending. If we associate budget with the government activities, governments usually request their various agencies to prepare their resource requirement so as to examining and approve the reasonableness and importance of the budget. Once the approved, the budget then will be used as a blue print for the agencies activities and means of controlling and limiting their spending.

In contrast, most business organizations use budget to focus attention on their companies’ operation and financial implication of their planned operation; not just to limit spending. Budget highlights potential problem and advantage early, allowing management to take steps to avoid these problems or use the advantages wisely.

Thus, a budget is a tool that helps managers in both their planning and control function. A budget is a formal written summery (statement) of management plan for a specific future time period expressed in financial terms. It normally represents primary means of communicating agreed up on objectives throughout the business organization. Once adopted, a budget becomes an important basis for evaluating performance. Thus it promote efficiency and serves as a deterrent to waste and inefficiency

☞ Budget in brief is a future plan of action expressed in quantitative terms which is also an aid to management control and performance evaluation

Budget can cover both financial and non financial aspects of the plan that can serve as blue print for the organization to follow in an upcoming period. A budget covers financial aspects and quantities of management expectation regarding income, cash flows, and financial position. Like financial statements present the historical financial condition and operating results of the business, budgeted balance sheet, cash flow and income statement are also prepare to show the future financial condition and operational performance. Budgeted financial statements are usually supported by detail schedule of

the various operation of the firm, so budget also include nonfinancial aspects of the plan such as units of output to be produced and sold, number of employee and working hours, etc.

Budget is a cyclical and sequential activity. In a well managed companies, budget usually cycles through the following steps:

1. Planning the performance of the company as a whole, as well as planning the performance of its subunits (such as department or divisions). Managers at all level agree on what is expected.
2. Providing a frame of reference, a set of specific expectations against which actual results can be compared
3. Investigating variations from plans. If necessary, corrective action follows investigation
4. Planning again, in light of feedback and changed condition.

2.2.3. Purposes of Budget

☞ ***Dear learners, based upon the explanations given above, can you mention why budget is prepared?***

Budget prepared as a formal business plan is used by all managers at different functional areas and managerial level. Further budget is used by all types of organizations, be it a business organization, government organization or NGO. When administered wisely budget can provide the following benefits:

i. Efficient Allocation of Resources

Resource available to meet the objective of any organization is generally limited; therefore efficient allocation of resource is one of the prerequisite for successful attainment of organizational goal. For example, an office of a city Administration must allocate its revenue among basic societal service such as security and protection, health, education, infrastructure etc. In the case of business organizations, the well designed business strategy hardly become successful without availability and efficient allocation of resource. Therefore, adopting formal budgetary process helps organization to identify the

resource requirement of the planed activity and allocate in accordance to the priority of each operation in achieving org-anizational objective.

ii. Compel strategic planning and implementation of plans

The budgeting process forces managers to plan ahead. The development of budget triggers managers to plan their operation ahead as well as to prepare on the ways of talking any change during the implementation of the plan.

Budget enable the successful implementation of strategy that is why in most business organization budget is considered as an integral part of strategic planning and implementation.

iii. Facilitating coordination and communication

For any organization to be effective, each manager throughout the organization must be aware of the plan made by other managers. In large and diverse organizations the problem of coordination becomes critical. An important role of budgeting is to improve the coordination among the various units of the organization. Planning or budgeting means establishing objectives in advance and identifying the steps by which the objectives are to be accomplished. The planning process initiates coordination and clarification of sub-goals to achieve major enterprise goals. The coordinated plan or budget provides a blue print for implementation and control.

 ***A good budgeting process facilitates communication in all direction in the organization and help coordinating the various resources, manpower and units of the organization so that goal of the organization is achieved.***

iv. Frame work for judging performance

Once plans are in Place, Company's performance can be measured against the budget established for those plans. Budget can overcome two limitations of using past performance as abases of judging actual results. one limitation is that past results incorporate past misuse and sub standard performance and the other limitation of

using past performance is that the future conditions may be expected to differ from the past.

☞ ***As a performance evaluation basis budgeted performance are better than actual results.***

V. Motivating Managers and Employees

Research shows that budgets that are challenging improve performance. An inability to achieve budget numbers is viewed as failure. Most individuals are motivated to work more intensely to avoid failure than to achieve success. As individuals get closer to goal they work harder to achieve it. For this reason many executives like to set challenging but achievable goal for their subordinates. Creating attitude of anxiety improves performance, but overly ambitious and unachievable budget increase anxiety without motivation that is because individuals see little chance of avoiding.

Activity 2.2.3: Discuss the benefits that a company may derive from a formal budgeting process_____

2.2.4. Types of budget and budgeting Techniques

The type of budget used by different organization differs based upon the nature of their business and the purpose of the budget; however, the general framework is the same. In this section we will try to see the different type of budget their advantage and disadvantage and in what circumstance organizations prefer to adopt a specific type of budget and budgeting techniques.

- (1) **Strategic Plan:** The most forward looking budget is the strategic plan, which sets the overall goals and objective of the organization. Some organization won't classify the strategic plan as an actual budget though because it does not deal with a specific time frame and it does not produce forecasted financial statement. In any case, the strategic plan leads to long range planning which produce forecasted financial statement for five or ten years. The financial statements are estimates of what management would like to see in the company's future financial statement.

Decisions made in long range planning include addition or deletion of department, acquisition of a new equipment or building and other long term commitment.

(2) **Capital Budget:** Capital budget is a budget that details the planned expenditure for facilities, equipment, new product, and other long-term investments.

(3) **Master budget:** A master budget is a short-term, comprehensive plan to achieve the financial and operational goals of an organization. Master budget comprises of the organizations overall plan for the given period and the budget for the various functional areas the make up the organization.

Activity2.2.4(A) : Can you explain the difference between long range and short term planning_____

☞ ***Long rang plane and budget gives an organization a direction and goals for the future while short term plane and budget guide the day to day operation. Both long term and short term budgets are relevant for archiving the overall goal of an organization, so managers are advised to give a reasonable attention to both short and long term budgets.***

Managers who pay attention to only short term budget will quickly lose sight of long term goals similarly managers who pay attention to only the long term budget could wind up mismanaging day to day operation. There has to be a happy medium that allows managers to pay attention to their short term budget while still keeping an eye on long term plan.

Master budget can be prepared as a standalone for one year or one operating cycle or in a continuous basis. Continuous budget or rolling budget or revolving budget are a very common form of a master budget that simply add a month in the future as the month just ended is dropped. Budgeting thus becomes an ongoing instead of periodic process. Continuous budgets for managers to allow think about the next twelve months not just the remaining month in fixed budgeting cycle. As they added a new twelfth month to continuous budget, managers may update the other month as well. They can compare actual monthly result with both the organization plan and the most recent revised plan. Continuous budgeting approach in preparing master budget is adapted mostly when the business environment is volatile to coup up with the change.

Different organizations prepare budget using different techniques that may be grouped as follows:

- (1) **Incremental budgeting:** is a budget set based on past year's actual performance. In this technique a budget for the coming year is simply this year budgeted or actual results plus or minus some amount for expected change on planned operation or change in the market price. This budgeting technique is easy and widely used, however it has its own draw back. As the base is the current year performance or budget any anomaly in the current year performance or budget may be incorporated in the budget.
- (2) **Zero based budgeting:** In a dynamic business it often makes sense to 'start afresh' when developing a budget, rather than basing ideas too much on past performance. In this technique each budget is therefore constructed without much reference to previous budgets. Preparing a budget afresh is usually required in most business organizations, where the business environment is volatile that require continues effort of incorporating changes in budget thinking.
- (3) **Rolling budgets:** Given the speed of change and general uncertainty in the external environment, shareholders seek quick results. US companies typically report to shareholders every three months, compared with six months in the UK. Rolling budgets involve evaluating the previous twelve months' performance on an ongoing basis, and forecasting the next three months' performance.
- (4) **Strategic budgeting:** This involves identifying new, emerging opportunities, and then building plans to take full advantage of them. This is closely related to zero based budgeting and helps to concentrate on gaining competitive advantage.
- (5) **Activity based budgeting:** This examines individual activities and assesses the strength of their contribution to **company** success. They can then be ranked and prioritized, and be assigned appropriate budgets.

Activities 2.2.4. (B):

2.2.1. differentiate the types of budget_____

2. List and explain the different types of budgeting techniques

3. *Can you explain how zero based budgeting can help in reducing the drawbacks of incremental budgeting?*

2.3. Budgeting in business organization

2.3.1. Introduction

The type of budget and the extent of the budgeting activity vary considerably from organization to organization. In smaller business organization, there may only be a sales forecast, a production budget or a cash budget, larger organization generally prepare a master budget or a comprehensive budget.

A master budget involves the development of a complete set of financial statement for the budget period with supporting schedule. The primary responsibility for developing a master budget is given to the controller and her or his staff. In large organization, a special budget committee will be formed.

The budget committee is usually composed of several key executive from various segment of the organization. People from finance, sales, purchasing, production, engineering and accounting are usually represented. The procedure followed by this committee in developing the budget is largely determined:

- By the authority it has over the finance budget
- By the amount of participation it allows from others within the organization.

The authority of the budget committee is determined by top management philosophy; top management may have a predetermined profit objective in mind and will look to the budget as a means to accomplish it. This objective may be stated in variety of ways, such as rate of return on net asset, earning per share, or a specific amount of net income. It may be based on operating results of previous years adjusted for expectations about the

coming year or some desired level of profitability. When top management has a predetermined profit objective, the budget committee must recognize it and develop a budget that will achieve it.

If top management has no specific profit level in mind, the budget committee must first develop some notion about what is fair and reasonable expectation for the budget period without this, the budget process often turn in to “game” and much of the benefit is lost.

The budget committee may or may not invite other members in the organization to participate in developing the budget. In estimating sales for the coming period, for example, sales people may be asked to project the number of units of each product they expect to sell in their territories.

The sales representative on the budget committee would use these as a basis for developing the sales forecast for the entire company participation could be carried to the extreme and every person in the organization could asked to estimate productivity in her or his individual area. On the other extreme, the budget committee may allow no participation. It merely may develop a budget that will achieve the desired profit and pass it on as the standard of performance for the budget period. More will be said about the behavioral considerations associated with employee participation in developing the budget.

Activity 2.2.1.

1. Who are the members of a budget committee in business organizations?

2. Is there any difference between the budgeting process of large and small business organizations? Explain _____

2.3.2. Process of Developing a Budget

Although each organization is unique in the way it puts together its budget, all budgeting process share some common elements. After organizational goals, strategies, and long range plans have been developed, work begins on the master budget, a detailed

budget for the coming fiscal year with some detail.

The master budget is a comprehensive financial plan for a business. It is made up of the Operating and Financial budgets, which are in turn made up of supporting schedules (budgets).

To envision the master budget process, picture the financial statements most commonly prepared by companies: The income statement, the balance sheet, the cash flow statement. Then imagine the preparation of these statements before the fiscal period operational period.

2.3.2.1. Parts of A Master Budget

What are the parts of a master budget?

As shown on figure on the next page master budget consists of two major parts, namely: the operating budget and financial budget.

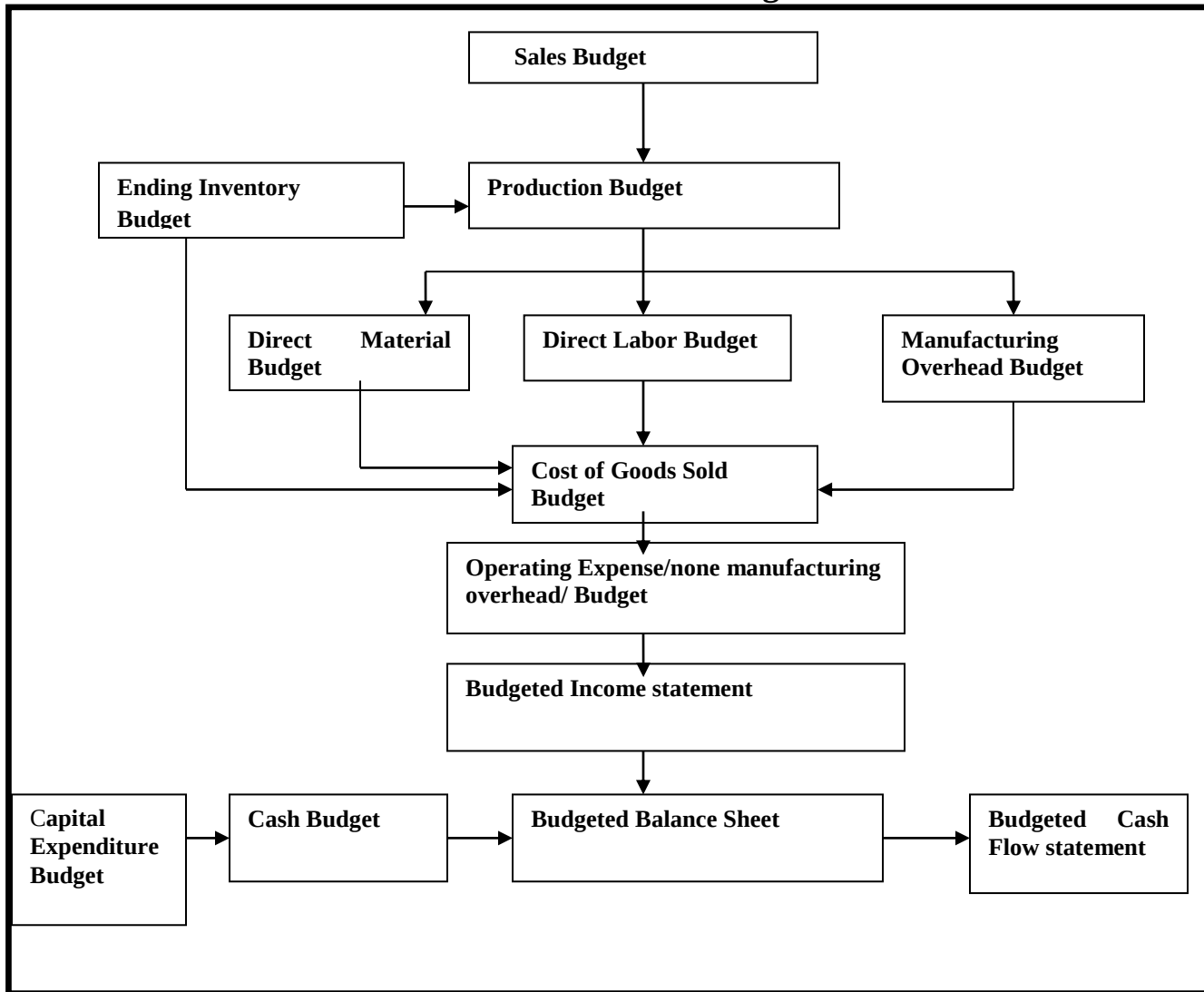
- i **Operating budget** refers to the budgeted income statement and the supporting budget schedules for various business functions in the value chain. The operating budget basically shows the expected operating result of the organization in the upcoming operational period.
- ii **The financial budget** is part of the master budget made up of the capital expenditures budget, the cash budget, the budgeted balance sheet, and the budgeted statement of cash flow.

Activities 2.3.

1. *What are the two major parts of a master budget?*

2. *What is the difference between an operating and financial budget?*

An Overview of a Master Budget



2.3.2.2. Steeps in Budget development

☞ *Dear learners, from the above parts of this unit you have seen how budget is useful in an organization, the types of budget and budgeting techniques. As you all understand in most business organizations there is a formal process of preparing budget for the coming fiscal in the form of a comprehensive master budget. If you are asked to prepare a master budget how can you start? Is it the operating budget or the cash budget that must be prepared? What are the relevant data and how they are processed and result the final master budget?*

Well, on the following section you are provided with answer for the above questions with illustrative examples

2.3.2.2.1. Steps in developing an operating budget

The Operating Budget refers to the budgeted income statement and all the supporting schedules.

☞ Dear learners, if you are given a responsibility of preparing an operating budget or coordinating the budgeting process, do you know where and how to start?

One way to think about this question is to understand that the organization has more control over some aspects of the business (for example how much to produce) and less control over other aspects, (the demand for its product and service).

For most organizations sales is uncertain. Therefore, beginning with sales forecast, the firm can plan the activities over which it has more control. As better information about sales becomes available, it is reasonably easy to adjust the rest of the budget. If, on the other hand, production is more uncertain than sales, the firm may want to begin with a raw material and production forecast so as to reduce the uncertainty related to production. To clearly understand the steps in development of an operating budget, consider the budget information gathered by the controller of Gibe Furniture Manufacturing company during the process of budgeting for the upcoming fiscal year, 2011.

The summary of required budget information obtained from different operating units, such as sales related information from the marketing department, production related information from production department, direct and indirect labor related information from the human resource department, and other manufacturing and non manufacturing overhead budgets from other departments as well as assumptions taken for the development of an operating budget are given as follows:

- (1) The only source of revenues is sales of tables and unit sold is the only revenue driver.
- (2) Work in Process inventory is negligible and is ignored.
- (3) Unit costs of direct materials purchased and finished goods sold remain unchanged throughout each budget year.
- (4) There are two types of Direct materials : Lumber and Metal
- (5) There are two types of direct labor: Laminating labor and Machine labor.

Direct labor rates remain unchanged throughout each budget year.

- (6) For computing inventor able costs, Gibe Furniture allocates all manufacturing overhead costs using manufacturing labor hours as the allocation base.

- (7) Numerical information

- (a) Each table has the following product specifications:

Direct materials:

Lumber----- 9 board feet/table

Metal----- 10 board feet/table

Direct labor:

Laminating labor-----0.25Hrs/table

Machine labor-----3.75 Hrs/table

- (b) Inventory information in physical units for 2011.

	<u>Beginning Inventory</u>	<u>Target Ending Inventory</u>
Direct materials:		
Lumber-----	20,000 board feet	18,000 board feet
Metal-----	25,000 board feet	22,000 board feet

Finished goods:

Tables----- 5,000 tables 3,000 tables

- (c) Revenue expected from sales of tables for 2011 are:

Selling price-----Br. 392/table

Units' sales----- 52,000 tables

- (d) Costs expected for 2011.

	<u>2010</u>	<u>2011</u>
Lumber/ board feet -----	Br. 3.90	Br. 4.00
Metal/ board feet-----	5.80	6.00
Laminating labor/ Hr. ----	24.00	25.00
Machine labor/ Hr. -----	29.00	30.00

- (e) Other budgeted costs and amounts for 2006 are:

- Variable non-manufacturing costs----- 13.5% of sales
- Fixed non-manufacturing costs----- Br. 1,400,000
- Variable MOH costs----- Br. 9.50/DL Hr.
- Fixed MOH costs----- Br. 1,600,000

- (f) The inventorable cost is Br.275/table in 2005.

Now you can see each step in the preparation of the operating budget using the budget information given above.

Step 1: Preparing Revenue budget

The starting point for operating budget development for most business organizations is a revenue budget. A revenue/Sales budget outlines the expected sales for each product in units and Birr. This budget will be developed after the firm made a forecast of the

demand for the company's product by taking into account. In forecasting sales different companies may adopt different forecasting techniques, however in most cases sales forecast take into account the following points:

- ☞ The sales volume in recent periods
- ☞ General industry and economic condition
- ☞ Market research studies
- ☞ Pricing policies
- ☞ Advertising & sales promotion
- ☞ Competition & regulatory policies

Based upon the forecasted sales the budgeted sale is prepared by a mere multiplication of forecasted sales volume and selling price.

Budgeted Sales = Budgeted sales volume X Budgeted Selling price
--

The Revenue budget for Gibe Furniture, based upon budget information 7(C) is,

Schedule (1) Sales Budget

Budgeted Unit	52,000tables
X Budgeted Unit Selling Price	Br 392/table
Budgeted Sales	Br. 20,384,000

Step 2: preparing the production budget (in units)

After the revenues are budgeted, you would then prepare a Production Budget. The production budget is prepared to show how many units must be produced in order to meet your budgeted sales need and the target level of ending inventory balance for finished goods.

The total number of budgeted production requirement, is therefore, the sum of budgeted sales in unit and target ending inventory. However, if the firm is not new in operation, usually some of its production requirement can be satisfied using the inventory kept of the beginning of the period. Therefore, the binging in vestry should be deducted from the total production requirement to determine the exact units in the production budget

Formula wise you can put the production budget in unit as follow:

Budgeted Production in unit Finished	= Budgeted sales in unit	+ Target Finished Goods inventory	- Begging goods inventory
---	---------------------------------	--	----------------------------------

The production budget for Gibe using the given information is prepared as follows

Schedule (2) Production budget

Budgeted unit sales	52,000
Add: Target Finished Goods Inventory	3,000
Total requirements	55,000
Less: Beginning FG Inventory	5,000
Units to be produced	<u>50,000</u>

After the sales and production budgets have been developed and the efforts of the sales and production groups have been coordinated, the next stage is the development of the production costs (direct material, direct labor and manufacturing overheads) at budgeted output level.

Step 3: Preparing the Direct Material Usage and Direct Material Purchase Budget

The number of units to be produced calculated in production budget Schedule is the key to computing direct materials in quantity and Birr. The direct materials budget ties the production to the Direct Materials that will need to be purchased in order to produce the estimated units. Direct materials purchases needed for the budget period are determined using this equation:

Required		material to be	+	Estimated	Estimated
Material	=	Used in Production		ending materials	- beginning
Purchase				inventory	materials inventory

The direct material usage budget in our case is prepared as follows:

Schedule 3A: DM Usage Budget

	<u>Lumber</u>	<u>Metal</u>	<u>Total</u>
<u>Physical Units Budget:</u>			
Lumber: 50,000 x 9 board feet	450,000b.f		
Metal: 50,000 x 10 board feet		500,000b.f	
To be used in production	<u>450,000b.f</u>	<u>500,000b.f</u>	
<u>Cost Budget:</u>			
-Available from Beginning Inventory.			
Lumber: Br. 3.90 x 20,000	Br. 78,000		
Metal: Br. 5.80 x 25,000		Br. 145,000	
-To be obtained from purchases			
Lumber: Br. 4.00 x 430,000	Br. 1,720,000		
Metal: Br. 6.00 x 475,000		<u>Br. 2,850,000</u>	
<i>DM to be used</i>	<i><u>Br. 1,798,000</u></i>	<i><u>Br. 2,995,000</u></i>	<i><u>Br. 4,793,000</u></i>

Schedule 3B: DM Purchases Budget

	<u>Lumber</u>	<u>Metal</u>	<u>Total</u>
<u>Physical Units Budget:</u>			
Production Budget	450,000b.f	500,000b.f	
Add: Target End. Inv.	<u>18,000b.f</u>	<u>22,000b.f</u>	
Total requirements	468,000b.f	522,000b.f	
Deduct: Beg. Inv.	<u>20,000b.f</u>	<u>25,000b.f</u>	
Units to be purchases	<u>448,000b.f</u>	<u>497,000b.f</u>	

Cost Budget:

Lumber: Br.4.00 x 448,000	Br. 1,792,000		
Metal: Br. 6.00 x 497,000		<u>Br. 2,982,000</u>	
Purchases	<u>Br. 1,792,000</u>	<u>Br. 2,982,000</u>	<u>Br. 4,774,000</u>

Sep 4: Preparation of direct manufacturing labor cost budget

This budget will show the number of employee and total hours required in producing the budgeted level of output along with the cost. The costs in this budget usually depend on wage rate, production method and human resource plan.

Schedule 4: DL Budget

	<u>Laminating Labor</u>	<u>Machine Labor</u>	<u>Total</u>
<u>Labor-hours Budget:</u>			
Laminating: 50,000 tables x 0.25Hrs.	12,500Hrs.		
Machine: 50,000 tables X 3.75Hrs.	<u> </u>	<u>187,500Hrs.</u>	
Total DL Hrs. required	<u>12,500Hrs.</u>	<u>187,500Hrs.</u>	

200,000Hrs

Cost Budget:

Machine: Br. 25/Hr. x 12,500Hrs.	Br. 312,500		
ML: Br. 30/Hr. x 187,500Hrs.	<u> </u>	<u>Br. 5,625,000</u>	
Total DL cost	<u>Br. 312,500</u>	<u>Br. 5,625,000</u>	<u>Br. 5,937,500</u>

Step 5: Preparing Manufacturing Overhead (MOH) Budget.

The Overhead budget shows the expected cost of all indirect manufacturing items. The total of these costs depends on how individual overhead costs vary with respect to the cost driver.

Gibe Furniture treats both variables MOH and fixed MOH as inventorable costs.

Schedule 5: MOH Budget

At the budgeted level of 200,000 DL-Hours

Total variables MOH [Br.9.50 x200, 000]	Br. 1,900,000
Total fixed MOH	<u>Br. 1,600,000</u>
Total MOH	<u>Br. 3,500,000</u>

Gibe Furniture inventories MOH at the budgeted rate of Br. 17.50/DL Hr.
[i.e. Br.3, 500,000/200,000Hrs.]. The budgeted MOH cost per table is Br. 70
[Br. 3,500,000/50,000tables].

Step 6: Preparing the Ending Inventory Budget

This budget is prepared for target ending raw material and ending finished goods inventory

Schedule 6A: Computation of Unit Costs of Ending Inventory of FG

	Cost/unit of		
	<u>Input</u>	<u>Input</u>	<u>Total</u>
<u>DIRECT MATERIAL:</u>			
Lumber:	Br. 4 /b.f	9 b.f	Br. 36
Metal:	Br. 6 /b.f	10b.f	<u>Br.60</u> Br. 96.00
<u>DIRECT LABOR:</u>			
Laminating Labor	Br. 25 /Hr.	0.25Hrs.	Br. 6.25
Manufacturing Labor	Br.30 /Hr.	3.75Hrs.	<u>Br.112.50</u> Br. 118.75
MOH	Br. 17.5/Hr.	4.00Hrs.	<u>Br. 70.00</u>
Total			<u>Br. 284.75</u>

Step 7: Preparing Cost of goods sold (CGS) budget

The following are inputs to prepare cost of goods sold budget

- Direct material usage budget
- Direct labor budget
- Manufacturing overhead budget
- Ending and beginning finished goods inventory
- Ending and beginning working in process inventory

Schedule 7: CGS Budget

Beg. FG Inv., Jan1, 2006 (275 x 5000)	Br. 1,375,000
DM Used (sch.3A)	4,793,000
DL (Sch. 4)	5,937,500

MOH (Sch. 5)	<u>3,500,000</u>
CGM	<u>14,230,500</u>
Cost of goods available for sale	15,605,500
Less: Ending FG Inv. (sch.6B)	<u>854,250</u>
CGS	Br. <u>14, 751, 25</u>

Step 8: Preparation of None Manufacturing overhead Cost Budget

The non manufacturing cost budget include the marketing and administrative departments' costs required to operate the company at its projected level of sales and production and to achieve long term company goals. Unless there is a change in the organizations production and sales or level of activity, the nonmanufacturing cost budget is easily prepared by taking previous year's actual or budgeted result after making the necessary adjustment for price change and otter similar changes between periods.

Schedule 8: Operating Expenses Budget

Variable non-manufacturing costs: 13.5% X 20,384,000	Br. 2,751,840
Fixed non-manufacturing costs	<u>1,400,000</u>
Total Operating Expense	<u>4,151,840</u>

Step 9: Preparing the Budgeted income Statement

The last effort in operational budget development is pulling all the budget schedules prepared in all the above steps in to the income statement. The budgeted income statement, which can also be called Performa income statement show the revenue costs of production, operating cost and the resulting operational profit envisage in the budget period.

BUDGETED INCOME STATEMENT

Revenues (Sch. 1)	Br. 20,384,000
Less: CGS (Sch. 7)	<u>14,751,250</u>
Gross Profit	5,632,750
Less: Operating Expenses (Sch. 8)	<u>4,151,840</u>
Operating Income	<u>Br. 1,480,910</u>

2.3.2.2.2. Financial Budgets

The remaining budgets that appear in the Master Budget make up the Financial Budget. The Financial Budget typically consists of the capital expenditure budget, the Cash Budget, the Budgeted Balance Sheet and the budgeted statement of cash flows. In this

section the focus is only on the cash budget and budgeted balance sheet, as the rest are discussed in detail in other course modules in financial accounting and financial managements.

i) **Cash budget**

Cash budget is a schedule of expected cash receipt and disbursement. It predicts the effects on the cash position at the given level of operation. Cash budget helps to avoid unnecessary idle cash and unexpected cash deficiencies. They thus, keep cash balance in line with needs, ordinarily; the cash budget has the following main sections.

The beginning cash balance plus cash receipt equals the total cash available before financing.

Cash receipts depend on the collection of accounts receivable, cash sales, and miscellaneous recurring sources such as rental royalty receipts. Information on expected collectable of account receivable is needed for accurate prediction.

Cash disbursement: Organizations make cash disbursement for various reasons such as:

- Payment for direct material purchased
- Salary paid for direct labor cost and other wages
- Other disbursements for property, equipment and other long term investment
- Interest on long term borrowing
- Income tax payment
- Others,

Short time financing requirement depends on how the total cash available for needs compare with the total cash disbursement plus the minimum ending cash balance desired. If there is a deficiency of cash, loan will be taken, if there is excess cash, an outstanding loan will be paid

Activities: 1. why is the cash budgeting so important in the master budgeting process? _____

3. Why cash budget is prepared after the completion of operating budget preparation?

Suppose **Gibe Furniture Company** had the balance sheet for the year ended December 31, 2010 as follows:

ASSETS		LIABILITIES & STOCKHOLDERS' EQUITY	
		Liabilities	
Cash	Br. 500,000	Account payable	Br. 384,000
Accounts Receivable	1,881,600	Tax payable	<u>20,460</u>
Direct Materials Inventory	223,000	Total current	404,460
Finished Goods Inventory	1,375,000	Long term debt	<u>2,400,000</u>
Land	1,200,000	Total current & Liability	2,804,460
Buildings & Equipment	2,300,000	Stockholders' Equity	
Accumulated Depreciation	<u>(800,000)</u> 1,500,000	Common Stock	3,000
Total Assets	Br. 6,679,600	Retained earnings	<u>3,872,140</u> 3,875,140
		Total Liabilities & SHE	Br. 6,679,600

The quarterly cash flow based on the budgeted cash effects of the operation formulated in operating budget above is given below:

	<u>Quarter 1</u>	<u>Quarter 2</u>	<u>Quarter 3</u>	<u>Quarter 4</u>
Cash Collection	Br. 5,331,200	Br. 4,704,000	Br. 4,704,000	Br. 6,272,000
Disbursements:				
Direct Materials	960,000	1,152,000	1,152,000	1,536,000
Payroll 1,	626 300	1,626 300	1,888,600	1,626,300
Other costs	1,580,460	1,580,460	1,580,460	1,580,460
Equipment purchase	- 0-	-0-	1,800,000	-0-
Interest expense	60,000	60,000	60,000	60,000
Income tax	<u>100,000</u>	<u>120,460</u>	<u>100,000</u>	<u>100,000</u>
Total	<u>Br. 4,326,760</u>	<u>Br. 4,539,220</u>	<u>Br. 6,581 060</u>	<u>Br. 4,902,760</u>

Additional information

- Long term debt is Br. 2.4 million at an annual interest rate of 10% with 60,000 interest payable every quarter
- The company wants to maintain a Br. 100,000 minimum cash balance at the end of each quarter
- The company borrows cash in multiple of Br. 1,000 at the beginning of each quarter and repayment is made at the end of each quarter
- The company can borrow or repay money at an interest rate of 12% per year.
- Management doesn't want to borrow any money more short term cash than is necessary
- Interest is computed and paid when the principal is repaid
- Tax rate is assumed to be 36%.
- During the year the company paid Br. 420,640 income tax. This amount is the remaining due for the year 2010, plus four quarter payment of each Br. 100,000
- Equipment amounting Br. 1,800,000 was bought in the 3rd quarter
- No land is bought or sold.
- Depreciation for the year is Br. 500,000.
- Long term notes payable is not repaid.
- No dividend is paid.

To prepare a cash budget you need to get adequate information about the cash receipt and disbursement made by the organization during the budget period. Most of this information is obtained from the different schedules prepared for the operating budget parts of a master budget. In addition to cash receipts and payments for operational activities, information is required on planned investing and financing activity of the firm on the budget period. Information on the companies desired minimum cash balance is also required.

☞ Dear learner in preparing a cash budget noticing the following points are essential:

1. The ending balance of cash in one quarter is the beginning balance of cash for the next quarter.
2. In the year for total column the receipt and disbursement are totaled for the four quarters, however the beginning balance in the column is the beginning balance in the column are the beginning balance in for quarter 4.
3. Depreciation is not a cash disbursement,
4. The cash receipt and disbursement for operational activity appears in all the quarter as the budgeted operation of the company will continue in without interruption. However when you come to cash receipt and disbursement for investing and financing activities appears only on some of the quarters. For example the firm acquired fixed asset costing Birr 1,800,000, that is why the cash disbursement for investing activity appears only once in Q3.
5. When you cash receipt and payments related to financing activity the firm borrowed Birr 308,000 to finance the expected short term cash shortage at quarter III, as a result cash receipt from financing activity appeared only in this quarter. The principal amount and the interest on borrowed money are paid at quarter IV when the firm has excess cash on hand beyond the required amount of cash for planned payments and minimum cash balance requirement of the quarter, as a result cash payments for financing activity appears only in the fourth quarter.

Using all the information given above on illustration to the cash budget for Gibe Furniture can be prepared as follow:

Cash Budget for Gibe Furniture

	Quarters	Total for
--	-----------------	------------------

Descriptions	I	II	III	IV	the year
Cash balance at the beginning	Br 500,000	Br 1,504,440	Br 1,669,220	Br 100,160	500,000
Add Receipts:					
Cash collection from customers	5,331,200	4,704,000	4,704,000	6,272,000	21,011,200
Total Cash available for needs (X)	5,831,200	6,208,440	6,372,220	6,372,160	21,011,200
Deduct Disbursements:					
Direct materials	960,00	1,152,000	1,152,000	1,536,000	4,800,000
Payroll	1,626,300	1,626, 300	1,888,600	1,626,300	6,767,500
Other costs	1,580,460	1,580,460	1,580,460	1,580,460	6,321,840
Interest cost(long term debts)	60,000	60,000	60,000	60,000	240,000
Machinery purchase#	0	0	1,800,000	0	1,800,000
Income tax	<u>100,00</u>	<u>120,460</u>	<u>100,000</u>	<u>100,000</u>	<u>420,460</u>
Total cash Disbursement(Y)	4,326,760	4,539,220	6,581,060	4,902,760	20,449,800
Minimum Cash Balance desired	<u>100,000</u>	<u>100,000</u>	<u>100,000</u>	<u>100,000</u>	<u>100,000</u>
Total Cash needed	4,426,760	4,639,220	6,681,060	4,902,760	20,349,800
Cash excess (deficiency)	1,404,440	1,569,220	(307,840)	1,369,440	1,061,400
Financing					
Borrowing at the beginning	0	0	308,000	0	308,000
Repayment at the end	0	0	0	(308,000)	(308,000)
Interest at 12% per annum	<u>0</u>	<u>0</u>	<u>0</u>	<u>(18,480)</u>	<u>(18,480)</u>
Total effect of financing	<u>0</u>	<u>0</u>	<u>308,000</u>	<u>(326,484)</u>	<u>(18,480)</u>
Ending cash balance	Br1,504,440	Br 1,669,220	Br100,160	Br1,142,920	1,142,920

ii) The Budgeted Balance Sheet

The budgeted balance sheet shows the financial position of the firm during the budget period. Each item in the budgeted balance sheet is projected in the light of the details of the business plane expressed in all the previous budget schedules.

The budgeted balance sheet of Gibe furniture prepared on the bases of previously developed budget is prepared as follows:

Budgeted Balance Sheet of Gibe furniture

Assets			
Cash		Br1,142,920	
Accounts Receivable		1,254,400	
Direct Materials		204,000	
Finished Goods		<u>854,250</u>	Br 3,455,570
Property plant and equipment			
Land		1,200,000	
Building & Equipment	4,100,000		
Ac.Depreciation	<u>(1,300,000)</u>	<u>2,800,000</u>	<u>4,000,000</u>
Total Assets			<u>7,455,570</u>
Liabilities and Stock Holders' Equity			
Current liabilities			
Account Payables		358,000	
Income Tax payables		<u>40,075</u>	398,075
Long term Debt(interest 10% per year)			2,400,000
Stock Holders Equity			
Common Stock		3,000	
Retained earnings		<u>4,654,495</u>	<u>4,657,495</u>
Total Liabilities & SHE			<u>Br 7,455,570</u>

- iii) **Budgeted statement of cash flow:** statement of cash flow presents cash flow from operating, investing and financing activity in detail.
- iv) **Capital budgeting:** is the process of making long term planning decision for investment capital budgeting also called long term investment. Long term investment involves the commitment of resource for projects which have long term consequence. Such decisions usually involve large investment of money. Capital budgeting decision have uncertain actual outcome that have long-term effect on the organization. Poor long-term investment decision can affect the future stability of an organization because it is often difficult for organization to recover money tied up in bad investment. Mangers need a long term planning tool to analyze and control investment with long-term consequence.

2.4. RESPONSIBILITY ACCOUNTINNG

Responsibility accounting is an underlying concept of accounting performance measurement systems. The basic idea is that large diversified organizations are difficult, if not impossible to manage as a single segment, thus they must be decentralized or separated into manageable parts. These parts or segments are referred to as responsibility centers. An underlying concept of responsibility accounting is referred to as controllability. Conceptually, a manager should only be held responsible for those aspects of performance that he or she can control.

The purpose of a responsibility accounting system is to ensure that each manager and worker in an organization strives towards the overall goals set by top management. The basis of a responsibility accounting system is the designation of each sub-unit in the organization as a particular type of responsibility center. A responsibility center is a sub-unit in an organization whose manager is held accountable for specified financial and non-financial results of the sub-unit's activities. Thus, responsibility accounting is a system that measures the plans and actions of each responsibility center.

There are four common types of responsibility centers:

- Cost centers,
- Revenue centers,
- Profit centers, &
- Investment centers.

(i) **Cost center** is an organizational unit whose manager is responsible for costs only. Examples: Production Department of Manufacturing Companies.

(ii) **Revenue center** is a responsibility center whose manager is held accountable for the generation of revenues. Examples: Sales departments, marketing departments, etc

(iii) **Profit centers** are sub-divisions of a business assigned responsibility for both costs and revenues. Example: - A restaurant of a large hotel

(iv) **Investment centers** accountable for costs, revenues and the profitable utilization of invested capital. Example: Divisions of large companies.

The performance of each responsibility center is summarized periodically on a performance report. A performance report shows the budgeted and actual amounts key financial results appropriate for the type of responsibility center involved. Performance reports also typically show the variance between budgeted and actual amounts for the financial results conveyed in the report. The data in performance report help managers

use MBE (Management by Exception) to control an organization's operations effectively. MBE means that managers follow up on only the most significant variances between budgeted and actual results. This allows managers to use their time most effectively.

Activities 2.3:

1. *How do companies use responsibility centers and responsibility accounting?*

2. Define the following terms and provide appropriate example taking in to account a manufacturing type of business.

i) Cost Center _____

ii) Revenue center _____

iii) Profit center _____

iv) Investment center _____

2.5. Budget Administration

Large organization use a formal process to collect data and prepare a budget such organization usually designate a budget director or chief budget officer. The budget director specifies the process by which budget data will be gathered, and prepare budgets. To communicate budget procedure and deadlines to employees throughout the organization, the budget director often develops and disseminates a budget manual. The budget manual says who is responsible for the required and what form the information to take

A budget committee consisting of key senior executives is often appointed to a device the budget director during preparation of the budget. The authority to give final approval to a budget usually belongs to the board of directors in business organizations or board of trustees in many nonprofit organizations; usually the board has a subcommittee whose

task is to examine the proposed budget carefully and recommend approval or any changes deemed necessary. By exercising its authority to make changes in the budget and grant final approval, the board of directors or trustee, can wield considerable influence on the overall direction the organization takes

2.5.1. Behavioral impact of Budget

There is no other area where the behavioral implication is more important than in budgeting area. A budget affects virtually every one in an organization: those who prepare the budget, those who use the budget to facilitate decision making, and those who are evaluated using budget. The human reactions to the budgeting process can have considerable influence on an organization

2.5.1.1. Budgetary Slack: padding the budget

The implication upon which a budget is based comes largely from people throughout an organization. For example, the sales forecast relies on market research and analysis by market research staff but also incorporates the projection of sales personnel. If territorial sales manager's performance is evaluated on the basis of whether the sales budget for the territory is exceeded, what is the incentive for the sales manager in projecting sales? The incentive is to give a conservative, or cautiously low sales estimate. The sales manager's performance will look much better in the eye of top management when a conservative estimate is exceeded than when an ambitious estimate is not met. At least that is the perception of many sales managers, and in the behavioral area perception is what counts most.

When a supervisor provides a departmental cost projection for the budgetary purpose, there is an incentive to overestimate costs when the actual cost incurred in the department proves to be less than the inflated cost of projection, the supervisor appears to have managed in cost effective way.

This illustration is example of padding the budget. The budget padding means under estimating revenue or overestimating costs. The difference between the revenue or cost projection that a person provides and a realistic estimate of the revenue or cost is called budgetary slack.

For example, if a plant manager believes the annual utility cost will be Br.18, 000, but gives a budgetary projection of Br.20, 000, the manager has built Br.200 of slack in to the budget.

 Why do people pad budget with budgetary slacks?

There are three primary reasons. First people often perceive that their performance will look better in their supervisor eyes if they can beat the budget. A second budgetary slack is often used to cope with uncertainty. A department supervisor may feel confident in the cost projection for 10 cost items. However, the supervisor may also feel that some unforeseen event during the budgetary period could result in unanticipated cost. One way of dealing with that unforeseen event is to pad the budget. If some negative event does occur, the supervisor can use the budgetary slack to absorb the impact of the event and still meet the cost budget. The third reason why cost budgets are padded is that budgetary cost projection is often cut in the resource allocation process. Thus, we have a vicious circle, budgetary projection are padded because they will likely be cut and they are cut because they are likely to have need padded

 Dear learners, do you know how does an organization can solve the problem of budgetary slack?

First, it can avoid relying on the budget as negative evaluate tool. if a department supervisor is harassed by the budget director or some other top manager every time a budgetary cost projection is exceeded, the likely behavioral response will be to pad the budget. In contrast, if the supervisor is allowed some managerial discretion to exceed the budget when necessary there will be tendency toward budgetary padding. Second managers can be given incentive not only to achieve budgetary projections but also to provide accurate projection.

Activity: 2.5.

1. How can budget affect individuals in an organization._____?
2. What is budgetary slack_____

3. Why individuals tend to pad budget? Explain using example_____

4. How can organizations overcome the budgetary slack problem?

Unit summary

The unit introduces the important topic of budgets. Budgets are the primary financial planning tool used by businesses and non business organizations. If managed effectively budget is used for planning, performance e valuation, and facilitating coordination and communication as well as for better resource allocation.

Different organizations prepare various types of budget. The comprehensive set of budget that covers all phases of the organizations operation is called master budget. Master budget comprises the operation budget wish starts by sales forecasting and sale budgeting and end up by preparing other production and nonproduction cost budgets based on the sales budget and Performa income statement. The other main part of a master budget is a financial budget which includes the capital expenditure budget, cash budget and the budgeted balance sheet. The chapter explains how businesses use budgets and budgeting as part of the management process. The concept of responsibility centers and responsibility accounting is also discussed and related to the concept of controllability.

Model Examination Questions

Part I: Multiple Choice Questions

_____1. Which of the following would be included in the cash budget, but would not be included in the budgeted income statement?

- (i) Repayment of a bank loan
- (ii) Proceeds from the sale of a fixed asset
- (iii) Bad debts write off

- A (i) and (ii) only
- B (i) and (iii) only
- C (ii) and (iii) only
- D (i), (ii) and (iii)

_____2. An investment center manager is:

- 2.2.1.1.1. Responsible for costs.
- 2.2.1.1.2. Held accountable for revenues.
- 2.2.1.1.3. Held accountable for the profitable utilization of invested capital.

2.2.1.1.4. All of the above.

2.2.1.1.5. All except [A].

Based on the following information answered question number 4 and 5

Gedda super market, budgeted the following sales for indicating the month;

		<u>September 2000</u>	<u>October 2000</u>	<u>November 2000</u>
Sales on account	Birr	1800000	1920000	2040000
Cash sales	“	240000	250000	260000

All merchandise marked up to sell at its invoice cost plus 25%. Merchandise inventory at the beginning of each month 25% of the month estimated cost of goods sold.

____3. What is the cost of goods sold for the month of October 2000?

- a) Birr 1440000
- b) Birr 1870500
- c) Birr 2170000
- d) Birr 1627500
- e) None

____4. What is the projected merchandise purchase for September 2000?

- A) Birr 1912500
- B) Birr 1505625
- C) Birr 1530000
- D) Birr 2040000
- E) None

Part II: Discussion Questions

- 1) How does the process of budgeting assist managers in conducting the management control function?
- 2) The master budget contains both operational and financial budgets. What is the difference between an operating budget and a financial budget? How do they relate to each other?
- 3) Although managers are not clairvoyant, budgeting may assist in viewing the future. How might this be so?
- 4) Why is it useful to complete the budgeting process with a presentation of pro forma financial statements?
- 5) Why is the cash budget so important to an organization? If the cash budget identifies a period in which a cash shortage is expected, what actions can the organization take?

Part II: Workout Questions:

Serra Furniture is an Elite desk manufacture. It manufactures two products.

1. Executive desks – Oak desks
2. Chairman desks – Red Oak desks

The budgeted direct cost inputs for each production in 2008 are as follows:

Particulars	Executive line	Chairman line
Direct material:		
Oak top	16Sqf	-----
Red Oak top	---	25Sqf
Oak legs	4 legs	----
Red Oak legs	--	4 legs
Direct Labor:	3 Hr	5 Hr

Unit data pertaining to the direct materials for March 2008 as follows:

Actual beginning direct material's inventory (2008)

Particulars	Executive line	Chairman Line
Oak top	320 Sqf	-----
Red Oak top	---	150 Sqf
Oak legs	100 legs	----
Red Oak legs	--	40 legs

Target ending direct material inventory at March 31, 2008.

Particulars	Executive line	Chairman Line
Oak top	192Sqf	-----
Red Oak top	---	200Sqf
Oak legs	80 legs	----
Red Oak legs	--	44 legs

Unit cost data for direct cost inputs pertaining to February 2008 and March 1998 are:

Particulars	February 1998	March 1998
Oak top (per sqf)	Bir.18.00	Bir.20.00
Red Oak top (per sqf)	Bir.23.00	Bir.25.00
Oak top (per leg)	Bir.11.00	
Bir.12.00		
Red oak legs (per leg)	Bir.17.00	Bir.18.00

Manufacturing labor (per Hr)	Bir.30.00	Bir.30.00
------------------------------	-----------	-----------

Manufacturing overhead (both variable and fixed) is allocated to each desk on the basis of budgetary direct manufacturing labor hours per desk. The budgeting variable manufacturing overhead rate for March 1998 is Bir.35.00 per Direct manufacturing labor Hour. The budgeted fixed manufacturing overhead for March 1998 is Bir.42, 500. Both variable and fixed manufacturing overhead cost is allocated to each unit of finished goods.

Data relating to finished goods inventory for March 1998 are

Particulars	Executive line	Chairman Line
Beginning inventory	20 units	5 units
Beginning inventory	Bir.10, 480	Bir.4850
Target ending inventory	30 units'	15 units

Budgeted sale for March 1998 are 740 units of the executive line and 390 units of the chairman line. The budgeted selling prices per unit in March 1998 are Bir.1, 020 for an executive line desk and Bir.1600 for a chairman line desk.

Take the following assumptions in your answer:

- Works in Process inventories are negligible and ignored.
- Direct material inventory and finished goods inventory are determined using the FIFO method.
- Unit costs of direct material purchased and finished goods are constant in March 1998.

Required:

Prepare the following budget for March, 2008

- Revenue budget
- production budget in units
- Direct material usage budget and direct material purchase budget
- Direct manufacturing labor budget
- Ending inventory budget
- Cost of goods sold budget

Answer key for Model Exam Questions

Part III solutions for work out questions

Solution: 1

- (1) Revenue Budget for the year ended March31, 2008

Budgeter Revenue = Budgeted Sales in unit X Budgeted Selling Price

(1)	(2)	(3)	(2)x(3)=(4)
<u>Revenue</u>	<u>Units</u>	<u>selling price</u>	<u>Total Revenue</u>
Executive line	740	1020.00	754,800
Chairman line	390	1,600.00	624,000

(2) Production Budget in units for the year ended March 31, 1998

**Budgeted Production = Budgeted sales + Target finished goods inventory –
Beginning finished goods inventory.**

Executive line Budgeted production= 740+30-20 =750 units

Chairman Line Budgeted production=390+15-5 =400 units

(3) Direct material usage budget & Direct material purchase Budget:

a). Direct material Usage budget in units and Birr, at end of the year March 31, 2008.

Particulars	Oak	Oak	Red oak	Red oak
	<u>Top</u>	<u>legs</u>	<u>top</u>	<u>legs</u>
Direct material used in Production of executive l Line (750x16) (750x4)	12,000	3000	----	-----
Direct material used in Production of chairman Line (400x25) (400x4)	----	-----	10,000	1,600
Total Direct material Used (Sqf& Legs)	12,000	3,000	10,000	1,600
Less: Beginning Inventory (FIFO)	320	100	150	40
Cost direct material to used From purchases	11,680	2,900	9,850	1,560
Costs	<u>Executive line</u>		<u>Chairman line</u>	
A) Beginning Inventory:	<u>Oak top</u>	<u>Oak leg</u>	<u>Red top</u>	<u>Red leg</u>

1) Square feet's & Legs	320	100	150	40
2) Cost per Square Feet's & Legs (In Birr.)	20.00	12.00	25.00	18.00
3) Cost (1x2) (In Birr.)	6,400	1,200	3,750	720
B) Direct material to be used				
From current purchase:				
1) Square feet's & Legs	11,680	2,900	9,850	1,560
2) Cost per Square Feet's & Legs (In Birr.)	20.00	12.00	25.00	18.00
3) Cost (1x2) (In Birr.)	233,600	34,800	246,250	28,080
C) Total cost of Direct Material				
To be used (A3 + B3)	240,000	36,000	250,000	28,800
b) Direct material purchase budget:				

	<u>Executive line</u>		<u>Chairman line</u>	
	<u>Oak top</u>	<u>Oak leg</u>	<u>Red top</u>	<u>Red leg</u>
Direct material to be used in				
Production (Sqf & legs)	12,000	3,000	10,000	1,600
Add: Target ending Material				
Inventory:	<u>192</u>	<u>80</u>	<u>200</u>	<u>44</u>
Total Requirement:	12,192	3,080	10,200	1,644
Less: Beginning inventory				
Of Direct material (sqf & legs)	320	100	150	40
1) Direct material to be				
Purchased (Sqf * legs)	11,872	2,980	10,050	1,604
2) Cost per Sqf * legs (Birr)	20.00	12.00	25.00	18.00
3) Total direct material				
Purchase cost (In Birr)(1x2)	237,440	35,760	251,250	28,872
So Total Direct material purchase cost = 237,440 + 251,250 + 28,872 =Birr. 553,322				
(4) Direct manufacturing labor Budget:				

1 2 3 2x3=4 5 5x4=6

<u>Particulars</u>	<u>production</u>	<u>direct labor</u>	<u>total labor</u>	<u>labor</u>
total				
	<u>In units</u>	<u>Hours/unit</u>	<u>Hours</u>	<u>rate/Hr</u>
Executive line	750	3	2,250	30.00
Chairman line	400	5	<u>2,000</u>	30.00
Total			<u>4,250</u>	<u>127,500</u>

5) Manufacturing overhead budget at 4,250 labor Hours:

a) Variable Overhead costs:

<u>1</u>	<u>2</u>	<u>3</u>	<u>2x3=4</u>
Particulars	Direct labor Hr	Variable overhead	Total
variable		Cost per Hour	cost
Executive line	2,250	Bir.35.00	78,750
Chairman line	2,000	Bir.35.00	<u>70,000</u>
Total variable over head cost			148,750
b) Fixed manufacturing Overhead cost			<u>42,500</u>
c) Total manufacturing overhead cost (a+b)			191,250

6) Ending inventory Budget at March 31, 2008.

<u>1</u>	<u>2</u>	<u>3</u>	<u>2x3=4</u>
<u>Particulars</u>	<u>Sqf & Legs</u>	<u>Cost per Sqf & legs</u>	<u>Costs</u>
a) Direct material ending:			
Oak top	192	Bir.20.00	3,840
Oak legs	80	Bir.12.00	960
Red Oak top	200	Bir.25.00	5,000
Red Oak legs	44	Bir.18.00	<u>792</u>
Total costs			<u>10,592</u>

b) Finished goods ending inventory

<u>1</u>	<u>2</u>	<u>3</u>	<u>2x3=4</u>
Particulars	Units	Cost per unit	Costs
Executive line	30	Bir.593.00	17,790
Chairman line	15	Bir.1, 072.00	<u>16,080</u>
Totals			<u>33,870</u>

Total ending inventory budgeted cost (a+b) = 10.592 + 33,870= Bir.44, 462

Notes: computation of cost per unit of finished goods inventory in 1998.

Particulars	<u>cost per unit</u>		<u>Executive line</u>		<u>Chairman line</u>	
			<u>Input</u>	<u>amount</u>	<u>input</u>	<u>amount</u>
A) Direct materials:						
Oak top	Bir.20.00	units	16	Bir.320.00	---	-----
Oak legs	12.00		12	48.00	---	---
Red oak top	25.00				units 25	Bir.625
Red oak legs	18.00		---	---	4	72
b) Manufacturing labor	30.00		3	90.00	5	150
c) Manufacturing O.H						
(See 5 th requirement)	45.00		3	<u>135.00</u>	5	<u>225</u>
d) Total cost per unit				<u>593.00</u>		<u>1,072.00</u>

6) Cost of goods sold budget

<u>Particulars</u>	<u>Costs</u>	<u>Total Costs</u>
a) Beginning Finished goods inventory		15,330
b) Cost of goods manufactured:		
Direct material used	554,800	
Direct manufacturing labor	127,500	
Direct manufacturing Overheads	<u>191,250</u>	<u>873,550</u>
c) Cost of goods available for sale (a+b)		888,880
Less: ending finished goods inventory		<u>33,870</u>
Cost of goods sold		<u>855,010</u>

UNIT FOUR

FLEXIBLE BUDGET AND VARIANCE ANALYSIS

UNIT OUTLINE

- ✓ Unit Objectives
- 4.1.Introduction

4.2. Budget and Variance

4.2.1. Fixed or Static Budget

4.2.2. Flexible Budget

4.2.3. Fixed or Static Budget Variance

4.2.4. Flexible Budget Variance

- ✓ Summary
- ✓ Model Exam Questions

Unit Objectives:

Upon completion of this Unit you will be able to:

- Explain the concept of flexible budget;
- Explain the difference between master budgets and flexible budgets.
- Prepare flexible budget;
- Describe variance analysis
- Determine the static and the flexible budget variances

4.1. Introduction

Dear learners in the previous unit you have studied the benefit of budget as a planning tool. Hence, budgets are planning tools that are usually prepared prior to the start of the period being budgeted. However, the comparison of the budget to actual results provides valuable information about performance. Therefore, budgets are both planning tools and performance evaluation tools. In this unit, therefore the discussion focuses on how budget are used to evaluate feedback and variances aid managers in their control function.

In evaluating performance the budgeted performance are compared with actual operational results and the resulting variance will be examined so as to identify the causes for variance on the bases of which performance can be rewarded for favorable variance or corrective actions will be taken to avoid unfavorable variance on the coming operational periods.

The unit highlights the importance of variance analysis and show how the budget initially prepared at planning stage creates problem while comparing actual results with the budget. In this unit you are also introduced with the advantage of flexible budget over the static budget, steps in the preparation of flexible budget and evaluating performance using flexible budget.

4.2. Budget and Variance analyses

The use of budget as performance evaluation tool focuses on determining the discrepancies between the planned and actual performance at the end of the operating cycle.

Variance is the difference between an amount on an actual result and the corresponding budgeted amount i.e., the actual amount of something and the amount it was supposed to be according to the budget. The budgeted amount is a point of reference from which comparison may be made. The difference between budget and actual result can be favorable or unfavorable based upon the impact of the discrepancy on the overall profitability of the firm. If the variance has an increasing effect on the operating income as compared to the budgeted amount, it is said to be **favorable variance**. On the other hand **unfavorable or adverse variance** occurs when the variance has a decreasing effect on the operating income relative to the budgeted amount.

Variances assist managers in their planning and control decisions. It enables to exercise Management by Exception (MBE), which is the practice of concentrating attention on areas not operating as expected and giving less attention to areas operating as expected. Managers regularly pay attention to areas with large variances. Variances are also used in performance evaluation. For example Production line managers in a manufacturing company may have quarterly efficiency incentives linked to achieving a budgeted amount of operating costs.

<i>Activity4.1: Explain how budget can be used for performance evaluation._____</i>

4.2.1. Fixed or Static Budget

The **static budget** is the budget that is based on this projected level of output, prior to the start of the period. In other words, the static budget is the “original” budget. **The static budget variance** is the difference between any line-item in this original budget and the corresponding line-item from the statement of actual results. Often, the line-item of most interest is the “bottom line”: total cost of production for the factory and other cost

centers; net income for profit centers.

☞ ***Static budget is a budget that is based on one level of activity.***

Evaluating performance based upon the master budget which fixed and prepared at a single level of activity may not provide accurate picture of performance. This is because usually the planned and actual output or activities levels may not be equal, as a result the comparison is performed at two different levels of activity which hides the variance attributable to the actual performance units as well as overall organization. For example, if a company budgeted to produce and sell 12,000 units, but the actual performance showed only 10,000 units, the comparison of revenue, cost and profit at the budget and actual level of output do reveal only the variance resulted from the difference in the level of output. Therefore unless the analysis is redone by adjusting the budgeted level of output towards the actual units produced and sold, the variance is not helpful to the management as a performance evaluation tool.

☞ ***Static Budget Variance [SBV] is the difference between an actual result and the corresponding budgeted amount in a static budget.***

4.2.2. Flexible Budget

The flexible budget is a performance evaluation tool. It cannot be prepared before the end of the period. A flexible budget adjusts the static budget for the actual level of output so as to avoid the inherent limitation of using static budget for performance evaluation. The flexible budget asks the question: “If I had known at the beginning of the period what my output volume (units produced or units sold) would be, what would my budget have looked like?” The motivation for the flexible budget is to compare apples to apples. If the factory actually produced 10,000 units, then management should compare actual factory costs for 10,000 units to what the factory should have spent to make 10,000 units, not to what the factory should have spent to make 9,000 units or 12,000 units or any other production level.

The **flexible budget variance** is the difference between any line-item in the flexible budget and the corresponding line-item from the statement of actual results.

☞ ***Dear learners from the above discussion you realized that Static budget variance does not give much information about the variances between actual results and budgeted amounts for each line item. Hence one has to prepare flexible budget.***

To have a better understanding of causes for variance managers usually require variance calculated at different level. Variance according to the degree of detailed feedback on performance can be classified as:

- Level 0 variance analysis
- Level 1 variance analysis
- Level 2 variance analysis
- Level 3 variance analysis
- Level 4 variance analysis

In this unit the focus is on level 0, 1, and 2 variances, and the reaming will be discussed in length on the next unit.

Activities 4.2:

1. Describe static budget. _____

2. Define master budget variance _____

3. Differentiate between flexible budget variance a static budget variance _____

4. Define the term variance and differentiate between favorable and unfavorable variance

Now let see the preparation of flexible budget as well as analysis of variance using the following:

Illustration 4.1: **Jimma Garment Co.** manufactures and sells a jacket. Sales are made to distributors who sell to independent clothing stores. Jimma Garment's only costs are manufacturing costs. All units manufactured in April 2003 are sold in April 2003. There is no beginning or ending inventory. Jimma Garment has variable cost categories. The budgeted data for April 2003 are:

Cost category	Variable cost / jacket.
DM costs.....	Br. 60
DL costs.....	16
Variable MOH costs.....	<u>12</u>
Total variable costs	Br. 88

The number of units manufactured is the cost driver for all variable-manufacturing costs. The relevant range for the cost driver is from 0 to 12,000 jackets. Budgeted

manufacturing fixed costs are Br. 276,000 for production between 0 & 12,000 jackets. Budgeted selling price is Br.120/jacket. The static budget for April 2003 is based on selling 12,000 jackets.

The actual data for April 2003 are as follows:

Units sold	10,000 jackets
Revenues	Br. 1,250,000
<u>Variable costs:</u>	
DM	621,600
DL.....	198,000
Variable MOH.....	130,500
Fixed costs	285,000

1. Level 0 or Static Budget Variance for operating income

Level zero variance analysis the least detail analysis which simply compares the operating income at static budget income statement with the operating income at the actual income statement. The level zero variance for Jimma Garment from the above given data is determined as,

Actual operating income.....	Br. 14,900
Budgeted operating income.....	<u>108,000</u>
SBV for operating income.....	<u>Br. 93,100 U</u>

The analysis revealed unfavorable variance as the actual operating income is lower than the budgeted operating income by Birr 93,100. The result here couldn't provide the management useful information as it couldn't show the contribution revenue and each cost element to operating income variance.

2. Static Budget Variance (SBV)

Level one variance can offer management a better insight about their organizational performance than level zero analysis. At this level, operating income variance will be decomposed into revenue and cost component as a result the management will identify the responsibility center that demands attention.

Static Budget Variance

Items	Actual Results (1)	Static Budget Variance(SBV) (2) = (1) – (3)	Static Budget Result (3)
Unit sales	10,000	2,000U	12,000 .
Revenues	Br.1,250,000	Br. 190,000U	1,440,000
Variable costs:			

DM	621,600	98,400F	720,000
DL	198,000	6,000U	192,000
MOH	130,500	13,500F	144,000
Total variable costs	950,100	105,900F	1,056,000
Contribution Margin	299,900	84,100U	384,000
Fixed Costs	285,000	9,000U	276,000
Operating Income	<u>Br. 14,900</u>	<u>Br. 93,100U</u>	<u>Br. 108,000</u>
	↑		↑
	<u>Br. 93,100U</u>		
	SBV		

The static budget variance shows an unfavorable variance for revenue, fixed costs whereas favorable variance of total variable cost. These variances are due primarily to the fact that the static budget was built on an output level of 12,000 units, while the company actually made and sold 10,000 units. The revenue variance might also be due to an average unit sales price that differed from budget. The variable cost variances might also be due to input prices that differed from budget (e.g., the price of fabric), or input quantities that differed from the per-unit budgeted amounts (e.g., yards of fabric per jackets) that may be identified at the later stages of the variance analysis.

Level 2-variance analysis [Flexible Budget Variance (FBV) & Sales-Volume Variance (SVV)]

To identify the amount of variance attributed the difference in the level of output as well as to real performance of the company, at this level the static budget variance will be decomposed into the flexible budget variance and sales volume variance.

Flexible Budget Variance (FBV) is a better measure of operating performance because they compare actual revenues to budgeted revenues and actual costs to budgeted costs for the same output level.

Sales-Volume Variance (SVV) is the difference between the flexible budget amounts and static budget amounts. It represents the variance caused solely by the difference in the

actual output volume and budgeted quantity of output expected to be produced and sold in the static budget.

To determine the flexible budget variance and sales volume variance, first you need to develop a flexible budget. The flexible budget, for the example given above is prepared at the end of the period after the actual output level of 10,000 jackets is known. The flexible budget is that Jimma Garment would have prepared at the start of the budget period had it correctly forecasted the actual level of 10,000 jackets.

In preparing the flexible budget,

- (1) The budgeted selling price is the same Br. 120/ jacket.
- (2) The budgeted variable costs per unit are the same Br. 88/ jacket.
- (3) The budgeted fixed costs are the same Br. 276, 000, are used.

The only difference between the static budget and the flexible budget is that the static budget is prepared for the planned output level of 12,000 jackets, whereas the flexible budget is based on the actual output of 10,000jackets.

The following stapes are used to prepare a flexible budget:

Step 1.Identify the Actual Quantity of Output produced and sold.

10,000jackets.

Step 2. Calculate the flexible budget for revenues based on Budgeted Selling Price and Actual Quantity of Output.

$$\begin{aligned}\text{Flexible B for Revenues} &= \text{Br. 120 /jacket} \times 10,000\text{jacket} \\ &= \text{Br. 1,200,000}\end{aligned}$$

Step 3. Calculate the Flexible Budget for Costs based on Budgeted Variable Costs per Unit, Actual Quantity of Output and Fixed Costs.

Flexible Budget for Variable Costs:

DM:	Br. 60/j X 10,000j	Br. 600,000
DL:	Br. 16/j X 10,000j	160,000
MOH:	Br. 12/j X 10,000j	<u>120,000</u>
FB for TVC		Br. 880,000
FB for FC		<u>276,000</u>
FB for Costs		<u>Br. 1,156,000</u>

Step 4: Building the flexible budget based on the information from steps 1 and 2, and step 3 results a flexible budget presented on column 3 of the following table.

After the flexible budget is developed it is possible to determine the flexible budget variance by comparing the flexible budget and the actual operational results, and sales volume variance by comparing the flexible budget results and the static budget as shown on the following table.

	Actual Results	Flexible Budget Variance	Flexible Budget	Sales Volume Variance	Static Budget
	(1)	(2) = (1) – (3)	(3)	(4)	(5)
Unit sales	10,000	0	10,000	2,000	12,000
Revenues	1,250,000	50,000F	1,200,000	240,000U	1,440,000
Variable costs:					
DM	621,600	21,600U	600,000	120,000F	720,000
DL	198,000	38,000U	160,000	32,000F	192,000
MOH	<u>130,500</u>	<u>10,500U</u>	120,000	24,000F	<u>144,000</u>
Total variable costs	<u>950,100</u>	<u>70,100U</u>	880,000	176,000F	<u>1,056,000</u>
Contribution Margin	299,900	20,100U	320,000	64,000U	384,000
Fixed Costs	<u>285,000</u>	<u>9,000U</u>	<u>276,000</u>	<u>0</u>	<u>276,000</u>
Operating Income	<u>Br. 14,900</u>	<u>Br. 29,100U</u>	<u>Br. 44,000</u>	<u>Br. 64,000U</u>	<u>Br. 108,000</u>
	Br. 29,100U.		Br. 64,000U		
	FBV		SVV		
	Br. 93,100U				
	SBV				

From this table, **Jimma Garment** sees that after adjusting for sales volume, revenue was higher than would have been expected. The favorable Birr 50,000 variance must be due entirely to an average sales price that was higher than planned which was Bir125 per jacket compared to the original budget of Birr120 per jacket.

Materials costs were higher than would have been expected for a sales volume of 2,000 units. This unfavorable variance is due to higher material prices, or to inefficient utilization of fabric (more waste than expected), or a combination of these two factors. Labor and overhead were higher than expected, even after adjusting for the sales volume of 2,000 units. This unfavorable flexible budget variance implies that either wage rates were higher than planned, or labor was not as efficient as planned, or both. Similarly, the components of variable overhead were either more expensive than budgeted, or were used more intensively than budgeted. For example, electric rates might have been higher than planned, or more electricity was used than planned per unit of output.

The fixed cost variances are identical in this table to the previous table. In other words, the flexible budget and flexible budget variance provide no additional information about fixed costs beyond what can be learned from the static budget variance

 ***Dear learners from the above discussion, you realized that flexible budget, that is prepared after actual units outputs are known, can provide a better feedback about performance than the static budget prepared at the planning stage. To investigate the real causes of variance usually it is essential to perform a further detail variance analysis covered on the next unit of this module***

Unit Summary

In this unit you have studied that how budget can assist managers in controlling and performance evaluation function in addition to planning. Using budget as bench mark managers compare their planned and actual operational results or perform a variance analysis.

Variance analysis allows management to compare discrepancies from the original plan. The discrepancy or variance can be either favorable or unfavorable, depending upon its effect on operating income as compared to the budget. Favorable variances contribute positively to operating income; on the contrary unfavorable variances have a negative effect on the operating income as compared to the plan.

The master budget prepared before starting operation is less useful for performance evaluation as it is prepared at a single projected activity level. Variance resulted from the comparison of actual results and static budget is termed as static budget variance.

As static budget variance is less informative, it is essential to prepare a flexible budget, to obtain a better feedback of the variance analysis. Flexible budget is prepared by taking the budgeted unit selling price and unit variable costs and total fixed cost at the actual output level.

Using the flexible budget it is possible to decompose the static budget variance in to flexible budget variance and sales volume variance.

Model Examination Questions

Part I : Multiple Choice Questions

1. The system that “build in” the effect of fluctuations in the level of activity in the various budget is:

- a) Static budget.
- b) Master budget.
- c) Flexible budget.
- d) Operating budget
- e) None of the above

2. A static budget:

- A) is based totally on prior year's costs.
- B) is based on one anticipated activity level.
- C) is based on a range of activity.
- D) is preferred over a flexible budget in the evaluation of performance.
- E) Presents a clear measure of performance when planned activity differs from actual activity.

3. Which of the following is true?

- A. The major purpose of a fixed budget lies in its use at planning
- B. A flexible budget is designed to change as volume of activity changes
- C. Flexible budget enables us to compare actual level of activity
- D. The importance of fixed budget is very high in controlling as compared to its importance in planning
- E. All except “D”

4. Lanther Corporation recently prepared a manufacturing cost budget for an output of 50,000 units, as follows:

Direct materials	Birr100,000
Direct labor	Birr 50,000
Variable Overhead	Birr 75,000
Fixed overhead	Birr 100,000

Actual units produced amounted to 60,000. Actual costs incurred were: direct materials, Birr 110,000; direct labor, Birr 60,000; variable overhead, Birr 100,000; and fixed overhead, Birr 97,000. If Lather evaluated performance by the use of a flexible budget, a performance report would reveal a total variance of:

- A) Birr 27,000 unfavorable.
- B) Birr 42,000 unfavorable.
- C) Birr 3,000 favorable.
- D) Birr 23,000 favorable.
- E) None of the above amounts.

5. A variance is said to be unfavorable:

- A. When actual operating income decrease as compared to the budget

- B. When actual cost and expense exceeds the budgeted cost and expense
- C. When actual revenue is less than budget revenue
- D. When budgeted cost is greater than actual cost incurred
- E. All except “D”

Part II: Discussion Questions

- 1) What is meant by the process of “management by exception”? How is a standard cost system helpful in such a process?
- 2) Master budget prepared at the planning stage is less helpful for managerial control and performance evaluation. **Comment**
- 3) Explain the difference between fixed and flexible budget.
- 4) Explain the difference between flexible budget variance and sales volume variance.

Answer to Model Examination Questions

Part I: Multiple Choice Questions

1. C 2. B 3.E 4. C 5. E

Part II: Key to Discussion questions

1. Question no “1” is related to section 4.2.
2. Question no “2” is related to section 4.2.2.
3. Question no “3” is related to section 4.2.2
4. Question no “4” is related to section 4.2.2.

Unit Five

Cost-Volume-Profit (CVP) Analysis

UNIT OUTLINE

- ✓ Unit Objectives
 - 5.1.Introduction
 - 5.2.Meaning, underlying Assumptions and importance of CVP analysis
 - 5.2.1. Meaning of CVP analysis
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5.5. Not for Profit Organizations and Cost-Profit-Volume (CVP) Analysis

✓ Unit Summary

✓ Model Examination Questions

Unit Objectives:

After completing this unit you are expected to:

- What are the objectives of cost volume profit analysis (CVP Analysis)?
- Understand the assumptions underlying cost-volume-profit (CVP) analysis.
- Explain the features of CVP analysis.
- Determine the breakeven point and output level needed to achieve a target operating income
- Understand how income taxes affect CVP analysis
- Explain CVP analysis in decision making and how sensitivity analysis helps managers cope with uncertainty
- Apply CVP analysis to a company producing Multiple products
- Apply CVP analysis to service rendering Not for Profit organizations

5.1. Introduction

In this unit you will be introduced with one of the most powerful management accounting tool that helps managers in quick decision about the income generated at different activity level. The tool is termed as Cost-Volume-Profit (CVP) Analysis, as the relationship among cost, profit and the volume of output or the level of activity is considered in the model.

Understanding the relationship between a firm's costs, profits and its volume levels is very important for strategic planning. When you are undertaking a new project, you will

probably ask yourself, “How many units do I have to produce and sell in order to breakeven?” The feasibility of obtaining the level of production and sales indicated by that answer is very important in deciding whether or not to move forward on the project in question. Similarly, before undertaking a new project, you have to assure yourself that, you can generate sufficient profits in order to meet the profit targets set by your firm. Thus, you might ask yourself, “How many units do I have to sell in order to produce a target income?” Breaking Even? You could also ask, “If I increase my sales volume by certain percent, what will be the impact on my profits?” The topics in this unit are, therefore designed to acquaint you with the ability of applying Cost-Volume-Profit (CVP) Analysis in answering different questions while you are taking part in planning decision in different types of organizations.

Although the term “profit” is attached in the CVP model; it does not mean that, the application of CVP analysis is limited to business organizations only. It can be used by government and nongovernmental organizations (NGOs) as well in planning their activity levels in light of the recourse availability and constraints.

5.2. Meaning , underlying Assumptions and importance of CVP analysis

5.2.1. Meaning

Examining shifts in costs and volume and their resulting effects on profit is called **cost-volume-profit (CVP) analysis**. This analysis is applicable in all economic sectors, including manufacturing, wholesaling, retailing, and service industries and not-for-profit (NFP) organizations. CVP can be used by managers to plan and control more effectively because it allows them to concentrate on the relationships among revenues, costs, volume changes, taxes, and profits.

CVP analysis can be used to determine a **company’s break-even point (BEP)**, which is that level of activity, in units or Birr value, at which total revenues equal total costs. At breakeven, the company’s revenues simply cover its costs; thus, the company incurs neither a profit nor a loss on operating activities. Companies, however, do not wish merely to “break even” on operations.

The break-even point is calculated to establish a point of reference. Knowing **BEP**, managers are better able to set sales goals that should generate income from operations rather than produce losses. CVP analysis can also be used to calculate the sales volume necessary to achieve a **desired target profit**. Target profit objectives can be stated as either a fixed or variable amount on a before- or after-tax basis. Because profit cannot be achieved until the break-even point is reached, the starting point of CVP analysis is BEP. It also helps in conducting a **sensitivity** analysis to understand how the change in variables in the CVP model affects the profitability.

After going through the major assumptions upon which the CVP model is developed, you will be exposed through the application of CVP in determining the BEP, target profit; in a single and multiple product companies. You will also come across on how CVP is used to conduct a sensitivity analysis as well as its application in NFPs.

Notice that on advanced managerial accounting courses you will come across with diverse and advanced applications of CVP analysis in various financial and strategic decisions.

☞ ***Cost-volume-profit analysis determines how costs and profit react to a change in the volume or level of activity, so that management can decide the 'best' activity level.***

5.2.2. UNDERLYING ASSUMPTIONS OF CVP ANALYSIS

CVP analysis is a short-run model that focuses on relationships among several items: selling price, variable costs, fixed costs, volume, and profits. This model is a useful planning tool that can provide information on the impact on profits when changes are made in the cost structure or in sales levels. However, the CVP model, like other human-made models, is an abstraction of reality and, as such, does not reveal all the forces at work. It reflects reality but does not duplicate it. Although limiting the accuracy of the results, several important but necessary assumptions are made in the CVP model. These assumptions follow.

1. All revenue and variable cost behavior patterns are constant per unit and linear within the relevant range.

2. Total contribution margin (total revenue - total variable costs) is linear within the relevant range and increases proportionally with output. This assumption follows directly from assumption 1.
3. Total fixed cost is a constant amount within the relevant range.
4. Mixed costs can be accurately separated into their fixed and variable elements. Although accuracy of separation may be questioned, reliable estimates can be developed from the use of regression analysis or the high-low method.
5. Sales and production are equal; thus, there is no material fluctuation in inventory levels. This assumption is necessary because of the allocation of fixed costs to inventory at potentially different rates each year. This assumption requires that variable costing information be available. Because both CVP and variable costing focus on cost behavior, they are distinctly compatible with one another.
6. There will be no capacity additions during the period under consideration. If such additions were made, fixed (and, possibly, variable) costs would change. Any changes in fixed or variable costs would violate assumptions 1 through 3.
7. In a multiproduct firm, the sales mix will remain constant. If this assumption were not made, no weighted average contribution margin could be computed for the company.
8. There is either no inflation or, if it can be forecasted, it is incorporated into the CVP model. This eliminates the possibility of cost changes.
9. Labor productivity, production technology, and market conditions will not change. If any of these changes occur, costs would change correspondingly and selling prices might change. Such changes would invalidate assumptions 1 through 3.

These assumptions limit not only the volume of activity for which the calculations can be made, but also the time frame for the usefulness of the calculations to that period for which the specified revenue and cost amounts remain constant. Changes in either selling prices or costs will require that new computations be made for break-even and product opportunity analyses.

The nine assumptions listed above are the traditional ones associated with CVP analysis. An additional assumption must be noted with regard to the distinction of variable and fixed costs. Accountants have generally assumed that cost behavior, once classified, remained constant over periods of time as long as operations remained within the relevant range. Thus, for example, once a cost was determined to be “fixed,” it would be fixed next year, the year after, and 10 years from now.

The above assumptions thus, imply that cost-volume-profit analysis is always based on contribution per unit (assumed to be constant unless a question clearly says otherwise) and never on profit per unit because profit per unit changes every time a few more or less units are made.

Activity 5.1:

1. *Explain how CVP helps for management decision making process*

2. *List and explain the underline assumptions in a CVP analysis*

5.3. Applications of CVP Analysis

 ***Dear learners in the previous parts of this unit you have gone through the meaning and underlying assumptions of CVP. As you have learnt CVP can be used by managers in a variety of sectors as management decision tool.***

On the subsequent part of the module you will find an explanation and illustrations on the applications of CVP in different scenarios.

5.3.1. Break Even Analysis

CVP analysis has wide-range applicability. It can be used to determine a company’s break-even point (BEP), which is that level of activity, in units or dollars/Birr, at which total revenues equal total costs. At breakeven, the company’s revenues simply cover its costs; thus, the company incurs neither a profit nor a loss on operating activities.

 ***At the breakeven point total revenue is equal to total cost (both variable and fixed)***

For instance **Sebastopol Cinema** sold 4,800 tickets during a show one month run. The following contribution margined approach income statement show that the operating income for the month will be zero:

Sales Revenue (4800 X Br25) -----	Br 120, 000
Less Variable Cost (4800 X Br 15) -----	<u>72,000</u>
Total Contribution Margin-----	Br 48,000
Less Fixed Costs-----	<u>48,000</u>
Operating Income-----	<u>Br 0</u>

The income statement above highlights (1) the distinction between variable and fixed cost and (2) the total contribution margin, which is the amount that contributes towards covering Sebastopol Cinema's fixed cost and income generation. To state it differently, each ticket sold add Birr 10 to the firm's bottom line profit. The Birr 10 unit contribution margin is derived by deducting the unit variable cost Birr 15 from Birr 25 the unit selling price of a ticket.

How could you compute Sebastopol Cinema's breakeven point if you didn't already know it is Birr 4,800 tickets per month? Well on the following discussion you will get the basics of CVP analysis to determine BEP and other application. CVP analysis can be done using three alternative approaches, namely: contribution margin approach, equation approach and graphical approach.

In discussing CVP application we will assume that the following variables have the meanings given below:

- SP = Selling Price Per Unit
- Q = Units Produced and Sold
- VCU = Variable Cost Per Unit
- CMU=contribution margin per Unit
- CM%= contribution margin percentage ($CMU \div SP$)
- FC = Total Fixed Costs
- TOI =Target operating Income
- TNI=Target Net Income
- t = Tax rate

5.3.1.1. Equation Method

In using equation approach to CVP analysis, you need to convert your income statement in the following equation form.

$$\text{Revenue} - \text{Variable Costs} - \text{Fixed Costs} = \text{Operating Income}$$

Your Sales Revenue is equal to the number of units sold times the price you get for each unit sold:

$$\text{Sales Revenue} = \text{SPQ}$$

Assume that you have a linear cost function, and your total costs equal the sum of your Variable Costs and Fixed Costs:

$$\text{Total Costs} = \text{Variable costs} + \text{Fixed costs}$$

The profit equation can, therefore rewritten as:

$$(\text{SPXQ}) - (\text{VCUXQ}) + \text{FC} = \text{OI}$$

This equation provides the most general and easiest approach to remember approach to any CVP situation. The determination of breakeven level using this method can easily performed by making the operating income on the right hand side of the equation zero. Here the basic assumption is that, when you are at breakeven, your *Sales Revenue minus your Total Costs is zero*.

☞ **At breakeven point,**

$$(\text{SPXQ}) - (\text{VCUXQ}) + \text{FC} = 0$$

From the information given above for the Sebastopol Cinema, you can determine the breakeven point (BEP) using the equation approach as follows,

$$\begin{aligned} (\text{Br}25 \times \text{Q}) - (\text{Br}15 \times \text{Q}) - \text{Br}48,000 &= 0 \\ \text{Br}10\text{Q} &= \text{Br } 48,000 \\ \mathbf{Q} &= \mathbf{4,800} \end{aligned}$$

If Sebastopol sells fewer than 4800 tickets, it will have a loss, if it sells 4800 tickets, it will breakeven; and if the sales are more than 4800 units, it will make a profit.

The breakeven point stated in units can be stated in terms of Birr by multiplying the breakeven quantity and unit selling price. The breakeven point in Birr= BEQ x SP

$$= 4,800 \times \text{Br}25$$

$$= \underline{\text{Br120,000}}$$

5.3.1.2. Contribution Margin Approach

The contribution margin technique is merely a short version of the equation technique. The formulas used in the determination of the breakeven point in unit as well as in value are derived by the rearrangement of the terms in the equation method above, which is

$$(\text{SP} \times \text{Q}) - (\text{VCU} \times \text{Q}) - \text{FC} = \text{OI}$$

Re-written equation takes the following format:

$$(\text{SP} - \text{VCU}) \times \text{Q} = \text{FC} + \text{OI}$$

That is, $\text{Q} = (\text{FC} + \text{OI}) / (\text{SP} - \text{VCU})$

The difference between unit selling price and unit variable cost is termed as unit contribution margin. You can replace the (P-V) by the term "Contribution Margin per Unit (CMU),

$$\text{Q} = (\text{FC} + \text{OI}) / \text{CMU}$$

As you know from the discussion above at breakeven point the target operating income is Birr 0, so replacing OI by 0, you will get,

$$\text{Q} = \text{FC} / (\text{SP} - \text{VCU})$$

A. Breakeven point in units using contribution Margin Approach

Breakeven point in units using this formula is determined by dividing the total fixed cost by the contribution margin per unit due to the fact that this approach centers on the idea that each unit sold provides a certain amount of fixed costs. When enough units have been sold to generate a total contribution margin equals to the total expense and only after that point the units sold contributes for profit of the organization.

The breakeven number of tickets that Sebastopol Cinema must sold to reach at breakeven using this method can be determined as,

$$\text{Q} = \text{FC} / \text{CMU}$$

$$\text{Q} = \text{Br48,000} / \text{Br10}$$

$$\underline{\text{Q} = 4,800 \text{ units}}$$

B. Determining breakeven point in terms of sales value (Dollar/Birr)

To find the breakeven sales value, you can use contribution margin percentage in place of contribution margin per unit in the formula you used in the determination of breakeven units. Contribution margin percentage is simply the ratio of contribution margin to selling price.

$$\text{CM\%} = \text{CMU/USP}$$

It can also be determined by dividing the total contribution margin to total sales when the unit selling price and variable cost is not known. We will see how this formula is used in such a situation later. Before that let see how the formula works using the above example.

$$\begin{aligned}\text{CM \% on our example of Sebastopol Cinema} &= \text{UCM/USP} \\ &= \text{Br 10/ Br 25} \\ &= \underline{\underline{0.4}}\end{aligned}$$

This can be interpreted as, units sold cover variable costs and contribute 40 percent to cover fixed cost and increase in profit.

$$\begin{aligned}\text{The break even sales amount (Birr) for Sebastopol Cinema} &= \text{FC/CM\%} \\ &= 48,000/0.40 \\ &= \underline{\underline{\text{Br 120,000}}}\end{aligned}$$

The breakeven sales determined using equation method was a mere multiplication of the breakeven point in unit and the unit selling price (4800 units x Br 25) which is exactly the same with what you determined using a more complicated contribution margin approach.

Why do you think using the contribution margin approach is essential?

Well, sometimes you will be given only the income statement and asked to determine the breakeven sales without sufficient information about unit selling price and unit variable cost. In this case, only using the contribution margin ratio you can determine the breakeven point in sales by dividing total fixed cost by CM ratio gives the break-even point in sales dollars.

For example, **Jimma Electronics Co.** produces portable radios. It has released the following Variable Costing Income Statement. This is the only financial information that we have regarding the company's operations:

Sales Revenue-----	100,000
Less Variable Costs-----	<u>30,000</u>

Contribution Margin-----	70,000
Less Fixed Costs-----	50,000
Operating Income -----	<u>Br 20,000</u>

What is the Break-Even point for Jimma Electronics Co? You do not know the number of units that Jimma Electronics sold in a year. You do not also know the Price or the Variable Cost per unit. Although you do not know the price or the Variable Cost per unit, you are still able to calculate the Contribution Margin Ratio.

Contribution ratio can be determined by dividing total contribution margin to total sales.

CM% in this case is, therefore, determined as follows:

$$\begin{aligned}
 &= \text{Br } 70,000 \div \text{Br } 100,000 \quad \text{or} \quad 1 - \text{variable cost ratio} \\
 &= \underline{\underline{0.7 \text{ Or } 70\%}} \qquad \qquad \qquad 1 - (\text{variable cost} \div \text{sales}) \\
 &\qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad 1 - (30,000 \div 100,000) \\
 &\qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \underline{\underline{0.7 \text{ Or } 70\%}}
 \end{aligned}$$

The breakeven point in sales is also determined by dividing the total fixed cost by the contribution margin ratio determined above as follows:

$$\begin{aligned}
 &= \text{FC/CM\%} \\
 &= 50,000/0.7 \\
 &= \underline{\underline{\text{Br } 71,428.57}}
 \end{aligned}$$

Keep in mind the reason that Jimma Electronics' Sales Revenue is lower than it was before is because the company sold fewer units. Keeping the price and Variable Cost per unit remained unchanged. Let's check if Breaks Even at this Sales Revenue figure:

Sales Revenue: -----	Br 71,428.57
Less Variable Costs (30% of 71,428.57) -----	21,428.57
Contribution Margin-----	Br 50,000
Less Fixed Costs-----	<u>50,000</u>
Operating Income -----	<u>Br0</u>

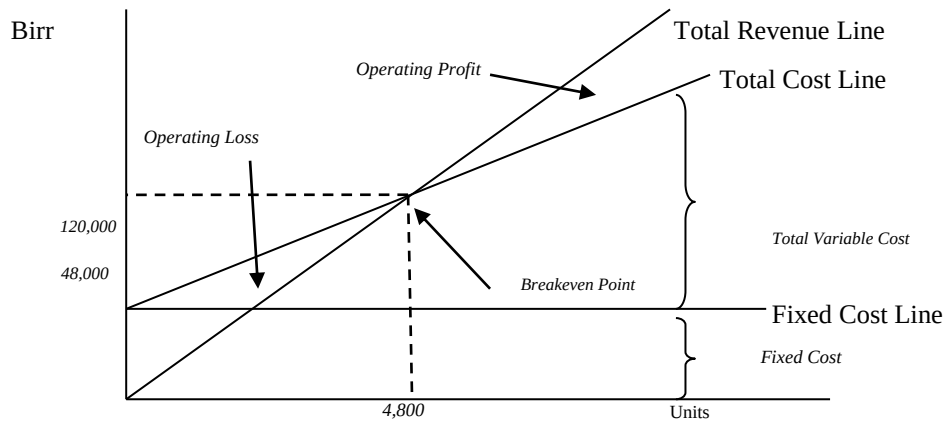
5.3.1.3. Graphical Approach

You have seen how solutions to break-even problems are determined using an algebraic formula. However, sometimes managers need information in more visual format, such as graphs. In this approach you can construct a CVP graph by plotting total cost and total revenue graph at different activity level. The breakeven point is found at the intersection point of total revenue and total cost lines.

The chart or graph is constructed as follows:

1. Plot fixed costs, as a straight line parallel to the horizontal axis
2. Plot sales revenue and variable costs from the origin
3. Total costs represent fixed plus variable costs.
4. The breakeven point represents the intersection point of total revenue and total cost lines

Now you can draw a CVP graph to Sebastopol Cinema following the steps given above and locate the breakeven point.



The breakeven points for Sebastopol cinema can be determined from the graph by identifying the level of output and sales value where the total revenue and total cost line cross to each other. In the case of Sebastopol cinema this happened when 4800 tickets are sold for birr 120,000 in which both the unit and the Birr value are similar with what have been determined in the equation and contribution margin approaches. The graphical approach is usually preferred by managers as it can provide a detail insight of the cost volume and profit relationship pictorially.

Activities 5.2.: 1. what are the different approaches in solving problems in a CVP model

5.3.2. Target operating Income Analysis

Using CVP analysis managers can determine the total sales in unit and Birr/Dollar needed to reach the target profit level. The computation of sales volume in unit and/ or in amount to attain the targeted profit is similar with that of the break even analysis, except that the targeted profit is more than offsetting the cost. For instance, the management of

Sebastopol Cinema desires to get Br 9000 profit for the coming month, instead of operating at breakeven point, *how many tickets must be sold?* Managers want to answer this question to provide the necessary resource support to attain the desired profit at already determined volume of activity. The analysis can be performed using equation method or contribution margin approach on the basis of personal preference.

See both the methods using the information given for Sebastopol Cinema to determine the target sales in unit and Birr to that enable the firm to earn the targeted profit of birr 9,000 can be determined using the equation method and contribution margin method as follows:

Equation Method to Determine (Target sales unit)	Contribution Margin Approach (Target sales unit)
$(SP \times Q) - (VCU \times Q) - FC = TOI$ $(25 \times Q) - (15 \times Q) - 48,000 = 9000$ $10Q = 57,000$ <u>Q = 5,700</u>	$Q = FC + TOI / CMU$ $Q = (48,000 + 9,000) / 10$ <u>Q = 5,700</u>
Equation Method to Determine (Target sales in birr)	Contribution Margin Approach (Target sales in birr)
$TR = SP \times Q$ $= 25 \times 5700$ <u>= Br 142,500</u>	$\text{Target Sales in Birr} = (FC + TOI) / CM\%$ $= (48,000 + 9000) / 40\%$ <u>= Birr 142, 500</u>

Bothe the methods provided similar result that the cinema must sell 5,700 tickets at a total of Birr 142, 500 to meet its target profit goal of birr 9000.

 *Here you can recognize that each ticket sales beyond the breakeven point contributes to the firms operating income.*

5.3.3.The impact of Income Tax on CVP Analysis

Profit seeking enterprises must pay tax on their profit, meaning that target income figures are set at high enough to cover the firm's tax obligation to the government. The

relationship between an organization's before tax income and after tax income is expressed in the following formula:

After Tax income = Before Tax Income – Income Taxes

$$\begin{aligned}\text{NIAT} &= \text{NIBT} - (\text{NIBT} \times t) \\ &= \text{NIBT} \times (1-t)\end{aligned}$$

Dividing both sides by (1-t) you can get:

$$\text{NIAT} / (1-t) = \text{NIBT}$$

Which gives you the desired before tax income that will generate the desired after tax income, given the company's tax rate.

For instance, if the target profit given for Sebastopol Cinema above is expressed on after tax basis and the firm is subject to a 40 percent income tax rate, what will be the required sales of ticket in units and Birr? If you want to know how many units that you need to produce and sell in order to generate a target Net Income (or after-tax profit), just convert the after-tax number into a before-tax number.

$$\begin{aligned}\text{NIBT} &= \text{NIAT} / (1-t) \\ &= \text{Br } 9000 / (1-0.4) \\ &= \textbf{Br 15,000}\end{aligned}$$

You can then substitute the before-tax profit figure in the formulas used in target operating income analysis. The analysis of sales of tickets and amount of Sebastopol cinema to earn the desired profit after tax can be determined using the equation and contribution margin approach as shown on the following table:

Equation Method to Determine (Target sales unit)	Contribution Margin Approach (Target sales unit)
$(\text{SP} \times \text{Q}) - (\text{VCU} \times \text{Q}) - \text{FC} = \text{TOIBT}$ $(25 \times \text{Q}) - (15 \times \text{Q}) - 48,000 = 15,000$ $10\text{Q} = 63,000$ $\text{Q} = 6,300$	$\text{Q} = \text{FC} + \text{TOIBT} / \text{CMU}$ $\text{Q} = (48,000 + 15,000) / 10$ $\text{Q} = 6,300$
Equation Method to Determine (Target sales in birr)	Contribution Margin Approach (Target sales in birr)

$TR = SP \times Q$ $= 25 \times 6300$ = Br 157,500	Target Sales in Birr = $(FC + TOIBT)/CM\%$ $= (48,000 + 15,000)/$ 40% = Birr 157, 500
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5.3.4. Sensitivity Analysis

The entire example above assumes a definite amount of certainty as far as knowledge or assumptions of the cost structure and the sale price are concerned. However, in the real-world, day to day problems may cause some or all of the information to change. The question to be answered in this section is *what if a change occurs in our assumption in the previous CVP analysis?* Obviously the analysis must be re-done.

Sensitivity analysis is a “what if” technique that managers use to examine how a result will change if the original predicted data are not achieved or if an underlying assumption changes.

In the context of CVP analysis, sensitivity analysis answers the following questions:

- What will be the operating income if units sold decrease by 15% from original prediction?
- What will be the operating income if variable cost per unit increases by 20%?

The sensitivity of operating income to various possible outcomes broadens the perspective of management regarding what might actually occur before making cost commitments.

To make the concept clear let you take the Income statement of Sebastopol Cinema prepared when the firm planned an operating profit of Birr 9000 from the sales of 5,700 tickets.

Sales Revenue :(Br25x5, 700) -----	Br 142,500
Less Variable Costs (Br 15 x 5,700) -----	85,500
Contribution Margin-----	Br 57,000
Less Fixed Costs-----	<u>48,000</u>
Operating Income -----	<u>Br 9,000</u>

The management has different proposals that are assumed to increase the operating income. To make decisions that have implication on the cost, volume and profit, the

management needs to conduct a sensitivity analysis. For example the following proposals are given:

The marketing manager proposed that increasing a monthly advertising budget by Birr 4000 may result a Birr 13,000 increase on a monthly sales. This proposal, if implemented will change the fixed cost as well as the volume of sales. As you all know total variable cost also varies with any change in volume, therefore the proposed change also changes the variable cost as well. So before accepting the proposal the general the manager needs to conduct a sensitivity analysis to realize the ultimate result of changes on fixed cost, variable cost and sales on operating income.

	(1)	(2)	(3) = (2)-(1)
	Existing Plan	Proposed Plan	Difference
Sales Revenue	Br 142,500	155,500	13,000
Variable cost (5,700 x 15; 6220 x Br 15)	<u>85,500</u>	<u>93,300</u>	<u>7,800</u>
Contribution Margin	57,000	62,200	5,200
Fixed Cost	<u>48,000</u>	<u>52,000</u>	<u>4,000</u>
Operating Income	<u>9,000</u>	<u>10,200</u>	<u>1,200</u>

As you can see from the above analysis a Birr 4000 increase in the advertising budget, which is a fixed cost increased the number of ticket that the firm is selling using the existing plan from 5,600 to 6,220 as a result the sale revenue as well as the variable cost increased along with the increased in sales volume. As the analysis revealed that the operating income can increased by Birr 1,200, the manager can implement the proposed plan.

As you have seen on the example above, Sensitivity analysis allows us to accommodate for the dynamic nature of organizational environments by adding in new information and adjusting our planning accordingly.

5.3.5. Margin of Safety

The margin of safety is the excess of the budgeted or actual sales over the breakeven sales level. This tells managers the margin between current sales and the breakeven point. In a sense margin of safety indicates the risk of losing money that a company

faces; that is the amount by which sales can fall before the company is in the loss area. The margin of safety is determined using the following formula:

$$\text{Margin of Safety} = \text{Sales Volume} - \text{Breakeven Sales volume} \\ \text{(Actual or Budgeted)}$$

As an example, consider the case of Sebastopol Cinema, that planned to attain a Birr 9000 before tax profit at a sales volume of 5700 units or Birr 142, 500. The breakeven point for this company as determined on section 3 above is attained at the sales volume of 4,800 units or Birr 120,000. The margin of safety for this company is the difference between the budgeted sales and the breakeven sales, which is 900 units (5,700 units - 4,800). The margin of safety in terms of sales value is also determined as Birr 22, 500 (Birr 142,500 - Birr 120, 000). The interpretation is that, sales volume of Sebastopol can drop by 1100 tickets or Birr 22,500 before the firm incurs a loss, if all other things remain constant.

In practice the margin of safety is determined in sales amount (Birr) and as a percentage of current sales. To determine the margin of safety in percentage the following formula can be used:

$$\text{Margin of Safety (\%)} = \frac{\text{Margin of safety in Birr}}{\text{Sales}}$$

For Sebastopol Cinema the margin of safety percentage is,

$$\frac{22,500}{142,500} \\ = \underline{\underline{19.3\%}}$$

Activities 5.3:

1. How can cost Volume profit (CVP) analysis be used by a company?
2. List and explain the three approaches to analyze a CVP problems
3. Explain the following terms:
 - i. Contribution margin
 - ii. Breakeven point
 - iii. Sensitivity analysis
 - iv. Margin of safety

5.4. Multiple Product Company and Cost-Profit-Volume Analysis

Our illustration of CVP Analysis on the previous part of this unit assumes that the company has only one product; however in real life, companies produce a range of products, not just one kind as was assumed in the example above. In case when a company offers more than one product, obviously different products will have different selling prices, variable costs per unit and, as a result, different contribution margins and contribution margin ratios.

 ***Dear learners, Can CVP analysis help managers in a multi product companies with complication of some CVP assumptions?***

The answer is 'Yes', you can perform the same similar analysis if you just obtain some data about the product mix and the constant sales mix assumption discussed above holds true. The term sales/revenue mix, which is essential in such a situation, is defined as the relative proportions or combinations of quantities of products that comprise total sales.

In performing the analysis, first, you need to know proportion in which each of the products is sold. Then you can calculate contribution margin for each product. After that you define weighted average contribution margin, which is used in the determination of **break-even point** or the amount of sales required to gain **desired income** (profit).

Let's illustrate it on Ethio.Tire Corporation an example. Assume, the company produces 3 types of tire: for trucks, cars and motor-bikes. The following data is available:

	Truck Tire	Car Tire	Motor-bike Tire
(A) Share in physical volume sold, %	30%	45%	25%
(B) Selling price per unit, Birr	Birr10	Birr8	Birr7
(C) Variable cost per unit, Birr	Birr7	Birr6	Birr5
(D) Contribution per unit, Birr (B - C)	Birr3	Birr2	Birr2
(E) Contribution margin ratio (D ÷ B)	0.30	0.25	0.29
(F) Fixed costs total, Birr	Birr10,000		

From the given above you can calculate weighted average contribution margin. To calculate weighted average contribution margin you need to "weight" the contribution margin per unit of these three products and present it as "three-in-one". To state it in the form of formula,

$$\text{Weighted Average Contribution Margin per Unit} = \text{CMU} \times \text{Percentage of sales Mix}$$

Using information given above you can calculate Weighted Average contribution margin as follows:

$$\begin{aligned} \text{Weighted Average Contribution Margin per Unit} &= (30\% \times \text{Birr3}) + (45\% \times \text{Birr2}) + \\ & (25\% \times \text{Br2}) \quad \quad \quad = \text{Birr2.3} \end{aligned}$$

Now break-even point may be calculated using the weighted average contribution margin as far as the given sales mixes remain as it is.

5.4.1. Contribution Margin Technique and Break-even point for a Multiple Product Company

The calculation of breakeven point in a multi-product company follows the same logic as in a single product company. While the numerator will be the same fixed costs, the denominator now will be weighted average contribution margin. The modified formula is as follows:

$$\text{Break-even Point (in units)} = \frac{\text{Fixed Costs}}{\text{Weighted Average Contribution Margin per Unit}}$$

For our example break-even point (in units) for Ethio.Tire Corporation, can be determined using the formula as,

$$\begin{aligned} \text{BEP (in units)} &= \text{FC} / \text{WA.CM} \\ &= \text{Br } 10,000 / 2.3 \\ &= \text{Br } \underline{4347.86} \end{aligned}$$

The company will be at breakeven if it approximately produces 4,348 units. The total numbers of outputs required to breakeven are then split in accordance with the proportion defined in row 1 in the above table:

Truck tires:	4,348 units x 30% = 1,304 units
--------------	---------------------------------

Car tires:	4,348 units x 45% = 1,957 units
Motor-bike tires:	4,348 units x 25% = 1,087 units

Ethio.Tire Corporation will be at break-even (i.e., will get neither profit nor loss) if it sells the above volumes of Tires at the given proportion 30%:45%:25%.

However, it is important to remember that each proportion of product mix will have different break-even points. For example, if market situation changes and Ethio.Tire Corporation switches to larger production of more expensive truck tires with proportion 45%:30%:25%, break-even point will change. In this case, the weighted average contribution margin per unit will be Birr2.45 and zero profit will be earned at total level of sales of around 4,082 valves (compared to 4,348 units).

The change occurred due to the fact that contribution ratio per unit of truck tires is the highest (Birr3 for truck tires versus Birr2 per car or motor-bike tires). Thus, more income can be generated by producing and selling truck tires and break-even point is reached faster (with less total items produced and sold).

☞ *A shift in sales-mix from high-margin items to low-margin items can cause total profit to decrease even though total sales may increase. Conversely, a shift in the sales mix from low margin items to high margin items can cause the reverse effect-total profit may increase even though total sales decrease.*

Break-even point for multiple product production can be calculated **in Birr value** as well. Here also, the numerator is the same fixed costs. The denominator now will be weighted average contribution margin ratio. The modified formula is as follows:

$$\text{Break-even Point (in Birr)} = \frac{\text{Fixed Costs}}{\text{Weighted Average Contribution Margin Ratio}}$$

Based on figures from the earlier table with information about three Tire types, the break-even point will be reached at:

$$\text{Break-even Point (in dollars)} = \frac{\text{Birr}10,000}{30\% \times 0.3 + 45\% \times 0.25 + 25\% \times 0.29} =$$

Therefore, to achieve the break-even point, Ethio.Tire Corporation has to sell tires for a total of Birr36, 364 (valid only when proportion of sales is 30%:45%:25%).

Let's check the results: $\sum \text{Sales Units} \times \text{Contribution} - \text{Fixed Costs}$, should approximately equal zero (rounding difference may arise): $\text{Br}3 \times 1,304 + \text{Br}2 \times 1,957 + \text{Br}2 \times 1,087 - \text{Br}10,000 = 0$

5.4.2. Contribution margin technique and sales required for desired income for a multi-product company

The logic to calculate sales required for designed income (profit) for a multi-product company is the same as for a single product company. In the formula for determining the sales required for desired profit, we just substitute contribution margin per unit for one product with weighted average contribution margin for multiple products.

The formula to calculate the required unit sale for a given **desired income** is shown below:

$$\text{Unit Sales} = \frac{\text{Fixed Costs} + \text{Profit}}{\text{Weighted Average Contribution Margin per Unit}}$$

And the amount of sales in dollars can be determined using the following formula:

$$\text{Sales (Birr)} = \frac{\text{Fixed Costs} + \text{Profit}}{\text{Contribution Margin Ratio}}$$

5.5. Not for Profit Organizations and Cost-Profit-Volume (CVP) Analysis

Managers in not-for-profit (NFP) organizations also routinely use CVP analysis to examine the effect of activity and others short run changes on revenue and costs. For example, activity administration must analyze the impact of increment in population on the different tax revenue and the cost of providing services such as education, transportation, and police protection etc. Managers in diverse NFPs such as, universities, Public hospitals, and charitable and development oriented nongovernmental organizations (NGOs) are also apply CVP as operational analysis model. For instance Jimma University can use CVP in analyzing the change in student population and its

impact on the cost of service provision such as, cafeteria, library teaching, tutorial, research guidance and the like.

Activity: Explain how do nonprofit organizations apply CVP model in their administrative decisions?

Unit Summary

Unit five presents the cost-volume-profit (CVP) analysis model and illustrates how managers use that model in their profit planning and answer important “what-if” business questions. CVP analysis helps management accountants alert managers to the risks and rewards of decisions they are considering, by illustrating how the “bottom-line” is affected by changes in activity levels and/or key pricing or cost components. CVP analysis is based on several assumptions, one of which is that fixed costs can be distinguished from variable costs. However, whether a cost is variable or fixed depends on the time period for the decision and also the range of activity (relevant range) being considered. You were also presented with a method for applying CVP analysis to a single product companies and companies with multiple products, and to the scenario of NFP.

Model Examination Questions

Multiple Choice Questions

1. A firm has a negative contribution margin. To reach breakeven, it must;
 - A. Increase Units selling Price
 - B. Increase sales volume
 - C. Decrease fixed cost
 - D. Decrease sales volume
 - E. Increase fixed cost
2. One of the following can increase breakeven point,
 - A. Decrease in Variable cost
 - B. Decrease in Fixed cost
 - C. An increase in selling price
 - D. An increase in variable cost
 - E. None of the above

3. The Environmental Filter Company is planning to sell air filter systems for Birr2, 500 per unit. Variable costs are Birr 1,500 per unit and total fixed costs are Birr1, 000,000. What is the dollar value of sales necessary to break even?

- A. Birr 1,000,000
- B. Birr 2,000,000
- C. Birr 2,500,000
- D. Birr 5,000,000

4. Efa Company reported sales of Birr 150,000 (20,000 units). Fixed costs amounted to Birr 20,000 and operating income for the period was birr 90,000. Determine the per-unit variable cost.

- A. Birr 1.00
- B. Birr 2.00
- C. Birr 4.50
- D. Birr 5.50

ANSWER KEY FOR MODEL EXAM QUESTIONS

1. A 2. D 3.C 4. B

UNIT FIVE

STANDARD COSTING AND VARIANCES

UNIT OUTLINE

5.1.Introduction

5.2.Standard cost system

5.2.1. Meaning of standard costing

5.2.2. Why standard costing are used

5.2.3. Budgetary control vs. standard costing

5.2.4. Advantage and limitation of standard costing

5.2.5. Development of standard cost system

5.2.6. Considerations in establishing standards

5.3.Variance analysis

5.3.1. introduction

5.3.2. classification of variances

5.3.2.1.direct material and labor variance

5.3.2.2.overhead cost variance

5.3.2.3.sales variance

5.3.3. causes and disposition of variances

UNIT OBJECTS

After completing this chapter, you should be able to answer the following questions:

- Why is standard cost systems used?
- How are standards for material, labor, and overhead set?
- How are material, labor, and overhead variances calculated and disposed?
- What are the benefits organizations derive from standard costing and variance analysis?
- How will standard costing be affected if a company uses a single conversion element rather than the traditional labor and overhead elements?
- How do multiple material and labor categories affect variances?

5.1. INTRODUCTION

One of the most important functions of management is control. The major aspect of managerial control is cost control. Control normally refers to the function which ensures the actual performance conforms to a given plan. Hence, the control function of management can be effective only when it is preceded by planning. Standard costing is a technique which helps management to plan and control business operations.

The word 'standard' means a bench-mark or yardstick. The standard cost is a predetermined or expected cost which determines what each product or service should cost under given conditions. In other words, it is the expected cost of producing one unit. It is, in effect, a budget for one unit. Standard costing may be defined basically as a technique of cost accounting which compares the standard cost of each product or service with the actual cost to determine the efficiency of operation so that remedial action may be taken immediately. The institute of costs and management accounts of England defines standard costing as "the preparation and use of standard costs, the comparison with the actual costs, and the analysis of variances to their cause and the points of incidence." Variance is the difference between a budget or standard amount and the actual amount during a given period.

5.2. STANDARD COST SYSTEMS

5.2.1. MEANING OF STANDARD COSTING

Standard costs are predetermined costs that are usually expressed on per unit basis. In other word, standard cost is a predetermined calculation of how much costs should be incurred under specified working condition. It is built up from an assessment of the value of direct material, direct labor and overhead items.

5.2.2. WHY STANDARD COST SYSTEMS ARE USED

 ***Dear learner! Do you know why managers use standard cost?***

"A standard cost system has three basic functions: collecting the actual costs of a manufacturing operation, determining the achievement of that manufacturing operation, and evaluating performance through the reporting of variances from standard." These basic functions result in six distinct benefits of standard cost systems.

Clerical Efficiency

A company using standard costs usually discovers that less clerical time and effort are required than in an actual cost system. In an actual cost system, the accountant must continuously recalculate changing actual unit costs. In a standard cost system, unit costs are held constant for some period. Costs can be assigned to inventory and cost of goods sold accounts at predetermined amounts per unit regardless of actual conditions.

Motivation

Standards are a way to communicate management's expectations to workers. When standards are achievable and when workers are informed of rewards for standards attainment, those workers are likely to be motivated to strive for accomplishment. The standards used must require a reasonable amount of effort on the workers' part.

Planning

Planning generally requires estimates about the future. Managers can use current standards to estimate future quantities and costs. These estimates should help in the determination of purchasing needs for material, staffing needs for labor, and capacity needs related to overhead that, in turn, will aid in planning for company cash flows. In addition, budget preparation is simplified because a standard is, in fact, a budget for one unit of product or service. Standards are also used to provide the cost basis needed to analyze relationships among costs, sales volume, and profit levels of the organization.

Controlling

The control process begins with the establishment of standards that provide a basis against which actual costs can be measured and variances calculated. Variance analysis is the process of categorizing the nature (favorable or unfavorable) of the differences between actual and standard costs and seeking explanations for those differences. A well-designed variance analysis system captures variances as early as possible, subject to cost-benefit assessments. The system should help managers determine who or what is responsible for each variance and who is best able to explain it. An early measurement and reporting system allows managers to monitor operations, take corrective action if necessary, evaluate performance, and motivate workers to achieve standard production.

Decision Making

Standard cost information facilitates decision making. For example, managers can compare a standard cost with a quoted price to determine whether an item should be manufactured in-house or instead be purchased. Use of actual cost information in such a decision could be inappropriate because the actual cost may fluctuate from period to period. Also, in making a decision on a special price offering to purchasers, managers can use standard product cost to determine the lower limit of the price to offer. In a similar

manner, if a company is bidding on contracts, it must have some idea of estimated product costs. Bidding too low and receiving the contract could cause substantial operating income (and, possibly, cash flow) problems; bidding too high might be uncompetitive and cause the contract to be awarded to another company.

Performance Evaluation

When top management receives summary variance reports highlighting the operating performance of subordinate managers, these reports are analyzed for both positive and negative information. Top management needs to know when costs were and were not controlled and by which managers. Such information allows top management to provide essential feedback to subordinates, investigate areas of concern, and make performance evaluations about who needs additional supervision, who should be replaced, and who should be promoted. For proper performance evaluations to be made, the responsibility for variances must be traced to specific managers.

Activity 5.1

1. *Why is standard cost systems used? -----*
2. *What are the basic functions of standard cost systems? -----*

4.2.3. BUDGETARY CONTROL vs. STANDARD COSTING

 ***Dear student! What makes the same and differentiate budget from standard? Why we need to use standard value rather than using budget amount?***

Budgetary control and standard costing are comparable system of cost accounting in that they are both predetermined of forward – looking. The common objective is of controlling business operations by establishing predetermined targets. However, there are a few differences between these two systems are which given below:

Budgetary control system	Standard costing system
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1. Budgetary control is related to all types of items of revenue and expenditure, whether they belong to the product or not, i.e. to all types of business activities. Hence, it is more extensive. 2. Budget is based on past experience and in most cases a projection of financial accounts. 3. Budgets are comparatively less rigid and 'should be' estimates. They fix limits. 4. Budgetary control can be operated without a standard costing system. It can be adopted in part. 5. The study of variances is not a subject of special study as in the case of standard costing	1. Standard costing is related to production and production costs. Hence, it is more rigorous and intensive. 2. Standard is established on the basis of technical estimates. It is the projection of accounts. 3. Standard are very rigid and 'ought to be' estimates. They fix targets. 4. Standard costing system cannot operate well without a budgetary control system. Also, it is not possible to operate the system in parts 5. Variance analysis is a subject of special study of standard costing.
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4.2.4. ADVANTAGES AND LIMITATIONS OF STANDARD COSTING

Advantages of standard costing

1. The weakness of the traditional costing system can be estimated by compiling standard costs more carefully.
2. Standard costs can be used as a yardstick against which actual costs can be compared. It is an effective tool for planning production costs. Hence, cost control is greatly facilitated.
3. Variance analysis helps management to have regular as well as better checks over costs incurred. It makes the application of the principle of management by exception more easy. That is, the management can concentrate its attention on variances only, leaving the other aspects of cost control to be taken care of at the lower level.

4. It is a valuable guide to management in the formulation of production and price policies in advance with certainty. It also assists management in the areas of profit – planning, product –pricing, and inventory pricing, etc.
5. Standard costing makes the reporting of operating data more meaningful and also fast. This makes the interpretation of management reports easy.
6. As the emphasis of the standard costing system is more on cost variations, it makes the entire organization cost conscious. It makes the employees recognize the importance of efficient operations so that costs can be reduced by joint efforts.
7. Labor, materials and machines can be effectively used, and economies can be affected in addition to increase productivity. Standards may also be used as the basis for introducing incentive schemes.

Limitations of standard costing

1. Setting of standards is a very difficult task. It requires a lot of scientific studies such as time –study, motion study, etc., and therefore it is very costly. Small firms may find it very difficult to operate such a system.
2. Standards are very rigid estimates and once set, are not changed for a considerable time. This makes the standards highly unrealistic in certain industries which face fluctuations in prices of products due to frequent changes.
3. The utility of variance analysis depends much more on the standard set. While a loosely set standard may be ridicule the standards which are set very high may create frustration in the minds of the workers. At the same time setting of correct standards is also, it is difficult to apply this system when production takes more than one accounting period.

4.2.5. DEVELOPMENT OF A STANDARD COST SYSTEM

 ***Dear learner! Do you know how to develop standards for production inputs?***

Although standard cost systems were initiated by manufacturing companies, these systems can also be used by service and not-for-profit organizations. In a standard cost system, both standard and actual costs are recorded in the accounting records. This dual recording provides an essential element of cost control: having norms against which actual operations can be compared. Standard cost systems make use of ***standard costs***,

which are the budgeted costs to manufacture a single unit of product or perform a single service. Developing a standard cost involves judgment and practicality in identifying the material and labor types, quantities, and prices as well as understanding the kinds and behaviors of organizational overhead.

A primary objective in manufacturing a product is to minimize unit cost while achieving certain quality specifications. Almost all products can be manufactured with a variety of inputs that would generate the same basic output and output quality. The input choices that are made affect the standards that are set. Some possible input resource combinations are not necessarily practical or efficient.

Once management has established the desired output quality and determined the input resources needed to achieve that quality at a reasonable cost, quantity and price standards can be developed. Experts from cost accounting, industrial engineering, personnel, data processing, purchasing, and management are assembled to develop standards. To ensure credibility of the standards and to motivate people to operate as close to the standards as possible, involvement of managers and workers whose performance will be compared to the standards is vital.

Material Standards

The first step in developing material standards is to identify and list the specific direct materials used to manufacture the product. This list is often available on the product specification documents prepared by the engineering department prior to initial production. In the absence of such documentation, material specifications can be determined by observing the production area, querying of production personnel, inspecting material requisitions, and reviewing the cost accounts related to the product. Three things must be known about the material inputs: types of inputs, quantity of inputs used, and quality of inputs used.

In making quality decisions, managers should seek the advice of materials experts, engineers, cost accountants, marketing personnel, and suppliers. In most cases, as the material grade rises, so does cost; decisions about material inputs usually attempt to balance the relationships of cost, quality, and projected selling prices with company objectives. The resulting trade-offs affect material mix, material yield, finished product quality and quantity, overall product cost, and product salability. Thus, quantity and cost estimates become direct functions of quality decisions.

Given the quality selected for each component, physical quantity estimates of weight, size, volume, or some other measure can be made. These estimates can be based on results of engineering tests, opinions of managers and workers using the material, past material requisitions, and review of the cost accounts.

Specifications for materials, including quality and quantity, are compiled on a bill of materials. Even companies without formal standard cost systems develop bills of materials for products simply as guides for production activity. When converting quantities on the bill of materials into costs, allowances are often made for normal waste of components. After the standard quantities are developed, prices for each component must be determined. Prices should reflect desired quality, quantity discounts allowed, and freight and receiving costs. Although not always able to control prices, purchasing agents can influence prices. These individuals are aware of alternative suppliers and attempt to choose suppliers providing the most appropriate material in the most reasonable time at the most reasonable cost. The purchasing agent also is most likely to have expertise about the company's purchasing habits. Incorporating this information in price standards should allow a more thorough analysis by the purchasing agent at a later time as to the causes of any significant differences between actual and standard prices.

When all quantity and price information is available, component quantities are multiplied by unit prices to obtain the total cost of each component. (Remember, the price paid for the material becomes the cost of the material.) These totals are summed to determine the total standard material cost of one unit of product.

Labor Standards

Development of labor standards requires the same basic procedures as those used for material. Each production operation performed by either workers (such as bending, reaching, lifting, moving material, and packing) or machinery (such as drilling, cooking, and attaching parts) should be identified. In specifying operations and movements, activities such as cleanup, setup, and rework are considered. All unnecessary movements by workers and of material should be disregarded when time standards are set.

To develop usable standards, quantitative information for each production operation must be obtained. Time and motion studies may be performed by the company; alternatively, times developed from industrial engineering studies for various movements can be used. A third way to set a time standard is to use the average time needed to manufacture a product during the past year. Such information can be calculated from employees' past

time sheets. A problem with this method is that historical data may include inefficiencies. To compensate, management and supervisory personnel normally make subjective adjustments to the available data.

After all labor tasks are analyzed, an **operations flow document** can be prepared that lists all operations necessary to make one unit of product (or perform a specific service). When products are manufactured individually, the operations flow document shows the time necessary to produce one unit. In a flow process that produces goods in batches; individual times cannot be specified accurately.

Labor rate standards should reflect the employee wages and the related employer costs for fringe benefits, FICA (Social Security), and unemployment taxes. In the simplest situation, all departmental personnel would be paid the same wage rate as, for example, when wages are job specific or tied to a labor contract. If employees performing the same or similar tasks are paid different wage rates, a weighted average rate (total wage cost per hour divided by the number of workers) must be computed and used as the standard. Differing rates could be caused by employment length or skill level.

Overhead Standards

To provide the most appropriate costing information, overhead should be assigned to separate cost pools based on the cost drivers, and allocations to products should be made using different activity drivers.

After the bill of materials, operations flow document, and predetermined overhead rates per activity measure have been developed, a standard cost card is prepared. This document summarizes the standard quantities and costs needed to complete one product or service unit.

Data from the standard cost card are then used to assign costs to inventory accounts. Both actual and standard costs are recorded in a standard cost system, although it is the standard (rather than actual) costs of production that are debited to Work in Process Inventory. Any difference between an actual and a standard cost is called a variance.

Activity.4.2

1. *How are standards for material, labor, and overhead set?* -----

4.2.6. CONSIDERATIONS IN ESTABLISHING STANDARDS

 *Dear learner! Do you know what should be taken into account when you are establishing standards?*

When standards are established, appropriateness and attainability should be considered. Appropriateness, in relation to a standard, refers to the basis on which the standards are developed and how long they will be expected to last. Attainability refers to management's belief about the degree of difficulty or rigor that should be incurred in achieving the standard.

Appropriateness

Although standards are developed from past and current information, they should reflect relevant technical and environmental factors expected during the time in which the standards are to be applied. Consideration should be given to factors such as material quality, normal material ordering quantities, expected employee wage rates, degree of plant automation, facility layout, and mix of employee skills. Management should not think that, once standards are set, they will remain useful forever. Current operating performance is not comparable to out-of-date standards.

Standards must evolve over the organization's life to reflect its changing methods and processes. Out-of-date standards produce variances that do not provide logical bases for planning, controlling, decision making, or evaluating performance.

Attainability

Standards provide a target level of performance and can be set at various levels of rigor. The level of rigor affects motivation, and one reason for using standards is to motivate employees. Standards can be classified as expected, practical, and ideal. Depending on the type of standard in effect, the acceptable ranges used to apply the management by exception principle will differ. This difference is especially notable on the unfavorable side.

4.3. VARIANCE ANALYSIS

4.3.1. Introduction

The main advantage of the standard costing system is variance analysis. The principle of "management by exception" is practiced easily with the help of variances. Variance may be defined as the difference between standard and actual for each element of cost and

sometimes for sales. And ‘variance analyses’ may be defined as the process of analyzing variance by sub –dividing the total variance in such a way that management can assign responsibility for off –standard performance. When the actual results are better than expected, a ‘favorable’ variance arises; where they are not up to the standard, an ‘adverse variance’ occurs.

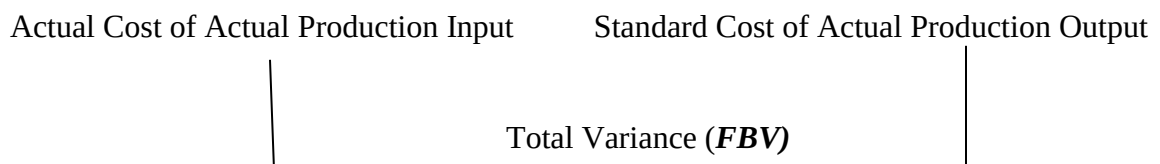
Variances help to fix the responsibilities so that management can ascertain the person responsible for the poor results. For example, an adverse material usage variance would indicate that excess material cost was due to inefficient use of materials. This would enable management to fix the responsibility on the supervisor in charge of a particular operation in which the inefficiency occurred. It may be discovered that the variance was caused by (say) inefficient handling, purchase of poor quality materials or employment of trainees. The important point is that the reason for the variance must be found, explained and wherever necessary, corrective measures taken.

4.3.2. Classification of variances

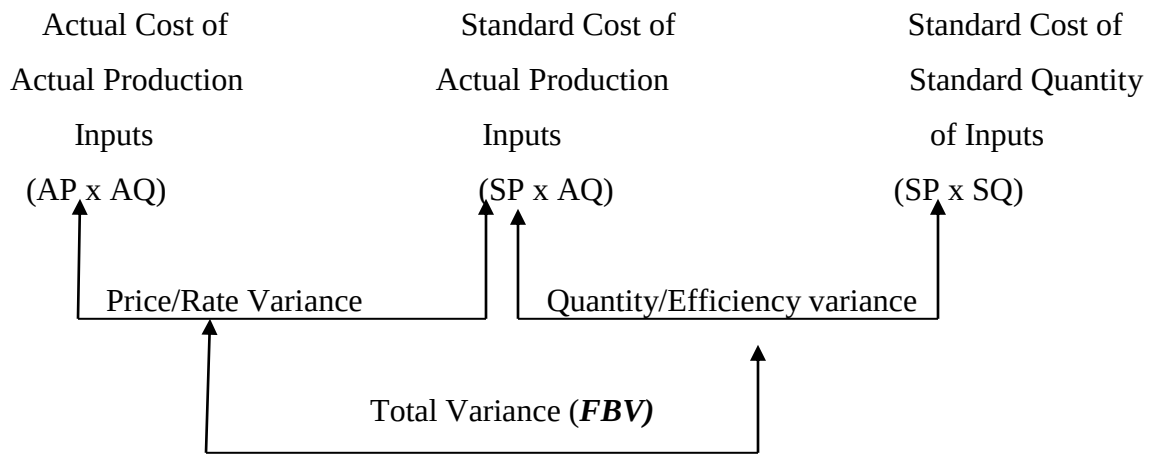
☞ ***Dear learner! What do you know about budget variance? And how do you classify it?***

Budget Variances indicate the total deviation of actual costs from expected costs. However, they do not give the complete story about deviations between budgeted and actual results; i.e. it is not yet clear what contributed to the variances unless further investigations are made.

Hence, a **total flexible budget variance (FBV)** is the difference between total actual costs incurred and total standard cost applied to the output produced during the period. This variance can be diagrammed as follows:



Since total variances do not provide useful information for determining why cost differences occurred; to help managers in their control objectives, total variances are subdivided into price and usage components. Hence, the total variance diagram can be expanded to provide a general model indicating the two sub-variances as follows:



A price/rate variance reflects the difference between what was paid for inputs and what should have been paid for inputs. A usage /Quantity//Efficiency variance shows the cost difference between the quantity of actual input and the quantity of standard input allowed for the actual output of the period. The quantity difference is multiplied by a standard price to provide a monetary measure that can be recorded in the accounting records. Usage variances focus on the efficiency of results or the relationship of input to output

As shown above, the diagram moves from actual cost of actual input on the left to standard cost of standard input quantity on the right. The middle measure of input is a hybrid of actual quantity and standard price. The change from input to output reflects the fact that a specific quantity of production input will not necessarily produce the standard quantity of output. The far right column uses a measure of output known as the standard quantity allowed. This quantity measure translates the actual production output into the standard input quantity that should have been needed to achieve that output. The monetary amount shown in the right-hand column is computed as the standard quantity allowed times the standard price of the input.

The price variance portion of the total variance is measured as the difference between the actual and standard prices multiplied by the actual input quantity:

$$\text{i.e. Price Element} = (AP - SP)(AQ)$$

Whereas, the usage variance portion of the total variance is measured as measuring the difference between actual and standard quantities multiplied by the standard price:

$$\text{i.e. Usage Element} = (AQ - SQ)(SP)$$

Dear student, the following sections illustrate variance computations for each cost element (i.e. price and usage components) on input of production.

4.3.2.1.DIRECT MATERIAL AND LABOR VARIANCES

☞ *Dear learner! How are material, labor, and overhead variances calculated? How materials and labor cost variance computed when there are is only one type (classes) of material (labor) input used in the production? What about if there are more than one type (classes) of material (labor) input used in the production?*

A. DIRECT MATERIAL COST VARIANCES:

To completely and meaningfully analyze the flexible budget variance for material, it should be analyzed in terms of the materials price standard and the materials quantity standard. This level of analysis resulted in: (i) material price variance that identifies the effect of differences in prices paid for materials. (ii) Material quantity (usage) variance that identifies the effect of difference in the quantities of materials used.

i) **Material Price Variance(MPV)**

A material price variance is the difference between the actual price of material/unit and the standard price of material per unit multiplied by the actual quantity of material purchased. In other words, the **material price variance** (MPV) indicates whether the amount paid for material was below or above the standard price. Material price variance can be calculated as:

$$\text{MPV} = (\text{SP} - \text{AP}) \text{AQ}$$

where: SP is the standard price of material per unit

AP is actual price of material per unit

AQ is the actual quantity of material purchased and consumed

If the actual price is larger than the standard price, this variance is unfavorable (U); if the standards are larger than the actual; the variance is favorable (F)

ii) **Materials Quantity (usage) Variance(MQV)**

The **material quantity variance** (MQV) indicates whether the actual quantity used was below or above the standard quantity allowed for the actual output. This difference is multiplied by the standard price per unit of material. i.e. A material quantity (usage) variance is the difference between the actual quantity of materials used and the standard

quantity of materials that should have been used to produce the actual output, multiplied by the standard price of materials per unit. It can be calculated as:

$$\text{MQV} = (\text{SQ} - \text{AQ}) \times \text{SP}$$

where: SP is the standard price of material per unit

SQ is the actual quantity of material used for
units produced

AQ= the actual quantity of material used

If the actual quantity amounts are larger than the standard quantity amounts, this variance is unfavorable (U); if the standards are larger than the actual; the variance is favorable (F)

Example 5.1 In August 2001, East Publishing Company's costs and quantities of paper consumed in manufacturing its 2002 Executive Planner and Calendar were as follow:

Actual unit purchase price	Br 0.16 per page
Standard quantity allowed for good production	195,800 pages
Actual quantity purchased during August	230,000 pages
Actual quantity used in August	200,000 pages
Standard unit price	Br 0.15 per page

Required:

- Calculate the total cost of purchases for August.
- Compute the material price variance on the bases of purchase
- Calculate the material quantity variance.
- Total FBV

Solution:

- Total cost of purchases for August would be:

Actual unit purchase price (a) ----- Br 0.16 per page

Actual quantity purchased during August (b) ----- 230,000 pages

Total cost (a x b) ----- Br.31, 220

- MPV = (SP – AP) AQ

= (Br 0.15 per page - Br 0.16 per page) 230,000 pages= Br. 2,300 (U)

- MQV = (SQ-AQ) x SP

= (195,800 pages - 200,000 pages) Br 0.15 per page= Br. 630(U)

- Total FBV= Br 2,300U + Br. 630U= Br.2,930 U

Activity4.3

Iron Eagle makes wrought iron table and chair sets. During April 2001, the purchasing agent bought 12,800 pounds of scrap iron at Br. 0.89 per pound. Each set requires a standard quantity of 35 pounds at a standard cost of Br 0.85 per pound. During April, the company used 10,700 pounds and produced 300 sets. For April, compute the direct material price variance and the direct material quantity variance.

B. LABOR COST VARIANCES:

Just like we have done for material inputs, we will do the same meaningful analysis for labor inputs. Hence, the variance investigation to flexible budget variance for labor resulted in: (i) labor rate variance that identifies the effect of differences in the rates paid to workers, and (ii) labor efficiently or usage variance that identifies the effect of differences in the quantities of labor used.

i) Labor Rate Variance(LRV)

The **labor rate variance** (LRV) shows the difference between the actual wages paid to labor for the period and the standard wages for all hours worked. Thus, Labor rate variance is the difference between the actual rate of labor per hour and the standard rate of labor per hour, multiplied by the actual hours of labor worked.

$$\text{LRV} = (\text{SR} - \text{AR}) \times \text{AH}$$

Where: SR is standard rate of labor per hour

AR is actual rate of labor per hour, AH is a total actual hour of labor worked

ii) Labor Efficiency Variance (LEV)

A labor efficiency variance is the difference between the actual labor hours worked and the standard labor hours that should have been worked to produce the actual output, multiplied by the standard rate of labor per hour. Thus, multiplying the standard labor rate by the difference between the actual minutes worked and the standard minutes for the production achieved results in the **labor efficiency variance** (LEV)

$$\text{LEV} = (\text{SH} - \text{AH}) \times \text{SR}$$

Where: SH is standard hours of labor for the actual unit produced

AH is actual labor hours used for the unit produced

SR is standard rate of labor per hour

Example 4.2 Sagittarius Corp. has established the following standards for the prime costs of one of its chief product, dart boards.

	<u>Standard Qty</u>	<u>standard Price (Rate)</u>	<u>Total Standard cost</u>
Direct material	8.5 pounds	Br.1.80/pound	Br.15.30
Direct labor	0.25 hour	8.00/hour	<u>2.00</u>
			Br. <u>17.30</u>

During May, Sagittarius purchased 160,000 pounds of direct material at a total cost of Br.304, 000. The total wages for May were Br.42, 000, 90% of which were for DL. Sagittarius manufactured 19,000 dart boards during May; using 142,500 pounds of direct material & 5,000 direct labor hours.

Required: Compute the following variances for May.

- Direct material price variances
- Direct material usage variance
- Direct material cost variance
- Direct labor rate variance
- Direct labor efficiency variance
- Direct labor cost variance

Solution

Material Variances

i). SQ = Standard Quantity of Direct material for Actual Output would be:

$$\begin{aligned} 1 \text{ dart board} &= 8.5 \text{ pounds Direct material} \\ 19,000 \text{ dart boards} &= ? \end{aligned}$$

$$\text{So, SQ} = 8.5 \text{ pounds Direct material} \times 19,000 = 161,500 \text{ pounds}$$

ii). SP = Standard price of Direct material = Br. 1.80/pounds.

$$\begin{aligned} \text{iii). AP} &= \text{Actual Price of Direct material} = \frac{\text{Total cost of Direct material purchased}}{\text{Total units of direct material purchased}} \\ &= \frac{\text{Br. 304,000}}{160,000 \text{ pounds}} = \text{Br.1.90/ pounds} \end{aligned}$$

iv). AQ=Actual Quantity of Direct material purchased or used = 142,500 pounds

- MPV = (SP – AP) AQ
= (Br.1.80/pds - Br.1.90/ pds) 142,500 pounds = Br. 14,250(U)
- MQV = (SQ-AQ) x SP
= (161,500 pounds - 142,500 pounds) Br. 1.80/pounds= Br. 34,200(F)
- Material cost variance = MPV + MQV
= Br. 14,250 (U) + Br. 34,200 (F) = Br. 19,950 (F)

Labor Variances

- i) SR= Standard Rate of DL per hour= Br.8.00/Hr
- ii) SH=Standard Hours of DL for Actual Output would be:
 $0.25\text{Hrs}=1\text{dart board}$
 $? = 19,000 \text{ dart boards}$
 $\text{SH} = 0.25 \times 19,000 = 4,750\text{Hrs}$
- iii) AH=Actual hrs of DL used = 5,000 hrs
- iv) AR= $\frac{0.9 \times 42,000}{5,000} = \frac{37,800}{5,000} = \text{Br.7.56/Hr}$
- d) LRV= (SR-AR) x AH
 $= (\text{Br.8.00/Hr} - \text{Br.7.56/Hr}) 5,000 \text{ hrs} = \text{Br. 2,200 (F)}$
- e) LEV = (SH – AH) x SR
 $= (4,750\text{Hrs} - 5,000 \text{ hrs}) \text{Br.8.00/Hr} = \text{Br.2, 000(U)}$
- f) labor cost variance = LRV + LEV= Br. 2,200 (F) + Br.2, 000(U) = Br. 200 (F)

c) MIX AND YIELD VARIANCES

 ***Dear learner! What do you know about production mix and yield?***

Most companies use a combination of many materials and various classifications of direct labor to produce goods. In such settings, the material and labor variance analysis presented in the above section are insufficient.

When a company's product uses more than one material, the goal is to combine those materials in such a way as to produce the desired product quality in the most cost-beneficial manner. Sometimes, materials can be substituted for one another without affecting product quality. In other instances, only one specific material or type of material can be used. For example, a furniture manufacturer might use either oak or maple to build a couch frame and still have the same basic quality. A perfume manufacturer, however, may be able to use only a specific fragrance oil to achieve a desired scent.

Labor, like materials, can be combined in many different ways to make the same product. Some combinations will be less expensive than others; some will be more efficient than others. Again, all potential combinations may not be viable.

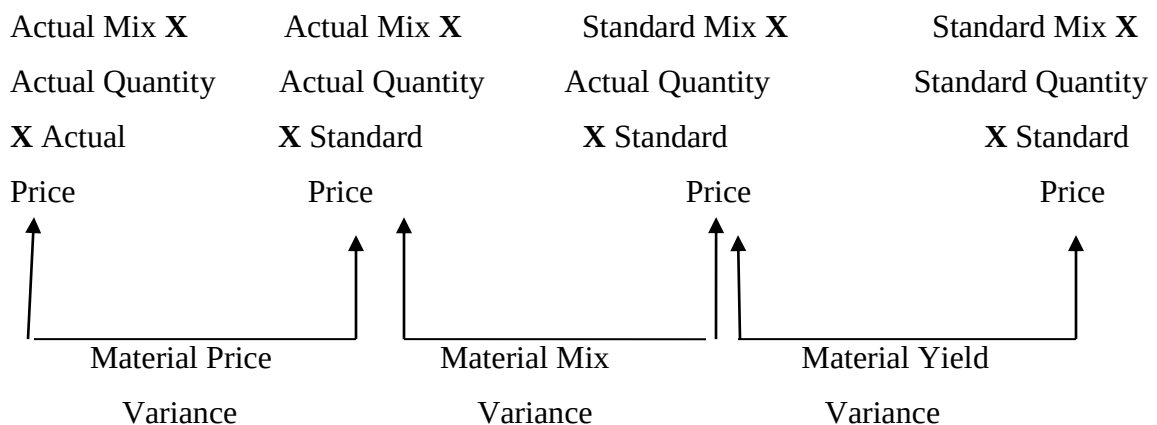
Management desires to achieve the most efficient use of labor inputs. As with materials, some amount of interchangeability among labor categories is assumed. Skilled labor is more likely to be substituted for unskilled because interchanging unskilled labor for skilled labor is often not feasible. However, it may not be cost effective to use highly skilled, highly paid workers to do tasks that require little or no training. A rate variance for direct labor is calculated in addition to the mix and yield variances.

Each possible combination of materials or labor is called a ***mix***. Management's standards development team sets standards for materials and labor mix based on experience, judgment, and experimentation. Mix standards are used to calculate mix and yield variances for materials and labor. An underlying assumption in product mix situations is that the potential for substitution exists among the material and labor components. If this assumption is invalid, changing the mix cannot improve the yield and may even prove wasteful. In addition to mix and yield variances, price and rate variances are still computed for materials and labor.

i) **Material Mix and Yield Variances**

A material price variance shows the Birr effect of paying prices that differ from the raw material standard. The ***material mix variance (MMV)*** measures the effect of substituting a nonstandard mix of materials during the production process. The ***material yield variance (MYV)*** is the difference between the actual total quantity of input and the standard total quantity allowed based on output; this difference reflects standard mix and standard prices. The sum of the material mix and yield variances equals a material quantity variance; the difference between these two variances is that the sum of the mix and yield variances is attributable to multiple ingredients rather than to a single one. A company can have a mix variance without experiencing a yield variance.

Computations for the price, mix, and yield variances are given below in a format similar to that used in the above section.



The formula to compute material mix and yield variance would be:

To compute material mix variance:

$\text{Material Mix Variance (MMV)} = (\text{RSQ} - \text{AQ}) \text{SP}$

Where, RSQ = revised standard quantity(i.e. actual quantity at standard mix)

$$= \frac{\text{SQ for each material}}{\text{Total SQ}} \times \text{Total AQ} \quad (\text{OR})$$

Total SQ

$$= \text{Total AQ} \times \text{standard mix ratio}$$

AQ= Actual Quantity at actual mix

SP = standard price

To compute material yield variance:

$$\text{MYV} = (\text{SQ} - \text{RSQ})\text{SP}$$

Where, SQ= Standard Quantity at standard mix

RSQ = Revised Standard Quantity

SP = standard price

Example4.3 The Scent Makers Company produces perfume. To make this perfume, Scent makers uses three different types of fluids: Dycone, Cycone, & Bycone are used in standard proportions of 4/10, 3/10, & 3/10 and their standard costs are Br. 6.00, Br. 3.50 & Br. 2.50 per unit, respectively. The chief engineer reported that for the past few months the standard yield has been 80% on 100 pints of mix. The Company maintains a policy of not carrying any direct material, as inventory storage space is costly.

Last week the company produced 75,000 pints of perfume at a total direct material cost of Br. 449,500. The actual number of pints used and costs per unit for the three fluids are as follows:

<u>Material</u>	<u>Actual Pints</u>	<u>Cost/Pint</u>
Dycone	45,000	Br. 5.50
Cycone	35,000	4.20
Bycone	<u>20,000</u>	2.75
	<u>100,000</u>	

Required

- 1) Compute the total direct material yield & mix variances for the last week..
- 2) Compute the total direct material price & usage variances for perfume made in the last week.

Solution:

i) SQ = Standard quantity for actual output

Standard yield (80% on 100 pints of mix)

i.e. 80pints required = 100 pints of mix

For actual production of 75,000 pints = ?

$$= 75,000 \times \frac{100}{80} = \underline{93,750 \text{ pints of mix}}$$

So, SQ for: Dycone: $0.4 \times 93,750 = 37,500$

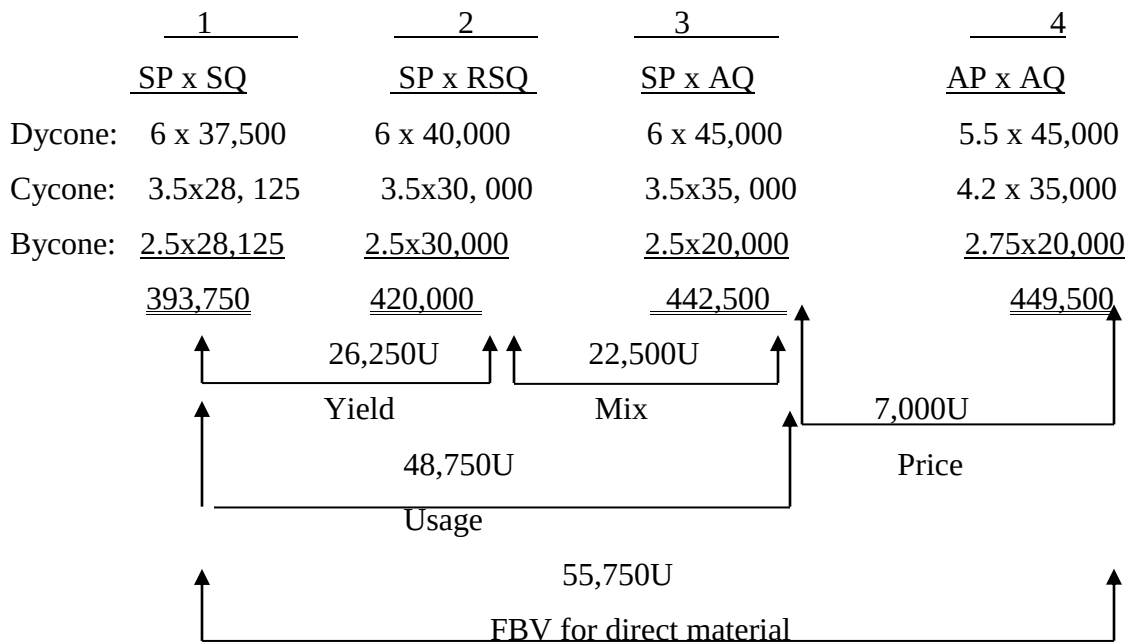
Cycone: $0.3 \times 93,750 = 28,125$

Bycone: $0.3 \times 93,750 = 28,125$

ii) RSQ for:

	Standard Mix	Actual quantity	Proportion	RSQ
Dycone :	0.4	45,000	$0.4 \times 100,000$	40,000
Cycone :	0.3	35,000	$0.3 \times 100,000$	30,000
Bycone :	0.3	<u>20,000</u>	$0.3 \times 100,000$	30,000
Total		<u>100,000</u>		

Thus, the direct material cost variances, in diagram, would be:

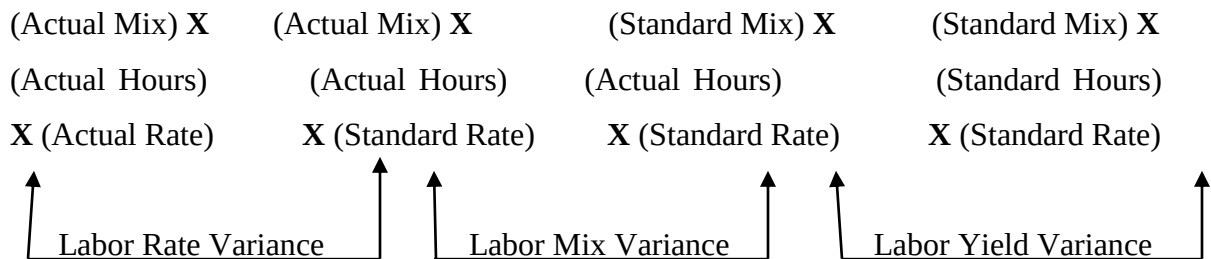


Note:

- $MYV = 1-2 = SP \times (SQ - RSQ) = \text{Br. } \underline{26,250U}$
- $MMV = 2-3 = SP \times (RSQ - AQ) = \text{Br. } \underline{22,500U}$
- $MQV = 1-3 = (SP \times SQ) - (SP \times AQ) = MYV + MMV = \text{Br. } \underline{48,750U}$
- $MPV = 3-4 = (SP - AP) \times AQ = \text{Br. } \underline{7,000U}$
- $FBV \text{ for DM} = 1-4 = (SP \times SQ) - (AP \times AQ) = MQV + MPV = \text{Br. } \underline{55,750U}$

ii. Labor Mix, and Yield Variances

The labor rate variance is a measure of the cost of paying workers at other than standard rates. The labor mix variance is the financial effect associated with changing the proportionate amount of higher or lower paid workers in production. The labor yield variance reflects the monetary impact of using more or fewer total hours than the standard allowed. The sum of the labor mix and yield variances equals the labor efficiency variance. The diagram for computing labor rate, mix, and yield variances is as follows:



The formula to compute labor mix and yield variance would be:

To compute labor mix variance:

Labor Mix Variance (LMV) = (RSH – AH)SR

Where: RSH= revised standard hour (i.e. actual hours at standard mix)

$$= \frac{\text{SH for each labor} \times \text{Total AH}}{\text{Total SH}} \quad (\text{OR})$$

$$= \text{Total AH} \times \text{standard mix ratio}$$

AH=Actual hours at actual mix SR= standard rate per hours

To compute Labor Yield Variance

Labor Yield Variance(LYV) = (SH-RSH)SR

Where: SH= standard hours at standard mix

RSH= revised standard hour SR= standard rate per hours

Example4.4 Buffon Legal Services has three labor classes: secretaries, paralegals, and attorneys. The standard wage rates are shown in the standard cost system as follows: secretaries, Br 25 per hour; paralegals, Br 40 per hour; and attorneys, Br 85 per hour. The firm has established a standard of 0.5 hours of secretarial time and 2 hours of paralegal time for each hour of attorney time in probate cases. The actual direct labor hours worked

on probate cases and the standard hours allowed for the work accomplished for one month in 2001 were as follows:

	Actual Labor Hrs	Standard Hours for Output Achieved
Secretarial	500	500
Paralegal	1,800	2,000
Attorney	<u>1,100</u>	<u>1,000</u>
Total:	<u>3,400hrs</u>	<u>3,500hrs</u>

Required: Calculate the amount of the direct labor efficiency variance for the month and decompose the total into the following components:

1. Direct labor mix variance
2. Direct labor yield variance

Solution:

i) SR= Standard Rate per DL Hr

For Secretarial: Br. 25, Paralegal: Br 40, and for Attorney: Br 85

ii) RSH: Revised Standard hours = $\frac{\text{SH for each labor}}{\text{Total SH}} \times \text{Total AH}$

RSH for Secretarial: $\frac{500 \text{ hrs}}{3,500\text{hrs}} \times 3,400\text{hrs} = 486\text{hrs}$

For Paralegal: $\frac{2,000 \text{ hrs}}{3,500\text{hrs}} \times 3,400\text{hrs} = 1,943\text{hrs}$

For Attorney: $\frac{1,000 \text{ hrs}}{3,500\text{hrs}} \times 3,400\text{hrs} = 971\text{hrs}$

a) LMV= (RSH – AH)SR

For Secretarial: (486hrs - 500 hrs) Br. 25 = Br. 350 (U)

For Paralegal: (1,943hrs - 1,800 hrs) Br 40 = 5,720(F)

For Attorney: (971hrs - 1,100 hrs) Br 85 = 10,965(U)

Total Br.5, 595(U)

b) LYV = (SH-RSH)SR

For Secretarial: (500 hrs - 486hrs) Br. 25 = Br. 350 (F)

For Paralegal: (2,000 hrs- 1,943hrs) Br 40 = 2,280(F)

For Attorney: (1,000 hrs - 971hrs) Br 85 = 2,465(F)

Total Br. 5, 095(F)

Activity 4.4

1. *Righting Moment Inc. is a mechanical engineering firm. The firm employs both engineers and drafts people. The average hourly rates are Br. 80 for engineers and Br.40 for drafts people. For one project, the standard was set at 375 hours of engineer time and 625 hours of draftsperson time. Actual hours worked on this project were: Engineers—500 hours at Br.85 per hour and Draftspeople—500 hours at Br.42.00 per hour. Determine the labor rate, mix, and yield variances for this project.*-----
2. *How do multiple material and labor categories affect variances?* -----

4.3.2.2.OVERHEAD COST VARIANCES

In developing overhead application rates, a company must specify an operating level or capacity. Capacity refers to the level of activity. Alternative activity measures include theoretical, practical, normal, and expected capacity. Because total variable overhead changes in direct relationship with changes in activity and fixed overhead per unit changes inversely with changes in activity, a specific activity level must be chosen to determine budgeted overhead costs.

A **flexible budget** is a planning document that presents expected overhead costs at different activity levels. In a flexible budget, all costs are treated as either variable or fixed; thus, mixed costs must be separated into their variable and fixed elements. The activity levels shown on a flexible budget usually cover the contemplated range of activity for the upcoming period. If all activity levels are within the relevant range, costs at each successive level should equal the previous level plus a uniform monetary increment for each variable cost factor. The increment is equal to variable cost per unit of activity times the quantity of additional activity.

The use of separate variable and fixed overhead application rates and accounts allows separate price and usage variances to be computed for each type of overhead.

a. VARIABLE OVERHEAD COST VARIANCE (VOHV)

This is the difference between standard variable overheads for actual production and the actual variable overheads.

Symbolically,

$$\text{VOHV} = \text{SC} - \text{AC}$$

Where: SC= standard variable overheads for actual production

AC= actual variable overheads

It can be sub –divided into ***Variable overhead expenditure variance, and Variable overhead efficiency variance.***

- i) ***VOH expenditure variance*** is the difference between the standard variable overheads for the actual hours worked, and the actual variable overheads incurred. The formula for computing it is as follows:

$$\text{VOH Exp. Variance} = \text{AVOH} - \text{SVOH}.$$

Where: AVOH is actual variable overheads incurred

SVOH is standard variable overheads for the actual hours worked,

- ii) ***VOH efficiency variance*** arises when the actual output produced differs from the standard output for actual hours worked. It is a measure of extra overhead (for saving) incurred solely because of the efficiency shown during the actual hours worked. The formula to compute it is as follows:

$$\text{VOH efficiency variance} = (\text{SHOV for actual hours worked}) - (\text{SHOV for actual output})$$

Example 4.5 From the following information, calculate VOH cost variances assuming labor hours as cost driver for variable manufacturing overhead.

Budget output	5000 units
Budgeted hours	10,000
Budgeted variable overheads	Br. 2,000
Actual variable overheads	Br. 3,000
Actual output	4,000 units
Actual hours	12,000 hours

Solution

- i) VOH cost variance = AVOH – SVOH for actual production
= Br.3, 000 - (4000 x Br 0.40*)
= Br...3,000 - Br. 1600= Br.1400 (U)
- ii) VOH expenditure variance = AVOH – SVOH for actual hours worked
= Br.3000 - (12000 x 0.20**) = Br. 600 (U)
- iii) VOH efficiency variance = SVOH for actual Hrs - SVOH for actual output

$$= \text{Br. } 2400 - \text{Br. } 1600 = \underline{\text{Br. } 800 \text{ (U)}}$$

Workings:

$$*\text{SVOH per unit of output} = \text{Br. } 2000 / 5000 = \text{Br. } 0.40 \text{ per unit}$$

$$**\text{ SVOH pre hours} = \text{Br. } 2000 / 10,000 = \text{Br. } 0.20 \text{ per hour}$$

b. FIXED OVERHEADS COST VARIANCE (FOHV)

This is the difference between the standard fixed overheads for actual output and actual fixed overheads. The reasons for the variance are over absorption or under –absorption of overheads for the actual production the budgeted production may be different from the actual production for the actual overheads incurred. The major sub –divisions of FOHV are FOH expenditure variance and FOH volume variance. The formula for FOHV is as follows:

$$\text{FOHV} = \text{AFOH} - \text{SFOH}$$

Where: AFOH = actual fixed overheads.

SFOH = standard fixed overheads for actual output

Note that, if the AFOH is less than the SFOH, the variance is favorable (F), and vice versa. This variance can be classified into two.

i) FOH expenditure variance (FOHEV)

This is the difference between Actual fixed overhead costs and Budgeted fixed overhead (Symbolically, FOHEV= AFOH –BFOH)

If the actual is greater than the budgeted, this variance is adverse (U), and vice versa

ii) FOH volume Variance (FOHVV)

This is the difference between the budgeted fixed overheads and the standard fixed overheads absorbed on actual production. The formula is as follows:

$$\text{FOHVV} = \text{BFOH} - \text{SFOH on actual production.}$$

If the BFOH is greater than the SFOH on actual production, the variance is adverse (U) and vice versa.

Example 4.6: From the following data calculate FOH cost variance.

Budgeted hours: 10,000 hours; Budgeted output: 5,000 units, Budgeted FOH: Br.3,000

Actual hours: 12,000hours; Actual output: 4,800 units; Actual FOH: Br.3,600

Solution:

$$\begin{aligned} \text{a.} \quad \text{FOHV} &= \text{AFOH} - \text{SFOH on actual output} \\ &= \text{Br. } 3600 - (0.60 * 4800) \\ &= \text{Br. } 3600 - \text{Br. } 2880 = \underline{\text{Br. } 720 \text{ (U)}} \end{aligned}$$

- b. $FOHEV = AHOH - BFOH$
 $= Br.3600 - Br.3000 = Br.600 (U)$
- c. $FOHVV = BFOH - SFOH \text{ on actual output}$
 $= Br.3000 - (0.60 \times 4800)$
 $= Br.3000 - 2880 = Br.120 (U)$

Workings:

*SFOH per unit = Br. 3000/5000 = Br.0.60 per unit

4.3.2.3.SALES VARIANCES

The standard costing system is complete only when sales variances are detailed with it as part of comprehensive information presented to management. Sales variances are calculated by two methods, viz., ***sales value method and sales margin or profit method.*** Basically, changes in price and changes in sales volume give rise to sales variances. A change in value may, in its own turn, result due to a change in quantity, or a change in sales mix

I) SALES VALUE METHOD

The value method is used to denote variance arising due to change in sales price, quantity mix, etc, or the sales value. The sales variances may be classified as: Sales price variance, and Sales volume variance.

- a) **Sales Value Variance (SVV)** is the difference between standard sales and the actual sales.

i.e. $SVV = (SS - AS) = (SP \times SQ) - (AP \times AQ)$ Where: SP= standard price AP = actual price SQ= standard quantity to be sold AQ= actual quantity sold
--

- b) **Sales Price Variance (SPV)** is the difference between standard price of actual quantity and actual price of the actual quantity of goods sold

i.e. $SPV = AQ (SP - AP)$ Where: SP= standard price AP = actual price AQ= actual quantity sold

If the actual price is greater than the standard price, the variance is favorable, and vice versa.

- c) **Sales Volume Variance (SVLV)** is the difference between the actual quantity or volume and the standard quantity or volume and the standard quantity or volume of sales. It shows the effect of a change in volume of total sales.

$$SVLV = SP(SQ - AQ)$$

Where: SQ= standard quantity to be sold AQ= actual quantity sold

SP= standard price

If the actual quantity sold is greater than the standard, the variance is favorable, and vice versa.

Sales volume variance may be further divided into sales mix variance and sales quantity or sub –volume variance

d) Sales Mix Variance (SMV)

Where two or more items are used in the composition of sales, the differences between the standard compositions of sales and the actual composition or mix is called the sales mix variance. It is a part of the value variance arising due to change in the mix. It highlights the fact that the actual mix of sales has not been in the same ratio as budgeted

$$SMV = SP (\text{Revised standard sales} - \text{Actual sales})$$

where: SP= standard price

Note that the **revised standard sales** are calculated by **dividing the total actual sales quantity in the standard proportion**. If the actual sales quantity is greater than the revised standard quantity, the variance is favorable, and vice versa.

e) Sales Quantity Variance (SQV)

This is the difference between the revised standard quantity of actual sales and the standard quantity budgeted. This variance shows the position of actual quantity of sales, as distinct from the mix of sales, in comparison with budgeted or expected sales.

$$\text{Thus, } SQV = SP (SQ - RSQ)$$

Where: SP= standard price SQ= standard quantity to be sold

RSQ = revised standard sales (quantity)

If the RSQ is greater than the SQ, the variance is favorable, and vice versa.

Example5.7: calculate (i) sales variance (ii) sales price variance; (iii) sales volume variances, (iv) sales mix variance and (v) sales quantity variance from the following

Product	Standard		Actual	
	quantity (units)	price (Br.)	quantity (units)	price (Br.)
A	5,000	5.00	5,000	5.00

B	4,000	6.00	6,000	6.25
C	<u>3,000</u>	7.00	<u>4,000</u>	6.75
Total	<u>12,000</u>		<u>15,000</u>	

Solution

- Sales value variance = (SP X SQ) – (AP X AQ)
 Product A = (5,000 x Br. 5) – (5,000 x Br. 5) = Br 0
 Product B = (4,000 x Br. 6) – (6,000 x Br. 6.25) = 13,500 F
 Product C = (3,000 x Br. 7) – (4,000 x Br. 6.75) = 6,000 F
 Total = Br 19,500 F
- Sales price variance = AQ (SP- AP)
 Product A = 5,000 (Br. 5- Br. 5) = Br 0
 Product B = 6,000 (Br. 6- Br. 6.25) = 1,500 F
 Product C = 4,000 (Br. 7- Br. 6.75) = 1,000 U
 Total = Br 500 F
- Sales Volume Variance = SP(SQ- AQ)
 Product A = Br. 5(5,000 - 5,000) = Br 0
 Product B = Br. 6(4,000 - 6,000) = 12,000 F
 Product C = Br. 7(3,000 - 4,000) = 7,000 F
 Total = Br 19,000 F
- Sales mix variance = SP(RSQ* - AQ)
 Product A = Br. 5(6,250 - 5,000) = Br 6,250 U
 Product B = Br. 6(5,000 – 6000) = 6,000 F
 Product C = Br. 7(3,750 – 4,000) = 1,750 F
 Total = Br.1, 500 F

*Calculation of RSQ: total AQ x standard ratio = 15,000units x (5:4:3)

Product A = 15,000x5/12	=6,250 units
Product B = 15,000x4/12	=5,000 units
Product C = 15,000x3/12	= <u>3,750 units</u>
Total	= <u>15,000 units</u>

- Sales quantity variance = SP(SQ- RSQ)
 Product A = 5(5,000- 6,250) = Br 6,250 (F)
 Product B = 6(4,000- 5,000) = 6,000 (F)
 Product C = 7(3,000- 3,750) = 5,250 (F)
 Total = Br 17, 500 (F)

II) SALES MARGIN METHOD OR PROFIT METHOD

Sales margin variance this is the difference between the standard margins appropriate to the quantity of sales budgeted for a period, and the margin between standard cost and the actual selling price of the sales effected. This variance arises due to the difference between total budgeted profit and total actual profit

$\text{Sales Margin Variance} = \text{budgeted profit} - \text{actual profit}$
--

Sales margin variance may be classified into the sales margin price variance and sales margin volume

- i) **Sales margin price variance** which is due to the difference between the standard price of the quantity of sales affected, and the actual price of the sales. It is the difference between the standard profit and actual profit. This variance is similar to the price variance calculated under value method, and the formula is as follows:

$$\text{Margin price variance} = \text{AQ (Standard profit - Actual Profit)}$$

If the actual profit is greater than the standard profit, the variance is favorable and vice versa.

- ii) **Sales margin volume variance.** This is the portion of total margin variance which is due to the difference between the budgeted quantity and actual quantity of sales. This is the amount by which standard profit differs from budgeted profit. The formula is as follows:

$$\text{Margin volume variance} = \text{standard profit (SQ - AQ)}$$

If the actual quantity is greater than standard quantity, the variance is favorable, and vice versa.

- iii) **Sales margin mix variance.** This is that portion of total margin variance which is due to the difference between the budgeted and actual quantities of each product of which the sales mixture is composed, valuing sales at the standard net selling prices and cost of sales at standard. That is, the difference between the revised standard profits is the mix variance.

$$\text{Margin mix variance} = \text{standard profit (RSQ - AQ)}$$

If the actual quantity is greater than the revised standard quantity, the variance is favorable, and vice versa

Example 5.8. ABC Ltd had budget the following sales for December 2007.

Product A	9,000 units at Br.5 per unit
Product B	6,500 units at Br.10 per unit
Product C	12,000 units at Br 7.5 per unit

As against this, the actual sales were:

Product A	10,000 units at Br.5 per unit
-----------	-------------------------------

Product B 7,000 units at Br.10per unit

Product C 11,000 units at Br 7.5per unit

The cost per unit of A, B and C was Br.4.5, Br.8.5, and Br.6.5 respectively. Compute the different variances to explain the difference between budgeted profit and actual profit.

Solution

i) Sales margin variance =(SM x SQ) – (AM x AQ)

Product A = (0.5 x9,000)-(1x10,000) = Br. 5,500 (F)

Product B = (1.5 x6,500)-(1x70,00) = Br. 2,750 (U)

Product C = (1x12,000)-(1.3x11,000)= Br. 2300 (F)

Total Br. 5,050 (F)

ii) Sales margin price variance = AQ(SM –AM)

Product A = 10,000 (Br. 0.5 _ Br. 1.00) = Br. 5,000 (F)

Product B =7,000 (Br. 1.5 _ Br. 1.0) =. Br. 3,500(U)

Product C = 11,000 (Br. 1.0_ Br. 1.3) =Br. 3,300 (F)

Total Br.4800 (F)

iii) Sales margin volume variance = SM (SQ –AQ)

Product A = Br. 0.5 (9,000- 10,000) = Br. 500 (F)

Product B = Br. 1.5 (6,500-7,000) = Br. 750 (F)

Product C = Br. 1.0(12,000-11,000) = Br. 1,000 (U)

Total Br. 250 (F)

iv) Sales margin mix variance =SM (RSQ –AQ)

Product A = Br. 0.5 (9,164 – 10,000) =. Br. 418(F)

Product B = Br. 1.5 (6,618 – 7,000) = Br. 573 (F)

Product C = Br. 1.0 (12,218 – 11,000) =Br. 1218(U)

Total Br.227 (U)

v) Sales margin quantity variance = SM (SQ-RSQ)

Product A = Br. 0.5(9,000 – 9,164) =Br 82 (F)

Product B = Br. 1.5(6,500 – 6,618) = Br 177 (F)

Product C = Br. 1.0(12,000 – 12,218) = Br 218(F)

TOTAL = Br. 477 (F)

*Working notes on profit per unit for budget and actual, and RSQ

BUDGET

Product	SQ	SP	Total sales	Cost per unit	Total cost	profit per unit	Total profit
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Br	Br	Br	Br	Br	Br	Br	Br
A	9,000	5	45,000	4.5	40,500	0.5	4,500
B	6,500	10	65,000	8.5	55,250	1.5	9,750
C	<u>12,000</u>	7.5	<u>90,000</u>	6.5	<u>78,000</u>	1.0	<u>12,000</u>
	<u>27,500</u>		<u>200,000</u>		<u>173,750</u>		<u>26,250</u>
ACTUAL							
A	10,000	5.5	55,000	4.5	45,000	1.0	10,000
B	7,000	9.5	66,500	8.5	59,500	1.0	7,000
C	<u>11,000</u>	7.8	<u>85,800</u>	6.5	<u>71,500</u>	1.3	<u>14,300</u>
	<u>28,000</u>		<u>207,300</u>		<u>176,000</u>		<u>31,300</u>

$$\text{RSQ} = \text{AQ} \times (\text{Standard ratio}) = 28,000 \times (18:13:24)$$

$$\text{Product A} = 28,000 \times 18/55 \quad 9,164$$

$$\text{Product B} = 28,000 \times 13/55 \quad 6,618$$

$$\text{Product C} = 28,000 \times 24/55 \quad 12,218$$

4.3.3. CAUSES AND DISPOSITION OF VARIANCES

In order to make the variance analysis a control instrument, the management should investigate the causes of variances and take the necessary corrective measures. There is no uniformity of opinion regarding the disposal of variances. **Method I:** where it is desired to maintain the same standard in future, the variances must be written –off to Profit and Loss Account. **Method II:** According to strict principle, the amount of variances resulting from incorrect standards or condition beyond control, like material price changes, wage rate increases due to fair legislation, etc., are allocated to inventories and cost of sales in proportion to their value of closing balances. **Method III:** another method is to carry forward the variances to the next financial year by crediting the same to a reserve account to be set off in the subsequent year or years. The favorable and adverse variances may cancel each other in the course of reasonable time and thus be disposed of.

1. **Materials price variances.** (a) *Causes.* Change in market price, delivery costs, purchase of non –standard materials, emergency purchases, incorrect shipping instruction, loss of discounts, etc. (b) *Disposition.* If all or a portion of the price variance is the result of inefficiencies or a saving has resulted from efficient purchasing, the amount may be adjusted to Profit and Loss Account. If it is due to incorrect standards or change in market price, the amount may be adjusted to inventories and cost of goods sold. It may be mentioned that under method I, such adjustment must be made for inventories under materials control, work –in –progress, finished goods, and cost of the

goods sold. Under Methods II and III, such adjustments are required for inventories under work –in –progress, finished goods, and cost of goods sold.

2. **Materials usage variance** (a) *Causes*. Poor quantity of materials: change in material mix, product or production methods: careless handling: excessive waste or scrap; incorrect setting of standards. (b) *Disposition*. The amount of usage variance resulting in inefficiency in handling and processing materials is transferred to profit and loss account. The amount of usage variance due to incorrect standards is apportioned to work –in –progress, finished goods and cost of goods sold.

3. **Direct wages rate variances**: (a) *Causes* general rise due to ward or agreement, non –standard grade, abnormal overtime or payment above or below standard rates during seasonal or emergency operations. (b) *Disposition*. The amount of variance arising out of inefficiency can be controlled if transferred to profit and loss account. The amount of variance resulting from the use of out –of –date standards or from conditions beyond the control of management is adjusted to work –in –progress, finished goods and cost of goods sold, on the basis of wages or time.

4. **Direct labor efficiency variance** (a) *Causes*, Poor working conditions, abnormal idle time, i.e. power failure, breakdown, go slow technique quality of supervision, non standard grade of material, or employee non cooperation in service departments. (b) *Disposition*. The amount of variance attributed to various forms of inefficiency which are controllable is transferred to profit and loss account. The amount of variance resulting from improperly prepared standard and from conditions beyond the control of management may be adjusted to work – in – progress, finished goods, and cost of goods sold.

5. **Overhead expenditure variance** (a) *Causes*. Under the over utilization of a service; seasonal conditions; inefficiency in the use of a service (e.g., electricity in lieu of gas) (b) *Disposition*. The amount of variance due to seasonal conditions should be treated as a deferred item. The amount arising out of inefficiency which is controllable is transferred to profit and loss account. The amount resulting from incorrectly prepared standard and from conditions beyond control is adjusted to work in progress, finished goods, and cost of goods sold.

UNIT SUMMARY

A standard cost is computed as a standard price multiplied by a standard quantity. In a true standard cost system, standards are derived for prices and quantities of each product component and for each product. A standard cost provides information about a product's standards for components, processes, quantities, and costs.

A variance is any difference between an actual and a standard cost. A total variance is composed of a price and a usage sub variance. The material variances are the price and the quantity variances. The material price variance can be computed on either the quantity of material purchased or the quantity of material used in production. This variance is computed as the quantity measure multiplied by the difference between the actual and standard prices. The material quantity variance is the difference between the standard price of the actual quantity of material used and the standard price of the standard quantity of material allowed for the actual output.

The two labor variances are the rate and efficiency variances. The labor rate variance indicates the difference between the actual rate paid and the standard rate allowed for the actual hours worked during the period. The labor efficiency variance compares the number of hours actually worked against the standard number of hours allowed for the level of production achieved and multiplies this difference by the standard wage rate.

If separate variable and fixed overhead accounts are kept (or if this information can be generated from the records), two variances can be computed for both the variable and fixed overhead cost categories. The variances for variable overhead are the VOH spending and VOH efficiency variances. The VOH spending variance is the difference between actual variable overhead cost and budgeted variable overhead based on the actual level of input. The VOH efficiency variance is the difference between budgeted variable overhead at the actual activity level and variable overhead applied on the basis of standard input quantity allowed for the production achieved.

The fixed overhead variances are the FOH spending and volume variances. The fixed overhead spending variance is equal to actual fixed overhead minus budgeted fixed overhead. The volume variance compares budgeted fixed overhead to applied fixed overhead. Fixed overhead is applied based on a predetermined rate using a selected measure of capacity. Any output capacity utilization actually achieved (measured in

standard input quantity allowed), other than the level selected to determine the standard rate, will cause a volume variance to occur.

Actual costs are required for external reporting, although standard costs may be used if they approximate actual costs. Adjusting entries are necessary at the end of the period to close the variance accounts. Standards provide a degree of clerical efficiency and assist management in its planning, controlling, decision making, and performance evaluation functions. Standards can also be used to motivate employees if the standards are seen as a goal of expected performance.

A standard cost system should allow management to identify significant variances as close to the time of occurrence as feasible and, if possible, to help determine the variance cause. Significant variances should be investigated to decide whether corrective action is possible and practical. Guidelines for investigation should be developed using the management by exception principle.

MODEL EXAMINATION QUESTIONS

PART I: MULTIPLE CHOICE QUESTIONS

1. A standard cost is
 - A. The planned unit cost of a product, component or service in a period.
 - B. The budgeted cost ascribed to the level of activity achieved in a budget centre in a control period.
 - C. The budgeted production cost ascribed to the level of activity in a budget period.
 - D. The budgeted non-production cost for a product, component or service in a period.

The following information is required for sub-questions 2 to 8

X Ltd operates a standard costing system. The following budgeted and standard cost information is available:

<i>Budgeted production and sales</i>	10,000 units
Selling price	Br 250 per unit
Direct material cost – 3 kg x Br.10.....	30 per unit
Direct labor cost – 5 hours x Br.8.....	40 per unit
Variable manufacturing overheads – 5 hours x Br.4...	20 per unit

Actual results for the period were as follows:

Production and sales	11,500 units
Sales value	<i>Birr</i> 2,817,500
Direct material – 36,000 kg	342,000
Direct labor – 52,000 hours	468,000
Variable manufacturing overheads	195,000

2. The direct material price variance is
A. *Br.*18, 000U. B. *Br.*3, 000 U C. *Br.*3, 000 F D. *Br* 18, 000 F E. none
3. The direct material usage variance is
A *Br.*15, 000 U. B. *Br.*14, 250 U. C. *Br.* 14,250 F D. *Br.*15, 000F. E. none
4. The direct labor rate variance is
A. *Br.*52, 000 U B. *Br.*8, 000 U C. *Br.*8, 000 F D. *Br.*52, 000 F E. none
5. The direct labor efficiency variance is
A. *Br.*49, 500 U B. *Br.*44, 000 U C. *Br.*44, 000 F D. *Br.*49, 500 F E. none

Part II: SHORT ANSWER QUESTIONS

1. Explain the concept of Standard Costing? What advantages are likely to be obtained from the application of standard costing as compared to historical costing?
2. What is standard costing? How does it differ from Budgetary Control?
3. Discuss the essential components of standard costing system.
4. What is Variance Analysis? Explain and illustrate the various types of material variances.
5. Distinguish between
 - a) Material price Variance and Material Mix variance
 - b) Labor Efficiency variance and Labor Rate Variance

Part III: WORK OUT QUESTIONS

1. Poly Containers makes 300-gallon plastic water tanks for a variety of commercial uses. The standard per unit material, labor, and overhead costs are as follows:

Direct material: 80 pounds @ <i>Br.</i> 2	<i>Br.</i> 160
Direct labor: 1.25 hours @ <i>Br.</i> 16 per hour	20
Variable overhead: 30 minutes of machine time @ <i>Br.</i> 50.00 per hour	25
Fixed overhead: 30 minutes of machine time @ <i>Br.</i> 40.00 per hour	20

The overhead application rates were developed using a practical capacity of 6,000 units per year. Production is assumed to occur evenly throughout the year.

During May 2001, the company produced 525 tanks. Actual data for May 2001 are as follows:

Direct material purchased: 46,000 pounds @ Br.1.92 per pound

Direct material used: 43,050 pounds (all from May's purchases)

Total labor cost: Br.10, 988.25 for 682.5 hours

Variable overhead incurred: Br.13, 770 for 270 hours of machine time

Fixed overhead incurred: Br.10, 600 for 270 hours of machine time

Required: Calculate the following:

- a. Material price variance based on purchases
- b. Material quantity variance
- c. Labor rate variance
- d. Labor efficiency variance
- e. Variable overhead spending and efficiency variances
- f. Fixed overhead spending and volume variances

2. Rock Solid Engineering Company compares actual results with a flexible budget. The standard DL rates used in the flexible budget are established each year at the time the annual plan is formulated and held constant for the entire year. The standard hours allowed for the actual output of insurance claims for April in a claims department are shown in the following schedule:

<u>Labor Classes</u>	<u>Standard Rate/Hrs</u>	<u>Standard hrs for Actual output</u>
Class III	Br. 8.00	500 Hrs
Class II	7.00	500
Class I	5.00	500

Required: compute

- 1) DL rate & efficiency Variances
- 2) DL mix & Yield Variances

ANSWER TO MODEL EXAMINATION QUESTIONS

A. SOLUTIONS FOR WORKOUT QUESTIONS

Q1.

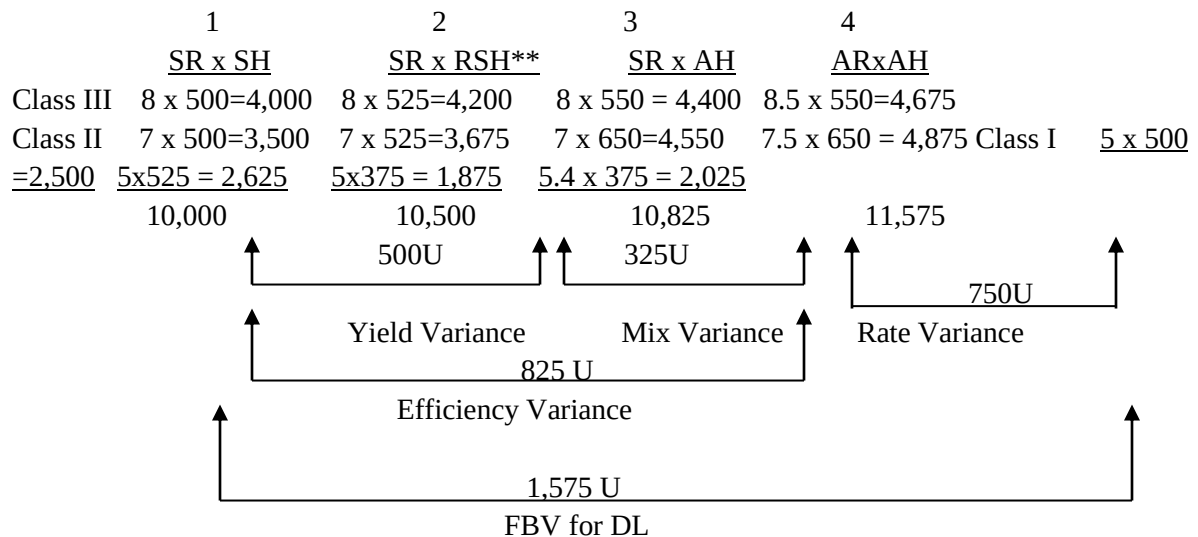
$$\begin{aligned} \text{a) } MPV &= (AP \times AQ) - (SP \times AQ) \\ &= (\text{Bir}1.92 \times 46,000) - (\text{Bir}2.00 \times 46,000) = \text{Bir}3,680 \text{ F} \end{aligned}$$

$$\begin{aligned} \text{b) } MQV &= (SP \times AQ) - (SP \times SQ) \\ &= (\text{Bir}2 \times 43,050) - (\text{Bir}2 \times 42,000^*) = \text{Bir}2,100 \text{ U} \end{aligned}$$

$$*SQ = 525 \times 80 \text{ pounds} = 42,000 \text{ pounds}$$

- c) $LRV = (AR \times AH) - (SR \times AH)$
 $= (Bir16.10 \times 682.5) - (Bir16 \times 682.5) = Bir68.25 \text{ U}$
**AR = Bir10,988.25 / 682.5 hours = Bir16.10 per hour*
- d) $LEV = (SR \times AH) - (SR \times SH^*)$
 $= (Bir16 \times 682.5) - (Bir16 \times 656.25) = Bir420 \text{ U}$
**SH = 525 x 1.25 hours = 656.25 hours*
- e) $VOH \text{ Spending Variance} = \text{Actual VOH} - (SP \times AQ)$
 $= Bir13,770 - (Bir50.00 \times 270) = Bir270 \text{ U}$
 $VOH \text{ Efficiency Variance} = (SP \times AQ) - (SP \times SQ^*)$
 $= (Bir50.00 \times 270) - (Bir50.00 \times 262.5) = Bir375 \text{ U}$
**SQ = 525 x 0.5 = 262.5 hours*
- f) $FOH \text{ Spending Variance} = \text{Actual FOH} - \text{Budgeted FOH}^{**}$
 $= Bir10,600 - Bir10,000 = Bir600 \text{ U}$
 $FOH \text{ Volume Variance} = \text{Budgeted FOH} - (SP \times SQ)$
 $= Bir10,000 - (Bir40 \times 262.5) = Bir500 \text{ F}$
***BFOH, annually = 6,000 x Bir20 = Bir120,000*
BFOH, monthly = Bir120,000 / 12 months = Bir10,000

Q2.



**RSH: Revised Standard hours = (Standard mix x Actual total DL Hrs used)

RSH for:

	Standard	Standard	Actual	Proportion	RSH
	Hours	Hours	Mix	Hours	
Class III	500	1/3	550	1/3 x 1,575	525
Class II	500	1/3	650	1/3 x 1,575	525
Class I	<u>500</u>	1/3	375	1/3 x 1,575	525
			<u>1,500</u>		

UNIT SIX

RELEVANT INFORMATION AND DECISION MAKING

UNIT OBJECTIVES

After completing this chapter, you should be able to answer the following questions:

- What factors are relevant in making decisions and why?
- How do opportunity costs affect decision making?
- What are sunk costs and why are they not relevant in making decisions?
- What are the relevant financial considerations in outsourcing?
- How can management make the best use of a scarce resource?
- How does sales mix pertain to relevant costing problems?
- How are special prices set and when are they used?
- How is segment margin used to determine whether a product line should be retained or eliminated?

UNIT OUTLINE

- 6.1 Introduction
- 6.2 Information and the decision process
- 6.3 The concept of relevance
- 6.4 Relevant information for special decision
 - 6.4.1 special order decision
 - 6.4.2 product line decision
 - 6.4.3 product mix decision
 - 6.4.4 make or buy decision
 - 6.4.5 keep or replace decision

6.1. INTRODUCTION

During the last decade, increasing competition has forced many companies to refocus their resources and to defend their core businesses against aggressors. In developing strategies to fight this war, managers have generally reached a consensus on two strategic criteria. First, to win a battle, the focus of organizations must be on delivering products and services in the manner most consistent with the desires of customers. Second, no company can do all things well. The strategies managers devise in this intensive struggle evolve from internal evaluations in which the managers identify the functions they must do well to survive. These functions are regarded as core competencies and maintaining leadership in these areas is regarded as vital. All other functions, although important to the organization, are regarded as noncore functions. By intensely focusing on core functions, managers try to maintain a competitive advantage. However, an undesirable consequence of focusing on only the core competencies is that the quality and capabilities of the noncore functions can deteriorate. This deterioration, in turn, can reduce a firm's ability to attract customers to its products and services. Outsourcing the noncore functions to firms that have core competencies in those functions frequently solves the dilemma of maintaining a focus on core competencies while also maintaining excellence in noncore functions. Managers are charged with the responsibility of managing organizational resources effectively and efficiently relative to the organization's goals and objectives. Making decisions about the use of organizational resources is a key process in which managers fulfill this responsibility. Accounting and finance professionals contribute to the decision-making process by providing expertise and information. Accounting information can improve, but not perfect, management understands of the consequences of decision alternatives. To the extent that accounting information can reduce management's uncertainty about economic facts, outcomes, and relationships involved in various courses of action, such information is valuable for decision-making purposes. Many decisions can be made using **relevant costing**, which focuses managerial attention on a decision's relevant (or pertinent) facts. Relevant costing techniques are applied in virtually all business decisions in both short-term and long-term contexts. This chapter examines their application to several common types of business decisions: replacing an asset, outsourcing a product or part, allocating scarce resources, determining the appropriate sales/production mix, and accepting specially

priced orders. In general these decisions require a consideration of costs and benefits that are mismatched in time; that is, the cost is incurred currently but the benefit is derived in future periods. In making a choice among the alternatives available, managers must consider all relevant costs and revenues associated with each alternative

6.2. INFORMATION AND THE DECISION PROCESS

 ***Dear student! What does decision making mean? What is relevant information for decision making?***

Decision making is the process of choosing the best course of action from alternatives available. Decision model is a method used by managers for deciding among courses of action. Accounting information (revenue and cost information) are basic inputs in to decision model. However, other quantitative as well as qualitative information can also be used. In general information is divided in to relevant and irrelevant information. Relevant information is information which is useful for decision making where as irrelevant information is not useful for decision making. The management accountant's role in the decision making process is to produce relevant information to the managers who make the decisions. Thus, the primary role of cost accountant in decision process is to: decide what information is relevant to each decision problem, and provide accurate and timely information (data)

Decision making process involves basically the following activities.

i. Identify and Define the Problem. The most important phase of decision making process because all other activities in the process depend on this phase. Incorrectly defined problems waste time and resources. That is why it is usually said that defining a problem is solving half of the problem.

ii. Specify the Criterion. The phase in which the purpose of decision is to be made. Is the objective to maximize profit, increase market share, minimize cost, or improve public service? For example, cost minimization, increase the quality of product, maximize profit, etc.

iii. Identify Possible Alternatives: Determining the possible alternatives is a critical step in the decision process.

iv. Gathering Relevant Information. Information could be subjective or objective, internal or external to the organization, historical (past) data, or future (expected) ones.

v. Making the Decision: Select the best alternative (course of action).

Activity 6.1

1. What is decision making -----
2. What are the activities to be performed in decision making process -----

6.3. THE CONCEPT OF RELEVANCE

Dear student! What are the features that should be fulfilled to say information is relevant or useful for decision you make? What are the costs and revenues relevant for decision making?

For information to be relevant, it must possess three characteristics. It must (1) be associated with the decision under consideration, (2) be important to the decision maker, and (3) have a connection to or bearing on some future endeavor.

Association with Decision

Costs or revenues are relevant when they are logically related to a decision and vary from one decision alternative to another. Cost accountants can assist managers in determining which costs and revenues are relevant to decisions at hand. To be relevant, a cost or revenue item must be differential or incremental. An **incremental revenue** is the amount of revenue that differs across decision choices and **incremental cost (differential cost)** is the amount of cost that varies across the decision choices.

To the extent possible and practical, relevant costing compares the incremental revenues and incremental costs of alternative choices. Although incremental costs can be variable or fixed, a general guideline is that most variable costs are relevant and most fixed costs are not. The logic of this guideline is that as sales or production volume changes, within the relevant range, variable costs change, but fixed costs do not change. As with most generalizations, some exceptions can occur in the decision-making process.

The difference between the incremental revenue and the incremental cost of a particular alternative is the positive or negative incremental benefit (incremental profit) of that course of action. Management can compare the incremental benefits of

alternatives to decide on the most profitable (or least costly) alternative or set of alternatives.

Some relevant factors, such as sales commissions or prime costs of production, are easily identified and quantified because they are integral parts of the accounting system. Other factors may be relevant and quantifiable, but are not part of the accounting system. Such factors cannot be overlooked simply because they may be more difficult to obtain or may require the use of estimates. For instance, **opportunity costs** represent the benefits foregone because one course of action is chosen over another. These costs are extremely important in decision making, but are not included in the accounting records.

Importance to Decision Maker

The need for specific information depends on how important that information is relative to the objectives that a manager wants to achieve. Moreover, if all other factors are equal, more precise information is given greater weight in the decision making process. However, if the information is extremely important, but less precise, the manager must weigh importance against precision.

Bearing on the Future

Information can be *based* on past or present data, but is relevant only if it pertains to a future decision choice. All managerial decisions are made to affect future events, so the information on which decisions are based should reflect future conditions. The future may be the short run (two hours from now or next month) or the long run (three years from now). Future costs are the only costs that can be avoided, and a longer time horizon equates to more costs that are controllable, avoidable, and relevant. *Only information that has a bearing on future events is relevant in decision making.*

Costs incurred in the past for the acquisition of an asset or resources are called sunk costs. They cannot be changed, no matter what future course of action is taken because past expenditures are not recoverable, regardless of current circumstances. Thus, the historical cost is not relevant to the decision.

Example6.1 Marina Company, a manufacturer of a line of ashtrays, is thinking of using aluminum instead of copper in the manufacture of its product. Historical direct material cost was Br. 0.50 per unit. The company expected future costs for aluminum is Br 0.40 and it is unchanged for copper. Direct labor cost were Br0.80 per unit and will not be affected by the switch in materials.

The analysis in a nutshell is as follows:

	Copper	Aluminum	Difference
Direct material	Br 0.50	Br 0.40	Br 0.10
Direct labor	0.80	0.80	-

In the foregoing analysis, the material cost (the expected future cost of copper compared with expected future cost of aluminum) is the only relevant cost. The material cost met both criteria for relevant information. That is, bearing on the future and an element of difference between the alternatives. However, the direct labor cost will continue to be Br 0.80 per unit regardless of the material used. It is irrelevant because the second criterion – an element of difference between the alternatives – is not met.

Example6.2 Consider a decision facing Home Appliances, a manufacturer of vacuum cleaner, whether to “reorganize” or “do not reorganize” its manufacturing operations to reduce manufacturing labor costs. The reorganization will eliminate all manual handling of materials. The current manufacturing line uses 20 workers- 15 workers operate machines and 5 workers handle materials. The 5 material-handling workers have been hired on contracts that permit layoffs without additional payments. Each worker puts in 2,000 hours annually. The cost of reorganization (consisting mostly of equipment leases) is predicted to be Br. 90,000 each year. The predicted production and sales of output of 25,000 units will be unaffected by the decision. Also unaffected are the predicted selling price of Br. 250, the direct materials cost per unit of Br. 50, MOH of Br. 750,000, and marketing costs of Br. 2,000,000. Historical labor costs were Br. 14 per hour. A recently negotiated increase in employee benefits of Br. 2 per hour will increase labor costs to be Br. 16 per hour in the future.

Required: Should Home Appliances reorganize its manufacturing operations to reduce manufacturing labor costs?

Solution:

<i>The comparative statement of for the given alternative</i>				
	<u>All Data</u>		<u>Relevant Data</u>	
	<u>Do Not Reorganize</u>	<u>Reorganize</u>	<u>Do Not Reorganize</u>	<u>Reorganize</u>
Revenues: 250 X 25,000	Br. 6,250,000	Br. 6,250,000	-	-
<u>Costs:</u>				
DM: 50 X 25,000	1,250,000	1,250,000	-	-
DL: 20 X 16 X 2,000	640,000		Br. 640,000	

15 X 16 X 2,000				
MOH:	750,000	480,000	750,000	Br. 480,000
Marketing	2,000,000	2,000,000	-	-
Reorganization costs	<u>-</u>	<u>90,000</u>	<u>-</u>	<u>-</u>
90,000				
Total costs	Br. 4,640,000	Br. 4,570,000	Br. 640,000	Br. 570,000
Operating Income	<u>Br. 1,610,000</u>	<u>Br. 1,680,000</u>	<u>Br. (640,000)</u>	<u>Br. (570,000)</u>
	↑	↑	↑	↑
	↓		↓	
	Br. 70,000 Difference		Br. 70,000 Difference	

In the above statement, the financial data underlies the choice between the do not reorganize and reorganize alternative. The first two columns present all data. The last two columns present only relevant costs...the Br640, 000 and Br 480,000 expected future manufacturing labor costs and Br 90,000 expected future reorganization costs that differ the between two alternatives. The revenues, direct materials, manufacturing overhead, and marketing items can be ignored because they do not differ between the alternatives and, therefore, are irrelevant.

And, also note that, the past (historical) manufacturing hourly wage rate of Br 14 and total past(historical) manufacturing labor costs of Br 560,000 (20 workers X 2,000 hours per worker per year X Br 14 per hour) do not use in solution, because historical costs themselves are past costs that, therefore, are irrelevant to decision making.

The analysis, in above solution, indicates that reorganizing the manufacturing operations will increase predicted operating income by Br, 70,000 each year and we reached the same conclusion whether we use all data or include only relevant data in analysis.

Activity6.2

1. What factors are relevant in making decisions and why? -----

2. How do opportunity costs affect decision making? -----

3. What are sunk costs and why are they not relevant in making decisions? --

.4. RELEVANT INFORMATION FOR SPECIFIC DECISIONS

Managers routinely choose a course of action from alternatives that have been identified as feasible solutions to problems. In so doing, managers weight the costs and benefits of these alternatives and determine which course of action is best. Incremental revenues, costs, and benefits of all courses of action are measured against a baseline alternative. In making decisions, managers must provide for the inclusion of any inherently non quantifiable considerations. Inclusion can be made by attempting to quantify those items or by simply making instinctive value judgments about nonmonetary benefits and costs. In evaluating courses of action, managers should select the alternative that provides the highest incremental benefit to the company. Rational decision-making behavior includes a comprehensive evaluation of the monetary effects of all alternative courses of action. The chosen course of action should be one that will make the business better off. Decision choices can be evaluated using relevant costing techniques.

6.4.1. Special Order Decisions

One type of decision that affects output level is accepting or rejecting a special order. A special order is a one-time order that is not considered part of the company's normal ongoing business. In general, a special order is profitable as long as the incremental revenue from the special order exceeds the incremental costs of the order. Thus, conditions to consider in a special order decisions are: (i) Customers must be from markets not ordinarily served by the company, and (ii) the company must operate below it maximum productive capacity

Example6.3. Consider the following details of the income statement, on absorption costing basis (that is, both variable and fixed manufacturing costs are included in inventoriable costs and cost of goods sold), of Samson Company for the year just ended December 31, 20X4

	<u>Total</u>	<u>per unit</u>
Sales (1,000,000 units)	Br 20,000,000	Br 20
Cost of Goods Sold	<u>15,000,000</u>	<u>15</u>

Gross Margin	Br 5,000,000	Br. 5
Selling and Administrative Expenses	<u>4,000,000</u>	<u>4</u>
Operating Income	<u>Br. 1,000,000</u>	<u>Br. 1</u>

Samson's fixed manufacturing costs were Br 3 million and fixed selling and administrative expenses were Br 2.9 million. Near the end of the year, Ethio Company offered Samson Br 13 per unit for 100,000 unit special order. The special order would not affect Samson's regular business in any way. Furthermore, the special sales order would not affect total fixed costs and would not require any additional variable selling and administrative expenses.

Required:

- Should Samson accept or reject the special order?
- Could the special order affect Samson's regular business?

Solution:

a). The correct analysis to the above problem employs the contribution approach to income statement, not the absorption or financial approach- that treats fixed costs, i.e., fixed manufacturing costs as if it were variable.

- Variable manufacturing cost per unit = $\frac{15,000,000 - 3,000,000}{1,000,000} = 12$ per unit
- Total Variable manufacturing cost = Br. 12 x 1,000,000 = Br. 12,000,000
- Variable selling and administrative cost per unit = $\frac{4,000,000 - 2,900,000}{1,000,000} = 1.1$ per unit
- Total Variable selling and administrative cost = Br. 1.1 x 1,000,000 = Br. 1,100,000 (the special order does not affect this cost)

The analysis would be as follows on **comparative contribution income statement**.

	Without special order 1,000,000 units to be sold	With special order 1,100,000 units to be sold	Difference: relevant amount for the 100,000 units of special order
Sales	<u>Br. 20,000,000</u>	<u>Br. 21,300,00</u>	<u>Br. 1,300,000</u>
Variable Expenses:			
Manufacturing	Br. 12,000,000	Br. 13,200,000	Br. 1,200,000
Selling and Adm.	<u>1,100,000</u>	<u>1,100,000</u>	
Total Variable Exp.	<u>Br. 13,100,000</u>	<u>Br. 14,300,000</u>	<u>1,200,000</u>
Contribution Margin	<u>Br. 6,900,000</u>	<u>Br. 7,000,000</u>	<u>Br. 100,000</u>

Fixed Expenses:				
Manufacturing	Br. 3,000,000	Br. 3,000,000		
Selling and Adm.	2,900,000	2,900,000		
Total Fixed Expenses	<u>Br.5,900,000</u>	<u>Br. 5,900,000</u>		
Operating Income	<u>Br.1,000,000</u>	<u>Br.1,100,000</u>	<u>Br. 100,000</u>	

The above comparative income statements for Samson illustrates two key complete to analyzing relevant revenues for decision: (1) distinguish relevant costs and revenues from irrelevant ones and (2) use the contribution income statement to focus on whether each variable cost and each fixed cost is affected by the alternatives(i.e. reject or accept) under consideration.

In this case, the relevant revenues and costs are the expected future revenues and costs that differ as a result of accepting the special offer ---- sales of Br 1,300,000(Br 13 per unit X 100,000 units) and variable manufacturing costs of Br. 1,200,000 (Br 12 per units X 100,000 units). The fixed manufacturing costs and selling and Administration costs (including variable costs) are irrelevant. That is because these costs will not change in total whether the special order is accepted or rejected. Based on the relevant data analyzed above, Samson would gain an additional Br100, 000(relevant revenues, Br 1,300,000 less relevant costs Br 1,200,000) in operating income by accepting the special order. In this example, comparing total amounts for 1,000,000 units versus 1,100,000 units or focusing only on the relevant amounts in the difference column in comparative income statement avoids misleading implication --- the implication that would result from comparing the Br 13 per unit selling price against the manufacturing cost per unit of Br 15 (from Samson's income statement on absorption costing basis) which includes both Variable and fixed manufacturing costs.

Thus, based on the relevant data analyzed above, Samson Company should accept the special order because it brings an additional income of Br. 100,000 for the company as:

Income with special order	Br. 1,100,000
Income without special order	<u>1,000,000</u>
Additional income if the order had been accepted	<u>Br. 100,000</u>

- b) Yes. Unless Samson Company has effectively segments its market so that the special order to the Ethio Company does not affect the regular business.

6.4.2. Product Line Decisions

This is a decision relating to whether old product lines or other segments of a company should be dropped and new ones added are among the most difficult decision that a manager has to make. Operating results of multiproduct environments are often presented in a disaggregated format that shows results for separate product lines within the organization or division. In reviewing these disaggregated statements, managers must distinguish relevant from irrelevant information regarding individual product lines. If all costs (variable and fixed) are allocated to product lines, a product line or segment may be perceived to be operating at a loss when actually it is not. The commingling of relevant and irrelevant information on the statements may cause such perceptions.

In classifying product line costs, managers should be aware that some costs may appear to be avoidable but are actually not. For example, the salary of a supervisor working directly with a product line appears to be an avoidable fixed cost if the product line is eliminated. However, if this individual has significant experience, the supervisor is often retained and transferred to other areas of the company even if product lines are cut. Determinations such as these need to be made before costs can be appropriately classified in product line elimination decisions.

For instance, mostly on add or delete decisions, fixed costs are divided into two categories, avoidable and unavoidable. Avoidable costs are costs that will not continue if an ongoing operation is changed, deleted or eliminated. These costs are relevant costs in decision making. Unavoidable costs are costs that continue even if a subunit or an activity is eliminated and are not relevant for decision.

Example.6.4 Eyoha Department store has three major departments: groceries, general merchandise, and drugs. Management is considering dropping groceries, which have consistently shown a net loss, as shown below *on statement of departments' profitability*

analysis of Eyoha.

Departments				
	Groceries	General merchandise	Drugs	Total
Sales	Br. 100,000	Br. 8,0000	Br. 10,000	Br.190,000

Variable CGS & Expenses	<u>80,000</u>	<u>56000</u>	<u>6,000</u>	<u>142,000</u>
Contribution margin	<u>Br. 20,000</u>	<u>Br. 24000</u>	<u>Br. 4,000</u>	<u>Br. 48,000</u>
Fixed expenses:				
Avoidable	Br. 15,000	Br. 10,000	Br. 1,500	Br. 26,500
Unavoidable	<u>6,000</u>	<u>10,000</u>	<u>2,000</u>	<u>18,000</u>
Total fixed expenses	<u>Br.21,000</u>	<u>Br. 20,000</u>	<u>Br.3,500</u>	<u>Br. 44,500</u>
Operating income (loss)	<u>Br. (1,000)</u>	<u>Br.4,000</u>	<u>Br. 500</u>	<u>Br. 3,500</u>

Required:

- Which alternative would be recommended if the only alternatives to be considered are dropping or continuing the grocery department? Assume that the total assets would be unaffected by the decision and the space made available by dropping groceries would remain idle.
- Refer the income statement presented above. Assume that the space made available by dropping groceries could be used to expand the general merchandise department. The space would be occupied by merchandise that increase sales by Br. 50,000, generate a 30% contribution margin percentage and have additional avoidable fixed costs of Br.7, 000. Should Eyoha discontinue grocery and expand merchandise department?

Solutions

- Analysis for dropping grocery department and leaving the space idle

	(A) Total Before change	(B) Effect of dropping grocery	(A – B) Total after change
Sales	Br. 190.000	Br 100.000	Br 90.000
Variable COGS and Expenses	<u>142.000</u>	<u>80.000</u>	62.000
Contribution margin	<u>Br 48.000</u>	<u>Br 20.000</u>	Br 28.000
Fixed expenses			
Avoidable	Br 26.500	Br 15,000	Br 11,500
Unavoidable	<u>18.000</u>	<u>-</u>	18,000
Total fixed expenses	<u>Br 44,500</u>	<u>Br 15,000</u>	<u>Br 29,500</u>
Operating income (loss)	<u>Br 3,500</u>	<u>Br 5,000</u>	<u>Br (1,500)</u>

In this analysis, column 2, presents the relevant –revenues and relevant-cost analysis using data from the grocery column in department profitability analysis of Eyoha. Eyoha’s operating income will be Br.5, 000 (income with grocery department, Br.3500 less loss assuming grocery is dropped, Br.1500 or it implies that the cost savings from dropping the grocery department, Br.95, 000 (Br.80, 000+ Br.15, 000), will not be enough to offset the loss of Br.100, 000 in revenues. So, under this condition Eyoha’s managers should decide to keep the grocery department rather dropping.

Notice that all of the grocery’s variable expenses are avoidable and relevant for decision making. If the grocery department is discontinued, the Br 6,000 of the fixed expenses will continue, which is irrelevant. And also note that there is no opportunity costs of using spaces for grocery because without grocery, the space and equipment will remain idle.

(b) Analysis for dropping the grocery department and expanding general merchandise.

	(A) Total before change	(B) Effect of Dropping Groceries	(C) Effect of Expanding General Merchandise	(A – B) + C Total after change
Sales	Br190,000	Br 100,000	Br 50,000	Br 140,000
Variable CGS and expense	<u>142,000</u>	<u>80,000</u>	<u>35,000</u>	<u>97,000</u>
Contribution margin	Br 48,000	Br 20,000	Br 15,000	Br 43,000
Fixed expenses				
Avoidable	Br 26,500	Br 15,000	Br 7,000	Br 18,500
Unavoidable	18,000	-		18,000
Total fixed expenses	<u>Br 44,500</u>	<u>Br 15,000</u>	<u>Br 7,000</u>	<u>Br 36,500</u>
Operating income (loss)	<u>Br 3,500</u>	<u>Br 5,000</u>	<u>Br 8,000</u>	<u>Br 6,500</u>

Effect of expanding general merchandise:

Incremental revenue = Br 50,000

Incremental cost

Variable cost = $(1-0.30) \times 500,000$ = (35,000)

Fixed cost = (7,000)

Incremental income = Br 8,000

Recommendation: As the above analysis shows, dropping grocery and using the vacated space to expand general merchandise will be a good decision.

Activity6.3

1. How is relevant information used to determine whether a product line should be retained or eliminated? -----

6.4.3. Optimal Use of Scarce Resources Decisions (Product Mix Decisions)

Managers are frequently confronted with the short-run problem of making the best use of scarce resources that are essential to production activity, but are available only in limited quantity. **Scarce resources** create constraints on producing goods or providing services and can include machine hours, skilled labor hours, raw materials, and production capacity and other inputs. Management may, in the long run, obtain a greater quantity of a scarce resource. For instance, additional machines could be purchased to increase availability of machine hours. However, in the short run, management must make the most efficient use of the scarce resources it has currently.

Determining the best use of a scarce resource requires managerial recognition of company objectives. If the objective is to maximize company profits, a scarce resource is best used to produce and sell the product having the highest contribution margin **per unit of the scarce resource**. This strategy assumes that the company is faced with only one scarce resource. A scarce resource or a limiting factor refers to any factor that restrict or constraint the production or sale of a product or service.

Example6.5 Jimma Computers manufactured two products, desktop computer and notebook computer. The Company's scarce resource is a data chip that it purchases from a supplier. Each desktop computer requires one chip and each notebook computer requires three chips. Currently, the firm has access to only 5,100 chips per month to make either desktop or notebook computers or some combination of both. Demand is above 5,100 units per month for both products and there is no variable selling or administrative costs related to either product. The desktop's Br. 650 selling price less its Br. 545 variable cost provides a contribution margin of Br. 105 per unit. The notebook's contribution margin per unit is Br.180 (Br.900 selling price minus Br.720 variable cost). Fixed annual overhead related to these two product lines totals Br. 6,570,000 and is allocated to products for purposes of inventory valuation. Fixed overhead, however, does not change with production levels within the relevant range

Instructions: on the bases of the above information which product is more profitable and on which products should the firm spend its resources?

Solution:

Present information on two products being manufactured by Jimma Computers and total contribution margin per unit and per chip would be:

Descriptions	Desktop	Notebook
Selling price per unit (a)	Br 650	Br 900
Variable production cost per unit:		
Direct material	Br.345	Br. 480
Direct labor	115	125
Variable overhead	<u>85</u> Br. 545	<u>115</u>
Total variable cost (b)		Br. 20
Unit contribution margin [(c) = (a) _ (b)]	Br 105	Br.180
Chips required per unit (d)	1	3
Contribution margin per chip of per unit [(c) /(d)]	Br.105	Br.60

In the above analysis, because fixed overhead per unit is not relevant in the short run, unit contribution margin rather than unit gross margin is the appropriate measure of profitability of the two products. Unit contribution margin is divided by the input quantity of the scarce resource (in this case, data chips) to obtain the contribution margin per unit of scarce resource. The last line in the above analysis table shows the Br. 105 contribution margin per chip for the desktop compared to Br. 60 for the notebook. Thus, it is more profitable for Jimma Computers to produce desktop computers than notebooks. At first glance, it would appear that the notebook would be, by a substantial margin, the more profitable of the two products because its contribution margin per unit (Br. 180) is significantly higher than that of the desktop (Br. 105). However, because the notebook requires three times as many chips as the desktop, a greater amount of contribution margin per chip is generated by the production of the desktops. If these were the only two products made by Jimma Computers and the company wanted to achieve the highest possible profit, it would dedicate all available data chips to the production of desktops. Such a strategy would provide a total contribution margin of Br. 535,500 per month (5,100 units * Br. 105), if all units produced were sold.

In addition to considering the monetary effects related to scarce resource decisions, managers must remember that all factors cannot be readily quantified and the qualitative aspects of the situation must be evaluated in addition to the quantitative ones. For example, before choosing to produce only desktops, Jimma Computers' managers would need to assess the potential damage to the firm's reputation and markets if the company limited its product line to a single item. Such a choice severely restricts its customer base and is especially important if the currently manufactured products are competitively related

Activity 6.4

1. How can management make the best use of a scarce resource? -----

6.4.4. Make or Buy (In source or out sourcing) decision

A concern with subcontracting or outsourcing has dominated business in recent years as the cost of providing goods and services in-house is increasingly compared to the cost of purchasing goods on the open market. Thus, a daily question faced by managers is whether the right components and services will be available at the right time to ensure that production can occur. Additionally, the inputs must be of the appropriate quality and obtainable at a reasonable price. Traditionally, companies ensured themselves of service and part availability and quality by controlling all functions internally. However, there is a growing trend toward "outsourcing" (buying) a greater percentage of required materials, components, and services.

This **outsourcing decision (make-or-buy decision)** is made only after an analysis that compares internal production and opportunity costs with purchase cost and assesses the best uses of available facilities. Consideration of an in source (make) option implies that the company has available capacity for that purpose or has considered the cost of obtaining the necessary capacity. The make versus buy decision should be based on which alternative is less costly on a relevant cost basis; that is, taking into account only future, incremental cash flows. In other words, in a make or buy situation with no limiting factors, the relevant costs for the decision are the differential costs between the two options.

For example, the costs of in-house production of a computer processing service that averages 10,000 transactions per month are calculated as Br. 25,000 per month. This comprises Br.0.50 per transaction for stationery and Br. 2 per transaction for labor. In addition, there is a Br. 10,000 charge from head office as the share of the depreciation charge for equipment. An independent computer bureau has tendered a fixed price of Br. 20,000 per month.

Based on this information, stationery and labor costs are variable costs that are both avoidable if processing is outsourced. The depreciation charge is likely to be a fixed cost to the business irrespective of the outsourcing decision. It is therefore unavoidable. The fixed outsourcing cost will only be incurred if outsourcing takes place.

The relevant costs for each alternative can be compared as shown in Table 6.1 below. The Br. 10,000 share of depreciation costs is not relevant as it is unavoidable. The relevant costs for this decision are therefore those shown in Table 6.2

Based on relevant costs, there would be a Br. 5,000 per month saving by outsourcing the computer processing service.

Table 6.1 Relevant costs – make versus buy

	Cost to make	Cost to buy
Stationery 10,000 @ Br. 0.50	Br. 5,000	
Labour 10,000 @ Br. 2	20,000	
Share of depreciation costs	10,000	10,000
Outsourcing cost	—	<u>20,000</u>
Total relevant cost	<u>Br. 35,000</u>	<u>Br. 30,000</u>

Table 6.2. Relevant costs – make versus buy, simplified

	Relevant cost to make	Relevant cost to buy
Stationery 10,000 @ Br. 0.50	Br. 5,000	
Labour 10,000 @ Br. 2	20,000	
Outsourcing		cost <u>20,000</u>

Total relevant cost	<u>Br. 25,000</u>	<u>Br.</u>
<u>20,000</u>		

Note that relevant information for make or buy decision includes both quantitative and qualitative factors. Such as:

Quantitative Factors

Buy

- the amount paid to supplier
- transportation costs
- costs incurred to process the part upon receipt

Make

- variable costs incurred to produce the component
- special equipment to produce the product
- hire additional supervisory personnel to assist with making the product

Qualitative Factors

- | | |
|---|---|
| <ul style="list-style-type: none"> • Advantage of long term relationship with suppliers • Possibility of shortage of material or labor for making the component • Uninterrupted supply of requisite quality from reliable supplies | <ul style="list-style-type: none"> • The quality of the product is decided to be controlled • If the purchase price is likely to rise due to increased demand in the market, it becomes uneconomical to buy • Where the technical know-how is to be kept secret and not to be passed on to the suppliers |
|---|---|

Activity6.5

1. *What are the relevant financial considerations in make versus buy decision? ---*

6.4.5. Keep or Replace Equipment Decisions

The usefulness of plant assets may be impaired long before they are considered to be worn out. Equipment may be no longer being efficient for the purpose for which it is

used. On the other hand, the equipment may not have reached the point of complete inadequacy. Decisions to replace usable plants assets should be based on studies of relevant costs. The relevant costs are the future costs of continuing to use the equipment versus replacement. The book values of the plant assets being replaced are sunk costs and are irrelevant.

As for example, assume that a business is considered disposing of several identical machines having a total book value of Birr 1,000,000 and an estimated remaining life of five years. The old machines can be sold for Birr 25,000. They can be replaced by a single high-speed machine at a cost of Birr 250,000. The new machine has an estimated useful life of five years and no residual value. Analyses indicate an estimated annual reduction in variable manufacturing costs from Birr 225,000, with the old machine to Birr 150,000 with the new machine. No other changes in the manufacturing costs or the operating expenses are expected. The relevant costs are summarized in the differential report are as follows:

Proposal for Replacement Equipment (Differential Analysis Report – Replacement Equipment):

Annual variable costs of present equipment (a)	Birr 225,000
Annual variable costs - new equipment (b)	<u>150,000</u>
Annual differential decrease in cost(c= a-b)	<u>Birr 75,000</u>
Number of years applicable (d)	5
Total differential decrease in cost (e=cxd)	Birr 375,000
Proceed from sales of present equipment (f)	<u>25,000</u>
total (g =e+f)	Birr 4, 00,000
Cost of new equipment (h)	<u>2, 50,000</u>
Net differential decrease in cost, 5 year total (i =g+h)	<u>Birr 1, 50,000</u>
So, annual net differential decrease in cost – new equipment (i÷d)	<u>Birr 30,000</u>

Additional factors are often involved in equipment replacement decisions. For example, differences between the remaining useful life of the old equipment and the estimated life of the new equipment could exist. In addition, the new equipment might improve the overall quality of the product, resulting in an increase in sales volume. Other factors that could be significant include the time value of money and other uses for the cash needed to purchase the new equipment.

In general, in deciding whether to replace or keep existing equipment, four commonly encountered items considered in relevance:

- i. Book value of old equipment: irrelevant, because it is a past (historical) cost. Therefore, depreciation on old equipment irrelevant.
- ii. Disposal value of old equipment: relevant, because it is an expected future inflow that usually differs among alternatives.
- iii. Gain or loss on disposal: this is the algebraic difference between book value and disposal value. It is therefore, a meaningless combination of irrelevant and relevant items. Consequently, it is best to think of each separately.
- iv. Cost of new equipment: relevant, because it is an expected future outflow that will differ among alternatives. Therefore, depreciation on new equipment is relevant.

5.3.4. PRICING DECISIONS

Companies are constantly making product and service pricing decision. These are strategic decision that affects the quantity produced and sold, and therefore cost and revenues. To make these decisions, managers need to understand cost behavior pattern and cost drivers. They can then evaluate demand at different prices and manage costs across the value chain and over a products life cycle to achieve profitability.

Major influences on pricing decision

How companies prices a product or a service ultimately depends on the demand and supply of it. Three influences on demand and supply are:-

- i. Customers: - customer influences price through their effect on the demand for a product or services, based on factors such as the features of a product and its quality.
- ii. Competitors: when there are competitors, knowledge of rivals' technology, plant capacity, and operating policies enables a company to estimate its competitors' costs-valuable information in setting its own prices.
- iii. Costs – costs influence prices because they affect supply. As companies supply more product the cost of producing each additional unit initially declines but then eventually increase managers who understand the cost of producing their companies product set policies that make the products attractive to customers. In computing the relevant costs for a pricing decision, the manager must consider relevant costs in all business functions of the value chain.

Costing and pricing for the short run

Short-run pricing decisions typically have a time horizon of less than a year and include decision such as (a) pricing one time only special order with no long run implications and (b) adjusting product mix and output volume in a competitive market.

Company's short run pricing decisions need identify a sufficiently low price at which company would still make a profit and assumed that (a) company has access to extra capacity and (b) a competitor with an efficient plant and idle capacity was likely to make a low bid. However, short run pricing does not always work this way. Companies may experience strong demand for their products in the short-run, but they may have limited capacity. In these cases, companies strategically increase prices in the short run to as much as the market will bear.

In general, short run pricing decisions are responses to short-run demand and supply condition, and the relevant costs are only those costs that will change in the short run.

Costing and pricing for the long run

Long run pricing decisions have a time horizon of a year or longer and include pricing a product in a major market in which there is some see way in setting price. Two key differences affect pricing for the long run versus the short run:-

1. Costs that are often irrelevant for short run pricing decisions, such as fixed costs that cannot be changed, are generally relevant in the long run because cost can be altered in the long run.
2. Profit margins in the long run pricing decision are often set to earn a reasonable return on investment. Short run is opportunistic, prices are decreased when demand is weak and increased when demand is strong.

Long run pricing is a strategic decision desired to build long run relationship with customers based on stable and predictable prices. But to change a stable price and earn the target long run return, a company must, over the long run, know and manage its costs of supplying product to customers. Thus, relevant costs for long run pricing decision include all future fixed and variable costs.

Long run pricing approaches

Two different approaches for pricing decision using product cost information are:-

1. Market based approach

2. Cost based/cost plus approach

1. Market based pricing

Market based pricing approach starts by management asking, given that our customers want and how our competitors will react to what we do, what price should we charge?

Companies operating in a very competitive market, for example, commodities such as steel, oil, and natural gas, use the market based pricing. An important form of market based pricing is target pricing. Target price is the estimated price for a product or service that potential customers will be willing to pay. This estimate is based on an understanding of customer's perceived value for a product or service and how competitors will price competing product or service.

Hence, target operating income is the operating income that a company wants to earn on each unit of a product or service sold and target price leads to a target cost, target cost per unit is the estimated long run cost per unit of a product or service that, when sold at the target price, enables the company to achieve the target operating income.

Thus, $\text{Target price} - \text{Target operating income} = \text{Target cost}$

Implementing target pricing and target costing

In developing target prices and target cost companies may require to follow the following five steps:

- Develop a product that satisfy the needs of potential customers
- Choose a target price based on customer's perceived value for the product and the price competitors charge, and target operating income per unit.
- Drive a target cost per unit by subtracting the target operating income per unit from the target price
- Perform cost analysis to analyze which aspects of a product or service to target for cost reduction.
- Perform value engineering to achieve target cost. Value engineering is a systematic evaluation of all aspect of the value chain business function with the objective of reducing cost while satisfying customers' needs. Value engineering can result in improvement in product design, change in material specification, and modification in process method. In this case, Costs can be value adding or non value adding. Value adding costs are costs that costumers perceive as adding utility or value while non value adding cost that do not add value to the product

and to customers. Value engineering will focus on eliminating non value adding cost and reduce as much as possible value adding cost without affecting quality of the product and customers satisfaction.

Example 6.6: Astel Company is a manufacturer of personal computer. Astel expects its competitors to lower prices of PC. Astel's management believes that it must respond by reducing price by 20% from Br. 1000 per unit to Br. 800 per unit. At this low price, Astel's marketing manager forecasts an increase in annual sales from 150,000 to 200,000 units. Astel management wants a 10% target operating income on sales revenue. The total production cost at the moment for 150,000 units is Br. 135 million.

Required compute

- a) The total target revenue
- b) Total target operating income
- c) Target operating income per unit
- d) Current target cost per unit

Solution

- a) Total target revenue = target price per unit x target annual unit sold
$$= \text{Br. } 800 \text{ per unit} \times 200,000 \text{ units} = \underline{\text{Br. } 160,000,000}$$
- b) Total target operating income = target rate x Total target revenue
$$= 10\% \times \text{Br. } 160,000,000 = \underline{\text{Br. } 16,000,000}$$
- c) Target operating income per unit = Total target operating income / annual unit sold
$$= \text{Br. } 16,000,000 / 200,000 \text{ units} = \underline{\text{Br. } 80}$$
- d) Current cost per unit = target price per unit less target operating income per unit
$$= \text{Br. } 800 \text{ per unit} - \text{Br. } 80 = \underline{\text{Br. } 720}$$

2. Cost-plus pricing

Accounting information may be used in pricing decisions, particularly where the firm is a market leader or *price-maker*. In these cases, firms may adopt **cost-plus pricing**, in which a margin is added to the total product/service cost in order to determine the selling price. In many organizations, however, prices are set by market leaders and competition requires that prices follow the market (i.e. the firms are *price-takers*). Nevertheless, even in those cases an understanding of cost helps in making management decisions about what product/services to produce, how many units to make and whether the price that exists in the market warrants the business risk involved in any decision to sell in that

market. An understanding of the firm's marketing strategy is therefore, essential in using cost information for pricing decisions.

In the long term, the prices that businesses charge must cover all of its costs. If it is unable to do so, it will make losses and may not survive. For every product/service, the full cost must be calculated, to which the desired profit margin is added. Full **cost** includes an allocation to each product/service of all the costs of the business, including producing and delivering a good or service, and all its marketing, selling, finance and administration costs.

The general formula for setting a cost based price adds a markup component to the *cost base to determine the prospective selling price*. One way to determine the markup percentage is to choose a markup to earn a target rate of return on investment.

The target rate of return on investment is the target annual operating income that an organization aims to achieve divided by invested capital (asset)

$$\text{i.e. TRR} = \frac{\text{Target operating income}}{\text{Invested capital}}$$

Therefore, Target operating income = TRR * Invested capital

Let illustrate a cost – plus pricing formula on top company. Assume top's engineers have redesigned product CD into 2CD and that top uses a 12% markup on the full unit cost of the product in developing the prospective selling price. The target product 2CD profitability for 2000 is as follows:

	Estimated total amounts for 200,000 units (1)	Estimated total amount per unit (2) = (1) ÷ 200,000
Revenues	Bir 160,000,000	Bir 800
Cost of goods sold	108,000,000	540
Operating costs	<u>36,000,000</u>	<u>180</u>
Total cost of product	<u>Bir 144,000,000</u>	<u>720</u>
Operating income	<u>16,000,000</u>	<u>Bir 80</u>

Suppose that top's target rate of return on investment is 18% and 2CD's capital investment is Bir 96 million. The target annual operating income for 2CD is:

Invested capital Bir 96,000,000

Target rate of return on investment.....	18%
Target Annual Operating income [$0.18 \times \text{Bir } 96\text{mln}$]...Bir17,280,000	
Target operating income per unit of 2A	
[Bir17,280,000 $\div$ 200,000 units]	Bir 86.40

This calculation indicates that top needs to earn a target operating income of Bir86.40 on each unit of 2A. The mark up of Bir 86.40 expressed as a percentage of the full production cost per unit of Bir720 equals 12% (Bir 86.40 $\div$ Bir 720]

Thus the prospective selling price of product 2A is Bir806.40 (Full unit cost of 2A, Bir 720 plus the markup component of 12% ($0.12 \times \text{Bir } 720 = \text{Bir } 86.40$)).

Activity 6.6.

Differentiate the different long term pricing approach -----

UNIT SUMMARY

Relevant information is logically related and pertinent to a given decision. Relevant information may be both quantitative and qualitative. Variable costs are generally relevant to a decision; they are irrelevant only when they cannot be avoided under any possible alternative or when they do not differ across alternatives. Direct avoidable fixed costs are also relevant to decision making. Sometimes costs give the illusion of being relevant when they actually are not. Examples of such irrelevant costs include sunk costs, arbitrarily allocated common costs, and no incremental fixed and variable costs.

Relevant costing compares the incremental revenues and/or costs associated with alternative decisions. Managers use relevant costing to determine the incremental benefits of decision alternatives. One decision is established as a base line against which the alternatives are compared. In many decisions the alternative of “change nothing” is the obvious base line case.

Common situations in which relevant costing techniques are applied include asset replacements, outsourcing decisions, scarce resource allocations, special price determinations, sales mix distributions, and retention or elimination of product lines.

The following points are important to remember:

1. In an asset replacement decision, costs paid in the past are not relevant to decisions being made currently; these are sunk costs and should be ignored.

2. In an outsourcing decision, include the opportunity costs associated with the outsource alternative; nonproduction potentially allows management an opportunity to make plant assets and personnel available for other purposes.
3. In a decision involving a single scarce resource, if the objective is to maximize company contribution margin and profits, then production and sales should be focused toward the product with the highest contribution margin per unit of the scarce resource.
4. In a special order decision, the minimum selling price that a company should charge is the sum of all the incremental costs of production and sales on the order.
5. In a product line decision, product lines should be evaluated based on their segment margins rather than on net income. Segment margin captures the change in corporate net income that would occur if the segment were discontinued.

Quantitative analysis is generally short range in perspective. After analyzing the quantifiable factors associated with each alternative, a manager must assess the merits and potential risks of the qualitative factors involved to select the best possible course of action. Some of these qualitative factors (such as the community economic impact of closing a plant) may present long-range planning and policy implications. Other qualitative factors may be short range in nature, such as competitor reactions. Managers must decide the relevance of individual factors based on experience, judgment, knowledge of theory, and use of logic.

MODEL EXAMINATION QUESTIONS

A) SHORT ANSWER QUESTIONS

1. Can a particular cost be relevant for one purpose, but not for other purposes? Give three examples in which this would be the case.
2. Are sunk costs ever relevant in decision making? If so, give one or more examples.
3. You are considering the sale of your old stereo system. According to your records, you paid Br. 500 for the stereo system. The current market value of the stereo is Br. 150. A new stereo of the same make and model could be purchased today for Br. 375. Which of these figures is relevant to your decision to sell or keep the stereo system? If any figures are not relevant, explain why.

4. Kelly XY, owner of Mexican Cafe, is trying to decide whether to make Enjera or buy them from a supplier. Kelly has come to you for advice. What factors would you tell her to consider in making her choice?
5. What is a scarce resource? Why will the resource that is most scarce in an organization be likely to change from time to time?
6. What is special order decision? What typical circumstances lead to the need to make this type of decision?
7. What are the differences among avoidable fixed costs, unavoidable direct fixed costs, and common fixed costs? Which are relevant and which are irrelevant in the decision to keep or eliminate a particular product line?

B) WORKOUT QUESTIONS

- 1) Belt and Braces Ltd makes a single product which sells for Br 20. It has a full cost of Br 15 which is made up as follows:

Direct Material	Br 4
Direct Labor	6
Variable Overhead	2
General Fixed Overhead	<u>3</u>
	Br. <u>15</u>

The labor force is currently working at 90% of capacity and so there is a spare capacity for 2,000 units. A customer has approached the company with a request for the manufacture of a special order of 2,000 units for which he is willing to pay Br. 25,000. Assess whether the contract should be accepted or not.

- 2) Buster Ltd makes four components, W, X, Y and Z, for which costs in the forthcoming year are expected to be as follows.

	<u>W</u>	<u>X</u>	<u>Y</u>	<u>Z</u>
Production (units)	1,000	2,000	4,000	3,000

Unit variable costs

Direct materials	Br. 4	Br 5	Br 2	Br 4
Direct labor	8	9	4	6
Variable production overheads	<u>2</u>	<u>3</u>	<u>1</u>	<u>2</u>
	<u>Br 14</u>	<u>Br 17</u>	<u>Br 7</u>	<u>Br 12</u>

Directly attributable fixed cost per annum and other fixed costs are as follows:

Incurred as a direct consequence of making W	Br 1,000
Incurred as a direct consequence of making X	5,000
Incurred as a direct consequence of making Y	6,000
Incurred as a direct consequence of making Z	8,000
Other fixed costs (committed)	<u>30,000</u>
Total fixed costs	<u>Br 50,000</u>

A subcontractor has offered to supply units of W, X, Y and Z for Br 12, Br 21, Br 10, and Br 14 respectively.

Required: Decide whether Buster Ltd should make or buy the components and mention some qualitative factors to be considered by Buster Ltd in decision making

3) Great Company manufacturers 60,000 units of part XL – 40:

	Total costs 60,000 units	Cost per unit
Direct material	Br 480,000	Br 8
Direct labor	360,000	6
Variable factory overhead (FOH)	180,000	3
Fixed FOH	<u>360,000</u>	<u>6</u>
Total manufacturing costs	<u>Br 1,380,000</u>	<u>Br 23</u>

Another manufacturer has offered to sell the same part to Great for Br 21 each. The fixed overhead consists of depreciation, property taxes, insurance, and supervisory salaries. The entire fixed overhead would continue if the Great Company bought the component except that the cost of Br 120,000 pertaining to some supervisory and custodial personnel could be avoided.

Required:

a. Should the parts be made or bought? Assume that the capacity now used to make parts internally will become idle if the parts are purchased?

- b. Assume that the capacity now used to make parts will be either (i) be rented to nearby manufacturer for Br 60,000 for the year or (ii) be used to make another product that will yield a profit contribution of Br 250,000 per year. Should the company purchase them from the outside supplier?

ANSWER TO MODEL EXAMINATION QUESTIONS

A. SOLUTIONS FOR SHORT ANSWER QUESTIONS

- I) For questions 1, 2, 3, 4, 5, and 6 refer to section 6.3
- II) For questions 7 refer to section 6.4.5
- III) For questions 8 refer to section 6.4.4
- IV) For questions 9 refer to section 6.4.3
- V) For questions 10 refer to section 6.4.1
- VI) For questions 11 refer to section 6.4.2

B. SOLUTIONS FOR WORKOUT QUESTIONS

Q1

	Br.	Br.
Value of order		25,000
Cost of order		
Direct materials (Br. 4 x 2,000)	8,000	
Direct labor (Br. 6 x 2,000)	12,000	
Variable overhead (Br 2 x 2,000)	<u>4,000</u>	
Relevant cost of order		(<u>24,000</u>)
Profit form order acceptance		<u>1,000</u>

Fixed costs will be incurred regardless of whether the special order is accepted and so are not relevant to the decision. The contract should be accepted since it increases contribution to profit by Br 1, 000.

Other factors to be consider in the special order decision.

- a) The acceptance of the special order at a lower price may lead other customers to demand lower prices as well.
- b) There may be more profitable ways of using the spare capacity.

- c) Accepting the special order may lock up capacity that could be used for future full-price business.
- d) Fixed costs may exist, in fact, if the contract is accepted.

Q2

a. The relevant costs are the differential costs between making and buying, and they consist of difference in unit variable costs plus differences in directly attributable fixed costs. Subcontracting will result in some fixed cost savings.

	<u>W</u>	<u>X</u>	<u>Y</u>	<u>Z</u>
Unit variable cost of making	Br. 14	Br 17	Br 7	Br 12
Unit variable cost of buying	<u>12</u>	<u>21</u>	<u>10</u>	<u>14</u>
	Br (2)	Br 4	Br 3	Br 2
	<u>W</u>	<u>X</u>	<u>Y</u>	<u>Z</u>
Annual requirements (units)	1,000	2,000	4,000	3,000
Extra variable cost of buying (per annum)	Br. (2,000)	Br. 8,000	Br.12, 000	Br. 6,000
Fixed costs saved by buying	<u>(1,000)</u>	<u>(5,000)</u>	<u>(6,000)</u>	<u>(8,000)</u>
Extra total cost of buying	(3,000)	3,000	6,000	(2,000)

b. The company would save Br 3,000 by subcontracting component W (where the purchase cost would be less than the variable cost per unit to make internally) and would save Br 2,000 by subcontracting component Z (because of the savings in fixed costs of Br 8,000).

c. In this question, relevant costs are the variable costs in-house manufacture the variable costs of subcontracted units, and the saving in fixed costs.

Q3

To approach the decision from a financial point of view, the manager must focus on the relevant or differential costs. The differential cost can be obtained by eliminating from the cost data those costs that are not avoidable – that is, by eliminating the sunk costs and the future costs will continue regardless of whether the parts XL – 40 are produced internally or purchased from outside. Thus, the relevant cost computation follows:

Cost To Make	Cost To Buy
---------------------	--------------------

	Per unit	Total	Per unit	Total
Purchase Cost	- 0 -	- 0 -	21.00	1,260,000
Direct materials	Br 8.00	Br 480,000		
Direct labor	6.00	360,000		
Variable FOH	3.00	180,000		
Fixed FOH, avoidable	<u>2.00</u>	<u>120,000</u>		
Total cost	<u>Br 19.00</u>	<u>Br 1,140,000</u>	<u>Br 21.00</u>	<u>Br 1,260,000</u>

Recommendation: Great Company should reject the outside supplier's offer because it costs Br 2 less per unit to continue to make the part – XL – 40.

Relevant costs	Per unit
Cost to buy	Br 21.00
Cost to make	<u>19.00</u>
Advantage of making the part internally	<u>Br 2.00</u>

Total advantage = Br 2.00 x 60,000units = Br 120,000

- a. Assuming the space now being used to produce part XL – 40 would be
- Rented to a nearby manufacture of Br 60,000 per annum or
 - Used to produce other product that contributes a profit of Br 250,000 per year, the relevant cost computation follows:

	Make	Buy and Leave Facility Idle	Buy and Rent out	Buy and Produce Other Product
Cost to obtain parts	Br1,140,000	Br1,260,000	Br1,260,000	Br 1,260,000
Contribution from other products	-	-	-	(250,000)
Rent revenue	=	=	(60,000)	
Net relevant costs	<u>Br1,140,000</u>	<u>Br1,260,000</u>	<u>Br 1,200,000</u>	<u>Br 1,010,000</u>

Great Company would be better off through accepting the supplier's offer and to using the available facility to produce the new product line. This move has the least net relevant cost of Br 1,010,000.

CHAPTER -7

DECENTRALIZATION AND TRANSFER PRICING

UNIT OUTLINE

- 7.1.Introduction
- 7.2.Decentralization
- 7.3.Transfer pricing

UNIT OBJECTIVES

- Describe management control system
- Describe the benefits and costs of decentralization
- Explain transfer pricing and the criteria used to evaluate
- Calculate transfer pricing using different methods

7.1 INTRODUCTION

A management control system is a means of gathering and using information to aid and coordinate the planning and control decisions throughout an organization and to guide the behavior of its managers and other employees.

In a small business, virtually all plans and decisions can be made by one individual. As a business grows or its operation become more diverse, it becomes difficult, if not impossible for one individual to perform these functions.

Management accounts, therefore, must have the interpersonal and analytical skills necessary to evaluate and implement management control system, as well as the ability to interpret outputs of these systems, and to be effective, management control systems should be closely aligned to the company's strategies and goals and also must fit an organization's structure.

7.2 DECENTRALIZATION

 ***Dear learner! Have you made any decision for any department before?***

As a business grows, it is difficult for one manager to manage the whole activity. Hence, they need to delegate responsibility for portions of operation. This separation of a business into more manageable units is termed as **Decentralization**, the freedom of manager at lower levels of the organization to make decision. The process of measuring

and reporting operating activity by area of responsibility is called responsibility accounting.

Benefit of Decentralization	Limitation of Decentralization
Create greater response to local needs Quicker decision making Increase motivation Increase creativity and productivity Aids management development Sharpens the focuses of managers	Suboptimal decision Duplication of asset & activity Decrease loyalty to the organization as a whole Increase cost of gathering information

Activity 7.1

1. *Define what decentralization means? And describe the merit and demerit of it.....*

7.3 TRANSFER PRICING

In decentralized organizations, much of the decision, making power resides in its individual submits. In these cases, the management control system often uses transfer prices to coordinate the actions of the subunit and to evaluate their performance.

A transfer price is the price one subunit (department or division) charges for a product or service supplied to another subunit of the same organization. If, for example, a car manufacturer has a separate division that manufactures engines, the transfer price is the price the engine division charges when it transfers engine to the assembly division. The transfer price creates revenues for the selling subunit (the engine division in our example) and purchase cost for the buying subunits (the assembly division), affecting each subunit operating income. These operating incomes can be used to evaluate subunits performance and to motivate their manager. The products or services transferred between sbunits of the organization are called intermediate products. These products may either be further worked on by the receiving subunit or, if transferred from production to marketing, sold to an external customer.

The three methods for determining transfer pricing are

1. Market-based transfer prices

In this case, top management may choose to use the price of a similar product or service publicly listed, say, a trade association web site. Also top management may select, for the internal price, the external price that subunit charges to outsider customers. This method is preferred, (a) when the intermediate market is perfectly competitive, (b) Interdependence of sub units is minimal, and (c) There are no other additional costs of using market price.

2. Cost-based transfer price:

Top management may choose a transfer price based on the cost of producing the product in question. The cost used in the cost based transfer price can be the actual cost or the budgeted cost. Sometimes, the cost based transfer price includes the mark up or profit margin that represents a return on subunit investment. This method is used when market price is unavailable, inappropriate or too costly to obtain. In this case, variable or full cost can be used as a base

3. Negotiated transfer price

In some cases, the subunit of the company are free to negotiate the transfer price between themselves and then to decide whether to buy and sell internally or dealing with external prices. Subunit may use information about costs and market price in these negotiations, but there is no requirement that the chosen transfer price bear any specific relationship to either cost or market price data. Thus, negotiated transfer price is the outcome of a bargaining process between the selling and buying divisions.

Example 7.1:

Horizon Petroleum Company has two divisions. Each division operates as a profit center. The transportation division manages the operation of pipeline that transfers crude oil from Mexico to Texas. The refining division manages a refinery at Texas that process crude oil into gasoline. Gasoline is the only salable product the refinery makes and that it takes two barrels of crude oil to yield one barrel of gasoline

Variable cost in each division is assumed to be variable with respect to single cost driver in each division: Barrels of crude oil transported by the transportation division, barrel of gasoline produced by the refining division. The fixed cost per unit is based on the budgeted annual output of crude oil to be produced and transferred and the amount of

gasoline to be produced. Horizon petroleum reports all costs and revenues of its non US operation in US dollars using the prevailing exchange rate.

	Transport	Refining
	<u>Division</u>	<u>Division</u>
VC per unit	Bir 1	Bir 8
FC per unit	<u>3</u>	<u>6</u>
Total	<u>Bir 4</u>	<u>Bir 14</u>

Additional information's

- The production division can sell crude oil to transport division in Mexico at Bir12 per barrel
- The transport division buys crude oil from the production division in Mexico and sells it to the refining division
- The refining division can buy crude oil in Texas from external supplier at Bir21 per barrel and can sell the gasoline it produces at Bir58per barrel.
- The three divisions have sufficient capacity

Assume that 100 barrel of crude oil produced by production division is transported to the refining division and assuming the following transfer pricing methods

Method A: Market based transfer price of Bir21 per barrel of crude oil based on the competitive market price

Method B:Cost based transfer price at 110% of full cost where full cost are the cost of transferred in product plus the divisions own variable and fixed cost.

Method C: Negotiated transfer price of Bir19.75 per barrel crude oil

Required

Compute the operating income for Horizon Petroleum Company and for each division under each transfer pricing method.

Solution

- i) *Horizon* Petroleum total operating income from purchasing, transporting and refining the 100 barrels of crude oil and selling of the 50 barrels of gasoline is the same, Bir600, regardless of the internal transfer price used.

Thus, Operating income would be:

Revenues: (Bir58x50 barrels of gasoline).....	Bir 2, 900	
Less: cost of crude oil purchases (Bir12x100) barrels of crude oil).....	Bir 1,200	
Transportation cost (Bir4(1+3) x100 barrels of crude oil).....	400	
Refining costs (Bir14(6+8) x50 barrels of gasoline).....	<u>700</u>	<u>2,300</u>

Operating income..... Bir600

ii) Division operating income of horizon petroleum for 100 barrels of crude oil under alternative transfer pricing methods would be:

	Internal transfers at market price of Bir21/barrel	Internal transfer at 110% of full cost = Bir17.60	Internal transfer negotiated at Bir19.25
<u>Transportation division</u>			
Revenues, (a)	Bir21x100 = Bir2,100	Bir17.60x100 = Bir1760	Bir19.25x100=Bir1,925
Costs:			
Crude oil purchase costs (Bir12x100)	Bir1200	Bir1200	Bir1200
Division VC (Bir1x100)	100	100	100
Division FC(Bir3x100)	300	300	300
Total division costs (b)	<u>Bir1600</u>	<u>1600</u>	<u>Bir1600</u>
Division operating income(a-b)	<u>Bir500</u>	<u>Bir160</u>	<u>Bir325</u>
<u>Refining Division</u>			
Revenues (Bir58x50 barrels of gas oil)	Bir2900	Bir2900	Bir2900
Costs:			
Transportation in cost	Bir21x100 = Bir2,100	1,760	1,925
Division VC (Bir8x50)	400	400	400
Division FC(Bir6x50)	<u>300</u>	<u>300</u>	<u>300</u>
Total division costs (b)	<u>Bir2800</u>	<u>Bir2460</u>	<u>Bir2625</u>
Division operating income (a-b)	<u>Bir100</u>	<u>Bir440</u>	<u>Bir275</u>
Operating income of both division	<u>Bir600</u>	<u>Bir600</u>	<u>Bir600</u>

MODEL EXAMINATION QUESTIONS

Discussion Question

1. What is management control system and how should it be desired?
2. What are the benefits and costs of decentralization?
3. What is a transfer price and what is it intended to achieve?
4. What methods can be used to calculate transfer prices?